

Data Communication & Computer Network

Lecture – 1 & 2

Presented By
Dr. S.Kanungo

A Communications Model

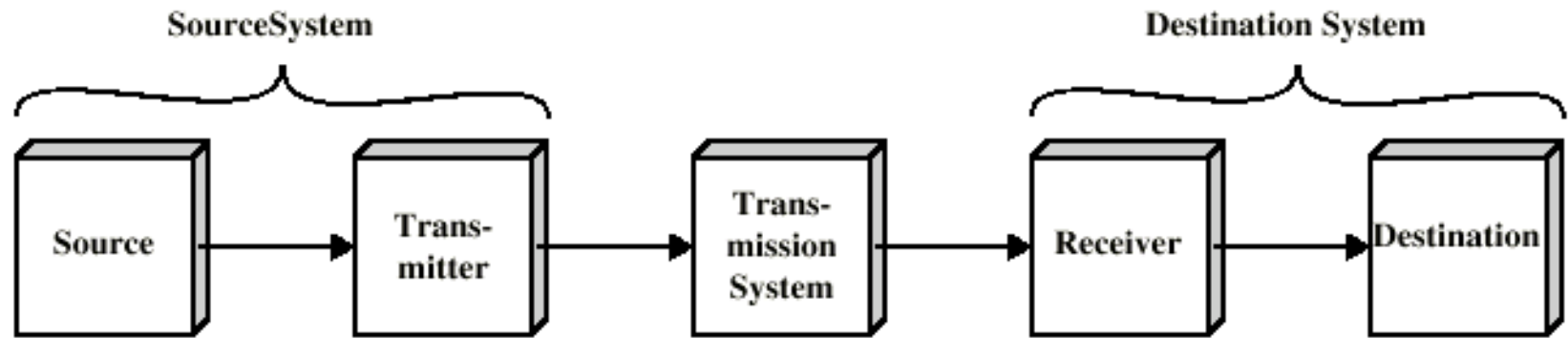
Key Elements of the Model:

- Source: - generates data to be transmitted
Examples: Telephones and PCs
- Transmitter: - Converts data into transmittable signals
- Transforms and encodes the information to produce electromagnetic signal. Example: Modem (transforms digital bit stream to analog signal)
- Transmission System: - Carries data
- Can be a single transmission line or a complex network connecting source and destination
- Receiver: - Converts received signal into data
- Example: Modem (transform analog signal to digital bit stream)
- Destination: Takes incoming data from receiver

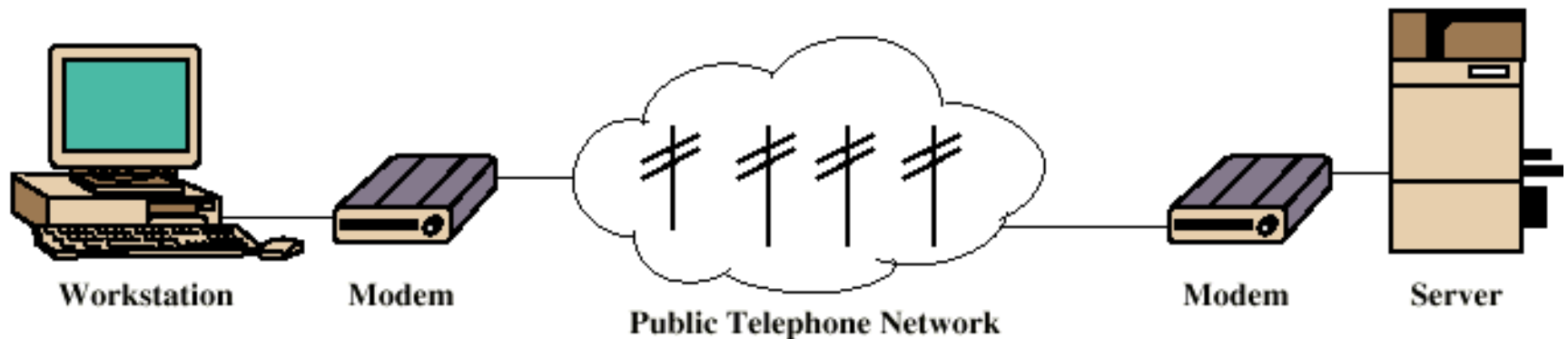
Communications tasks

- Transmission system utilization: efficient use of transmission facility
- Addressing
- Interfacing
- Routing
- Signal generation (Electromagnetic signal)
- Recovery(in case of interruption due to a fault somewhere in the system)
- Synchronization (identify begin/end of signal, duration of each signal element)
- Message formatting(agreement for form of data to be exchanged/transmitted: Eg. binary code for characters)
- Exchange management (conventions for transmission)
- Security
- Error detection and correction
- Network management(system configuration, monitoring, failure and overload handling)
- Flow control(to assure that the source does not overwhelm the receiver)

- **Simplified Communication Model**

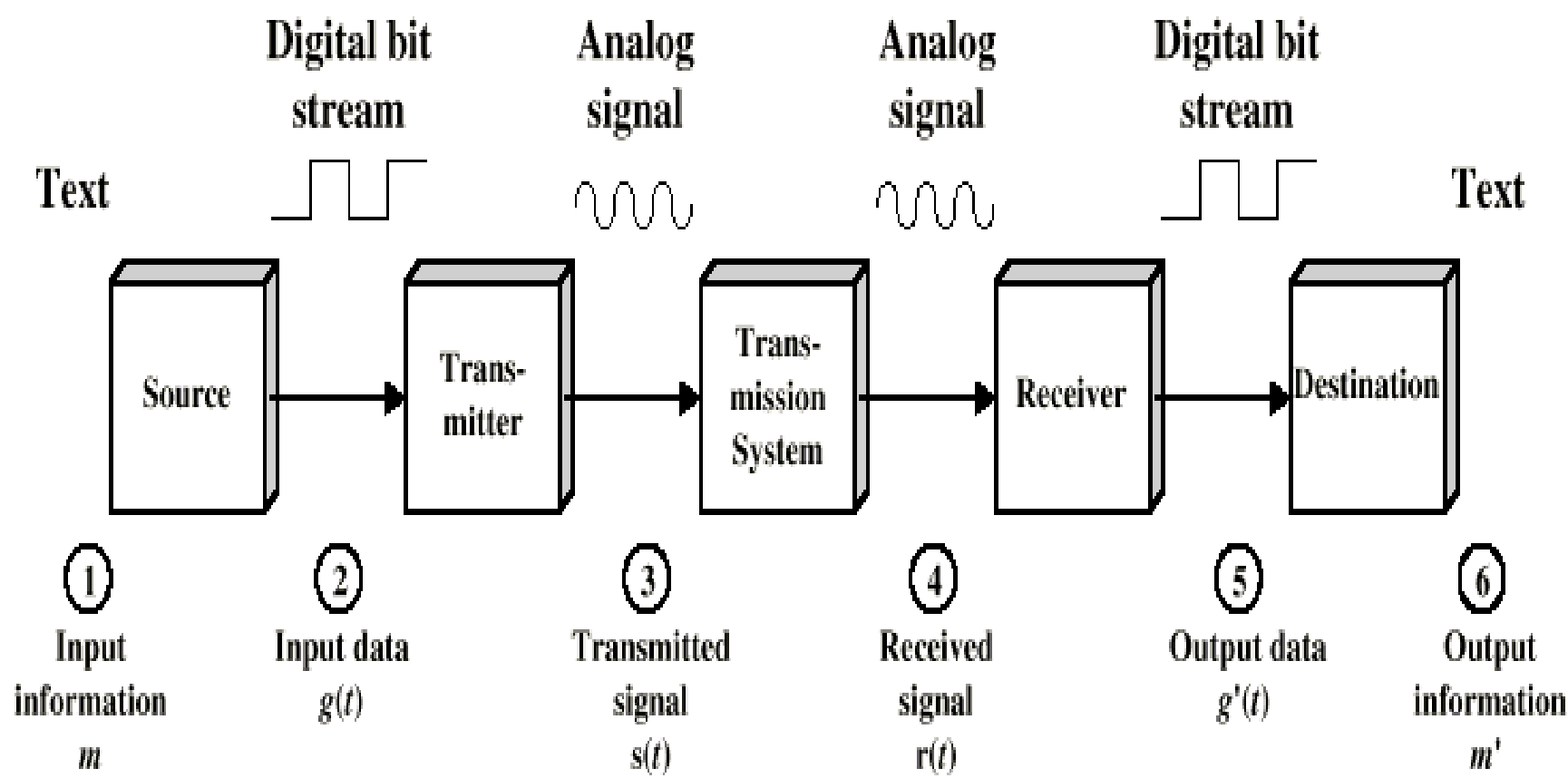


(a) General block diagram



(b) Example

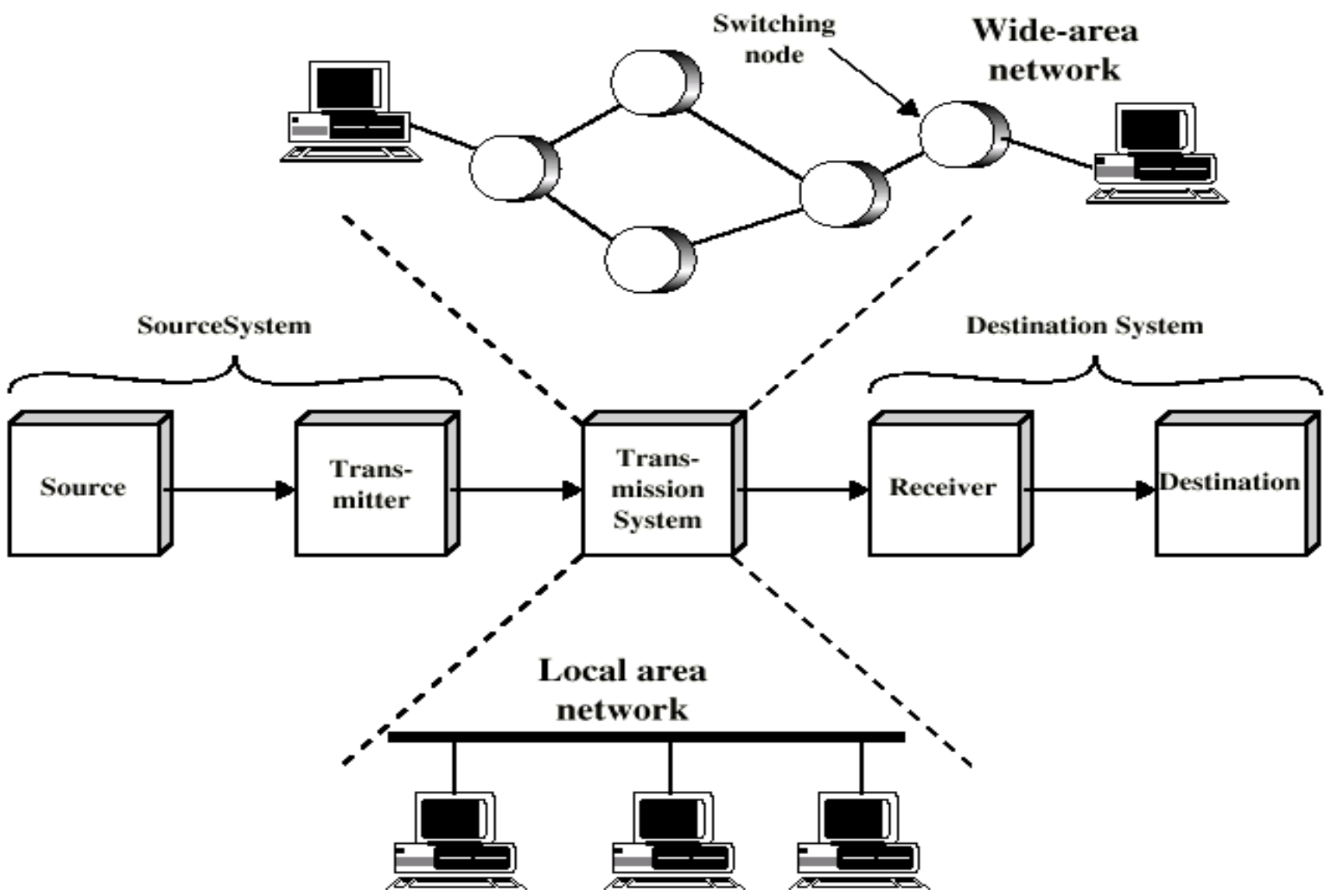
- # Simplified Data Communication Model



Networking

- Point to point communication not usually practical
 - Devices are too far apart
 - Large set of devices would need impractical number of connections
- Solution is a communications network
- Network: Interconnection of systems
- Computer Network: It is communication system that links **end systems** by communication lines and software protocols to exchange data between two processes running on different end systems of the Network. [Use: message transfer/ File transfer/Resource sharing]
- End Systems: nodes/sites/hosts/computers/machines

- Simplified Network Model:



WAN (Wide Area Network)

- Large geographical area
- Crossing public rights of way
- Rely at least in part on common carrier circuits
- Alternative technologies
 - Circuit switching
 - Packet switching
 - Frame relay
 - Asynchronous Transfer Mode (ATM)
- Circuit Switching: Dedicated communications path established for the duration of the conversation. e.g. telephone network

- **Packet Switching**

- Data sent out of sequence
- Small chunks (packets) of data at a time
- Packets passed from node to node between source and destination
- Used for terminal to computer and computer to computer communications

- **Frame Relay**

- Packet switching systems have large overheads to compensate for errors
- Modern systems are more reliable
- Errors can be caught in end system
- Most overhead for error control is stripped out

- **ATM (Asynchronous Transfer Mode)**
 - Evolution of frame relay
 - Little overhead for error control
 - Fixed packet (called cell) length
 - Anything from 10Mbps to Gbps
 - Constant data rate using packet switching technique
- **ISDN (Integrated Services Digital Network)**
 - Designed to replace public telecom system
 - Wide variety of services
 - Entirely digital domain
- **LAN (Local Area Network)**
 - Smaller scope
 - Building or small campus
 - Usually owned by same organization as attached devices

- Data rates much higher
- Usually broadcast systems
- Now some switched systems and ATM are being introduced
- **Protocols:** A set of rules and regulations using which a required service can be obtained
 - Used for communications between entities in a system
 - Must speak the same language
 - Entities: User applications, e-mail facilities, etc.
 - Key Features
 - Syntax: Includes things such as data format and signal levels
 - Semantics: Includes control information for coordination and error handling.
 - Timings: Includes speed matching and sequencing

ISO Open Systems Interconnection(OSI) Model

- This model has 7 layers. The principles that were applied to arrive at the layers are:
 - A layer should be created where a different level of abstraction is needed.
 - Each layer should perform a well-defined function.
 - The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
 - The boundaries should be chosen to minimize the information flow across the interfaces.
 - The no. of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity, and small enough that the architecture does not become complex.

Application

Provides access to the OSI environment for users and also provides distributed information services.

Presentation

Provides independence to the application processes from differences in data representation (syntax).

Session

Provides the control structure for communication between applications; establishes, manages, and terminates connections (sessions) between cooperating applications.

Transport

Provides reliable, transparent transfer of data between end points; provides end-to-end error recovery and flow control.

Network

Provides upper layers with independence from the data transmission and switching technologies used to connect systems; responsible for establishing, maintaining, and terminating connections.

Data Link

Provides for the reliable transfer of information across the physical link; sends blocks (frames) with the necessary synchronization, error control, and flow control.

Physical

Concerned with transmission of unstructured bit stream over physical medium; deals with the mechanical, electrical, functional, and procedural characteristics to access the physical medium.

- Computer Networks are implemented using the concept of layered protocols. The main reasons are:
 - The protocols of a network are complex. Designing them in layers makes their implementation more manageable.
 - Layering of protocols provides well-defined interfaces between the layers, so a change in one layer does not affect an adjacent layer.
 - Layering of protocols also allows interaction between functionally paired layers in different locations.
- Interfaces & Services
 - The function of each layer is to provide services to the layer above it.
 - Entities: S/W Entity (Process) or H/W Entity: I/O chips
 - Active elements in each layer
 - Peer Entities: Entities in the same layer on different machines
 - Entities in **layer n** (service provider) implement a service used by **layer n+1** (service user)

- Layer n may use the services of layer $n-1$ in order to provide its service.
- Services are available at SAPs (Service Access Points)
- The layer n SAPs are the places where layer $n+1$ can access the services offered. Each SAP has an address that uniquely identifies it.
- At a typical interface, the layer $n+1$ entity passes an IDU(Interface Data Unit) to the layer n entity through the SAP.
- IDU consists of SDU(Service Data Unit) and ICI(Interface Control Information)
- To transfer the SDU the layer n entity may have to fragment it into several pieces, each of which is given a header and sent as a separate PDU (Protocol Data Unit) such as a packet.

Data Communication & Computer Network

Lecture – 3, 4 & 5

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The Need for Protocol Architecture

Example: Transfer of file between two computer

Typical tasks to be performed:

1. The source system must either activate the direct data communication path or inform the communication network of the identity of the desired destination system.
2. The source system must ascertain that the destination system is prepared to receive data.
3. The file transfer application on the source system must ascertain that the file management program on the destination system is prepared to accept and store the file for this particular user.
4. If the file formats used on the two systems are incompatible, one or the other system must perform a format translation function.

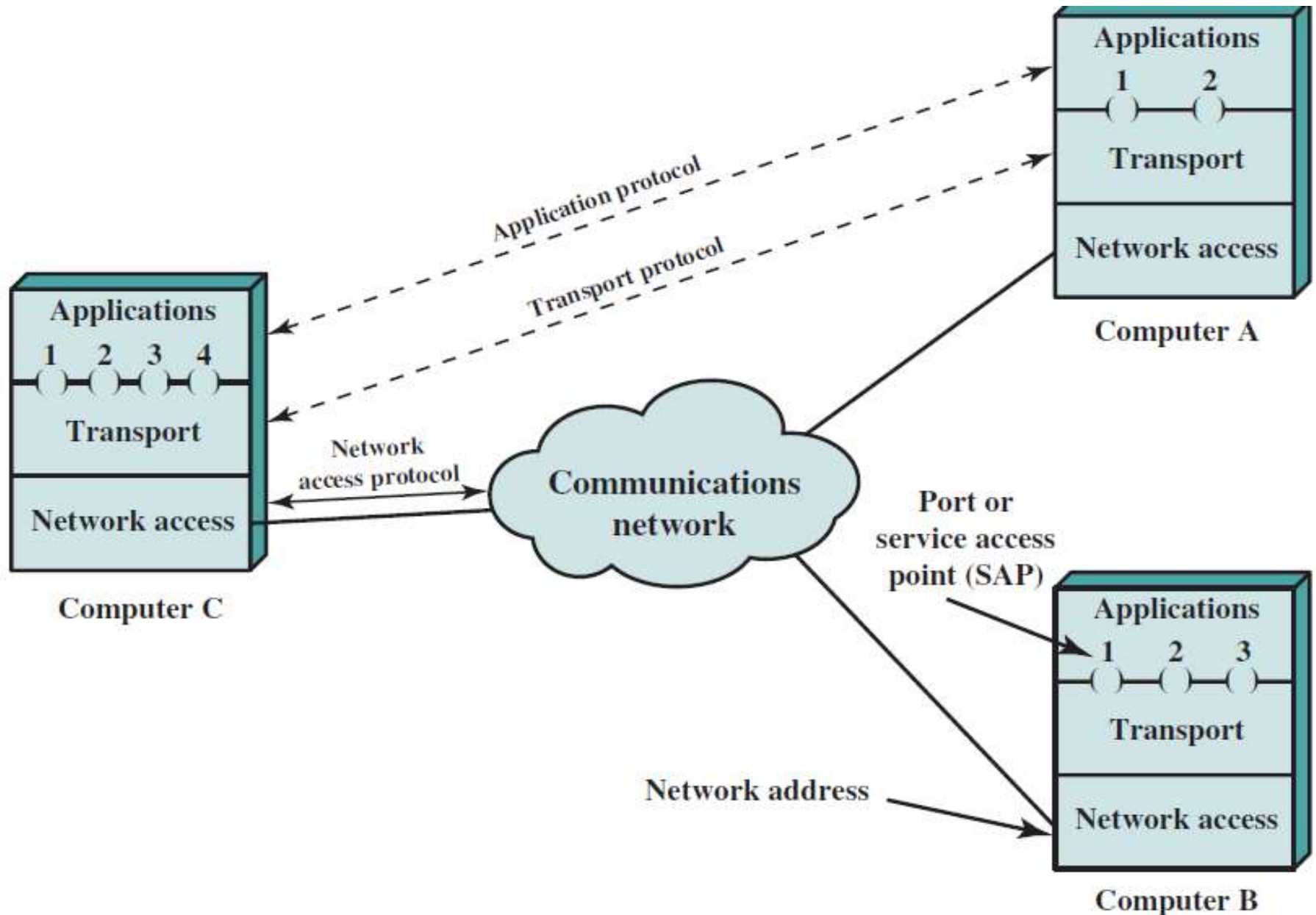
Thus, a high degree of cooperation is required between two computer systems

A Simple Computer Architecture

- Distributed data communication involve three agents:
 - Applications, Computers, Networks
- => Three Layers: Network Access Layer, Transport Layer, Application Layer
 - Network Access Layer: Concerned with exchange of data between a computer and the network to which it is attached.
 - Transport Layer: Reliable exchange of data
 - Application Layer: Contains the logic needed to support the various user applications

Let us trace a simple operation. Suppose that an application, associated with port 1 at computer A, wishes to send a message to another application, associated with port 2 at computer B. The application at A hands the message over to its transport layer with instructions to send it to port 2 on computer B. The transport layer hands the message over to the network access layer, which instructs the network to send the message to computer B. Note that the network need not be told the identity of the destination port. All that it needs to know is that the data are intended for computer B.

• Protocol Architectures and Networks



- **The sending application generates a block of data and passes this to the transport layer(TL).**
- **The TL may break this block into smaller pieces**
- **The TL appends a transport header to each of these pieces, containing protocol control information.(Encapsulation)**
- **Transport PDU (segment): Data from next higher layer + control information**
- **Control information used by the peer transport protocol**

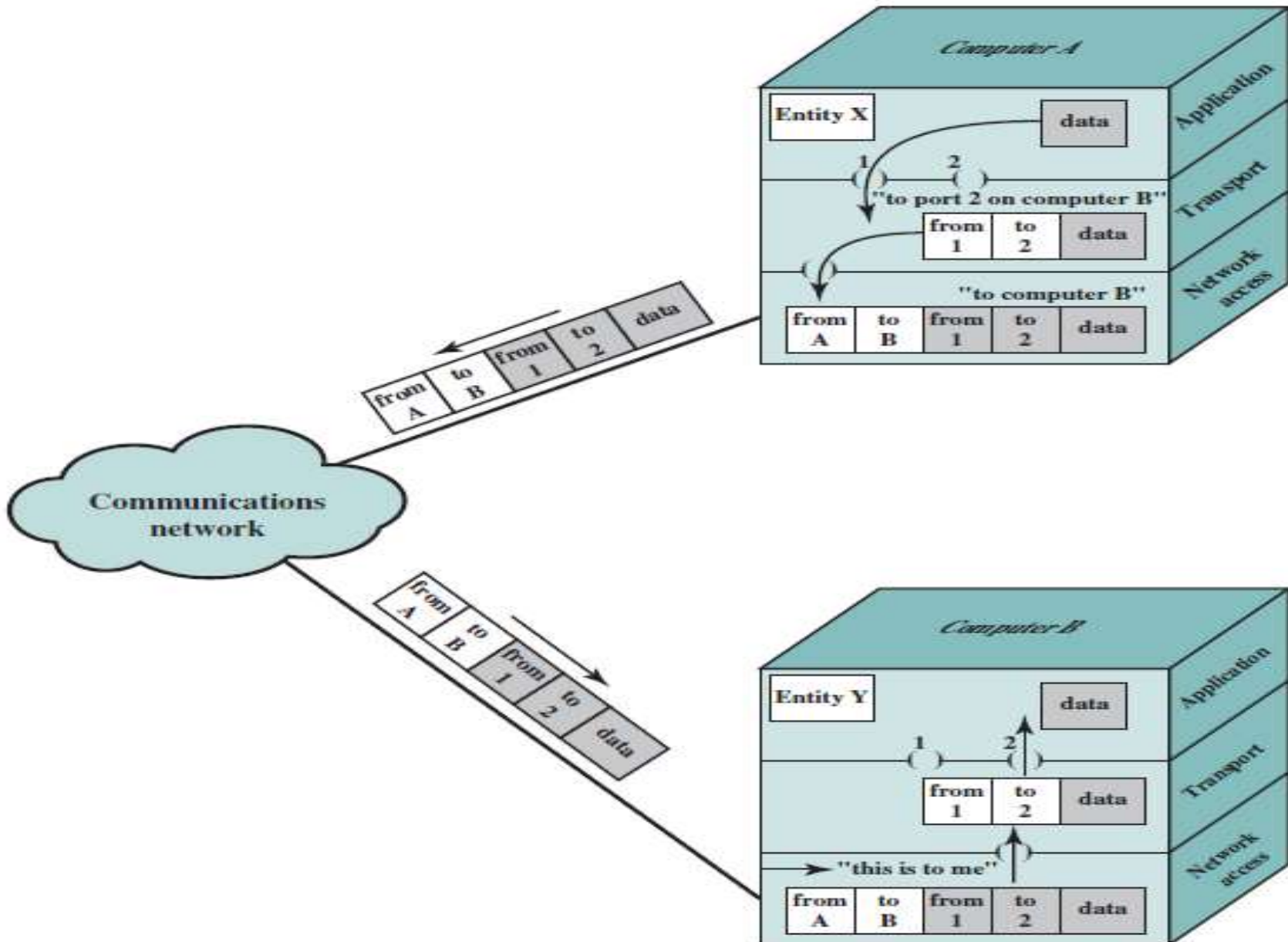
- **Contents of header:**

- **Source port:** This indicates the application that sent the data.
- **Destination port:** When the destination transport layer receives the segment, it must know to which application the data are to be delivered.
- **Sequence number:** Because the transport protocol is sending a sequence of segments, it numbers them sequentially so that if they arrive out of order, the destination transport entity may reorder them.
- **Error-detection code:** The sending transport entity may include a code that is a function of the contents of the segment. The receiving transport protocol performs the same calculation and compares the result with the incoming code. A discrepancy results if there has been some error in transmission. In that case, the receiver can discard the segment and take corrective action. This code is also referred to as a **checksum** or **frame check sequence**.

The next step is for the transport layer to hand each segment over to the network layer, with instructions to transmit it to the destination computer. To satisfy this request, the network access protocol must present the data to the network with a request for transmission. As before, this operation requires the use of control information. In this case, the **network access protocol (NAP)** appends a network access header to the data it receives from the transport layer, creating a network access PDU, typically called a **packet**. Examples of the items that may be stored in the header include the following:

- **Source computer address:** Indicates the source of this packet.
- **Destination computer address:** The network must know to which computer on the network the data are to be delivered.
- **Facilities requests:** The network access protocol might want the network to make use of certain facilities, such as priority.

- **Protocols in a Simplified Architecture**



- **TCP/IP Protocol Architecture**

TCP/IP is a result of protocol research and development conducted on the experimental packet-switched network, ARPANET, funded by the Defense Advanced Research Projects Agency (DARPA), and is generally referred to as the TCP/IP protocol suite. This protocol suite consists of a large collection of protocols that have been issued as Internet standards by the Internet Activities Board (IAB).

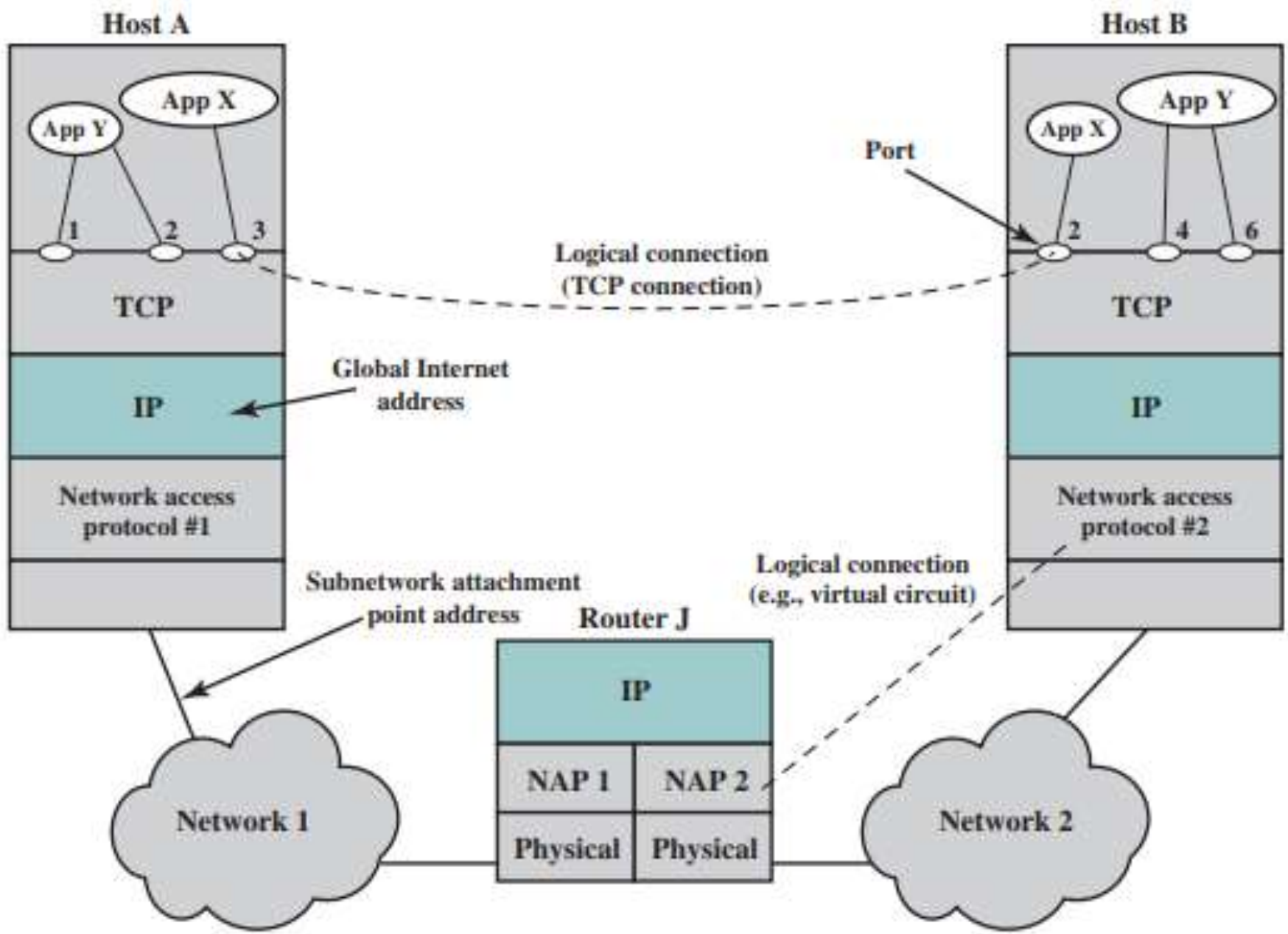
• TCP/IP Layers and Example Protocols

Application Provides access to the TCP/IP environment for users and also provides distributed information services.
Transport Transfer of data between end points. May provide error control, flow control, congestion control, reliable delivery.
Internet Shields higher layers from details of physical network configuration. Provides routing. May provide QoS, congestion control.
Network Access/ Data Link Logical interface to network hardware. May be stream or packet oriented. May provide reliable delivery.
Physical Transmission of bit stream; specifies medium, signal encoding technique, data rate, bandwidth, and physical connector.

SMTP, FTP, SSH, HTTP	
TCP, UDP	
IPv4, IPv6	ICMP, OSPF, RSVP ARP
Ethernet, Wi-Fi, ATM, frame relay	
Twisted pair, optical fiber, satellite, terrestrial microwave	

- **Physical Layer:** covers the physical interface between a data transmission device and a transmission medium or network
- **Network Access/Data Link layer:** concerned with access to and routing data across a network for two end systems attached to the same network.
- **Internet Layer:** Internet Protocol(IP) is used here to provide the routing function across multiple networks. This protocol is implemented in end-systems and **Routers.**
- **Host-to-Host Layer OR Transport Layer:** to provide reliable end-to-end delivery service
- **Application Layer:** contains the logic needed to support the various user application

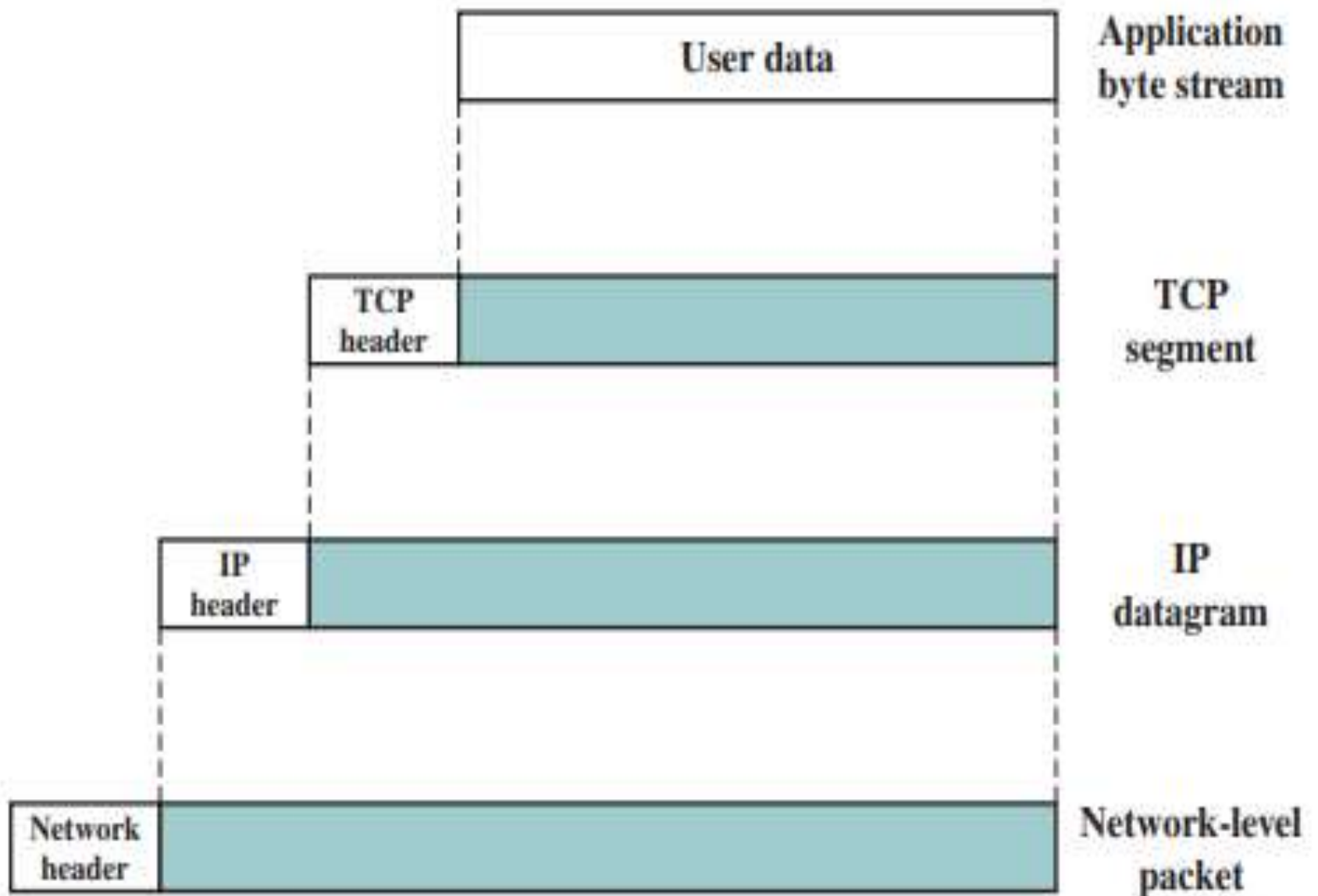
- TCP/IP Concepts



- **Example**

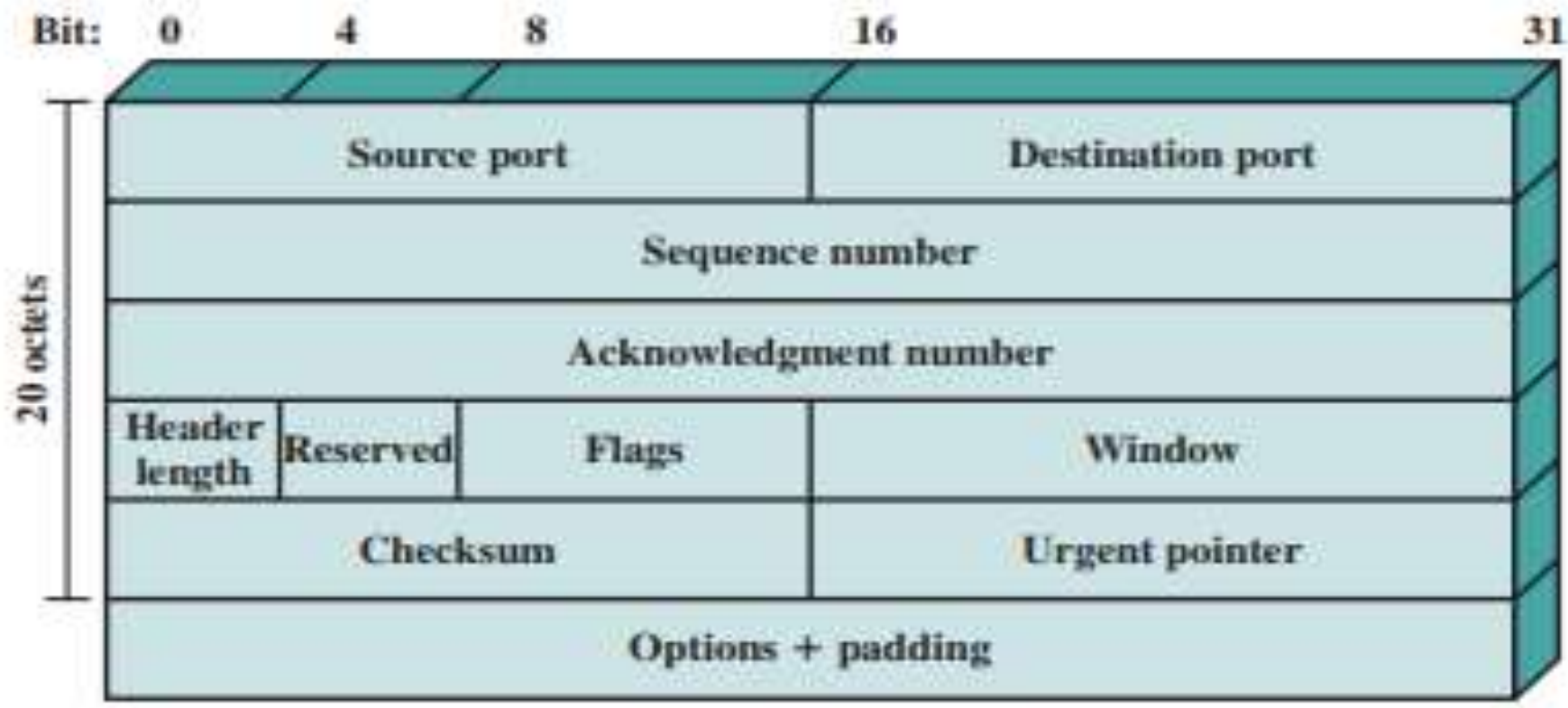
Let us trace a simple operation. Suppose that a process, associated with port 3 at host A, wishes to send a message to another process, associated with port 2 at host B. The process at A hands the message down to TCP with instructions to send it to host B, port 2. TCP hands the message down to IP with instructions to send it to host B. Note that IP need not be told the identity of the destination port. All it needs to know is that the data are intended for host B. Next, IP hands the message down to the network access layer (e.g., Ethernet logic) with instructions to send it to router J (the first hop on the way to B).

- **Protocol Data Units (PDUs) in the TCP/IP Architecture**

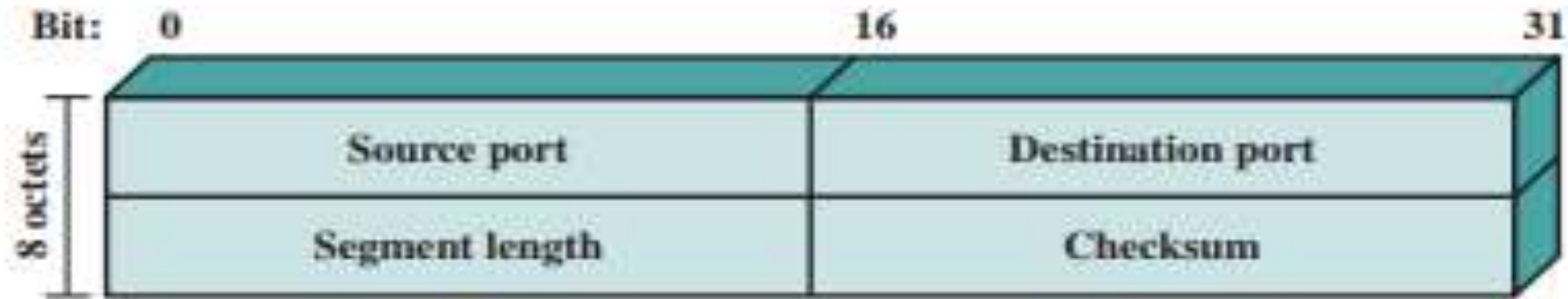


- **The TCP Header includes:**
 - **Destination port:** When the TCP entity at B receives the segment, it must know to whom the data are to be delivered.
 - **Sequence number:** TCP numbers the segments that it sends to a particular destination port sequentially, so that if they arrive out of order, the TCP entity at B can reorder them.
 - **Checksum:** The sending TCP includes a code that is a function of the contents of the remainder of the segment. The receiving TCP performs the same calculation and compares the result with the incoming code. A discrepancy results if there has been some error in transmission.

- TCP and UDP Headers



(a) TCP header



- Transmission Control Protocol is a transport layer protocol.
- It continuously receives data from the application layer.
- It divides the data into chunks where each chunk is a collection of bytes.
- It then creates TCP segments by adding a TCP header to the data chunks.
- TCP segments are encapsulated in the IP datagram.

TCP segment = TCP header + Data chunk

- A TCP connection is uniquely identified by using-

Combination of port numbers and IP Addresses of sender and receiver

- IP Addresses indicate which systems are communicating.
- Port numbers indicate which end to end sockets are communicating.

- **TCP Segment Header: Range 20-60 Bytes with 40 bytes for options**
- **Source Port Address: 16-bit port address of the application that is sending the data segment**
- **Destination Port Address: 16-bit port address of the application that is receiving the data segment**
- **Sequence Number: 32-bit that holds sequence no.**
 - **It contains the sequence no. of the 1st data byte**
- **Acknowledgement Number: 32-bit that holds Ack. no.**
 - **it contains the seq. no. of the data byte that receiver expects to receive next from the sender**
 - **It is always seq. no. of the last received data byte incremented by 1**

- **Seq. No., Ack. No. and Window:** provide flow control and error control
- **Header Length:** 4-bit that indicates the length of TCP header. It helps in knowing from where the actual data begins. Range – [20 bytes, 60 bytes]
 - The initial 5 rows of the TCP header are always used.
 - So, minimum length of TCP header = 5 x 4 bytes = 20 bytes.
 - The size of the 6th row representing the Options field vary.
 - The size of Options field can go up to 40 bytes.
 - So, maximum length of TCP header = 20 bytes + 40 bytes = 60 bytes.
 - So, to represent the header length, we use a scaling factor of 4.

Header length = Header length field value x 4 bytes

- **Reserved Bits: 6-bits are reserved and not used**
- **Flags**
 - **URG Bit: It is used to treat certain data on an urgent basis. When this bit is set to 1:**
 - It indicates the receiver that certain amount of data within the current segment is urgent.
 - Urgent data is pointed out by evaluating the urgent pointer field.
 - The urgent data has be prioritized.
 - Receiver forwards urgent data to the receiving application on a separate channel.
 - **ACK Bit: indicates whether ack. no. field is valid or not**
 - When ACK bit is set to 1, it indicates that acknowledgement number contained in the TCP header is valid.
 - For all TCP segments except request segment, ACK bit is set to 1.
 - Request segment is sent for connection establishment during Three Way Handshake.

- **PSH Bit: It is used to push the entire buffer immediately to the receiving application**
 - **When it is set to 1,**
 - All the segments in the buffer are immediately pushed to the receiving application.
 - No wait is done for filling the entire buffer.
 - This makes the entire buffer to free up immediately.
 - Unlike URG bit, PSH bit does not prioritize the data.
 - It just causes all the segments in the buffer to be pushed immediately to the receiving application.
 - The same order is maintained in which the segments arrived.
 - It is not a good practice to set PSH bit = 1.
 - This is because it disrupts the working of receiver's CPU and forces it to take an action immediately.

- **RST Bit: It is used to reset the connection**
 - **when it is set to 1,**
 - It indicates the receiver to terminate the connection immediately.
 - It causes both the sides to release the connection and all its resources abnormally.
 - The transfer of data ceases in both the directions.
 - It may result in the loss of data that is in transit.
- **Syn Bit: It is used to synchronize the seq. nos.**
 - **when it is set to 1,**
 - It indicates the receiver that the sequence number contained in the TCP header is the initial sequence number.
 - Request segment sent for connection establishment during Three way handshake contains SYN bit set to 1.

- **Fin Bit: It is used to terminate the TCP Connection.**
 - **When it is set to 1,**
 - It indicates the receiver that the sender wants to terminate the connection.
 - FIN segment sent for TCP Connection Termination contains FIN bit set to 1.
- **Checksum: 16-bit frame check sequence used to check error in TCP segment**
 - It verifies the integrity of data in the TCP payload.
 - Sender adds CRC checksum to the checksum field before sending the data.
 - Receiver rejects the data that fails the CRC check.

- **Urgent Pointer: It is a 16-bit field**

- It indicates how much data in the current segment counting from the first data byte is urgent.
- Urgent pointer added to the sequence number indicates the end of urgent data byte.
- This field is considered valid and evaluated only if the URG bit is set to 1.

$$\text{Number of urgent bytes} = \text{Urgent pointer} + 1$$

$$\text{End of urgent byte} = \text{Sequence number of the first byte in the segment} + \text{Urgent pointer}$$

- **Options: Used for several purposes; Size: 0-40 bytes**
 - **Purposes: Time stamp; Window size extension; Parameter negotiation, Padding**

- **Time Stamp**

When wrap around time is less than life time of a segment,

- Multiple segments having the same sequence number may appear at the receiver side.
- This makes it difficult for the receiver to identify the correct segment.
- If time stamp is used, it marks the age of TCP segments.
- Based on the time stamp, receiver can identify the correct segment.

- **Window size extension**

- Options field may be used to represent a window size greater than 16 bits.
- Using window size field of TCP header, window size of only 16 bits can be represented.
- If the receiver wants to receive more data, it can advertise its greater window size using this field.
- The extra bits are then appended in Options field.

- **Parameter Negotiation**

Options field is used for parameters negotiation.

Example- During connection establishment,

- Both sender and receiver have to specify their maximum segment size.
- To specify maximum segment size, there is no special field.
- So, they specify their maximum segment size using this field and negotiates.

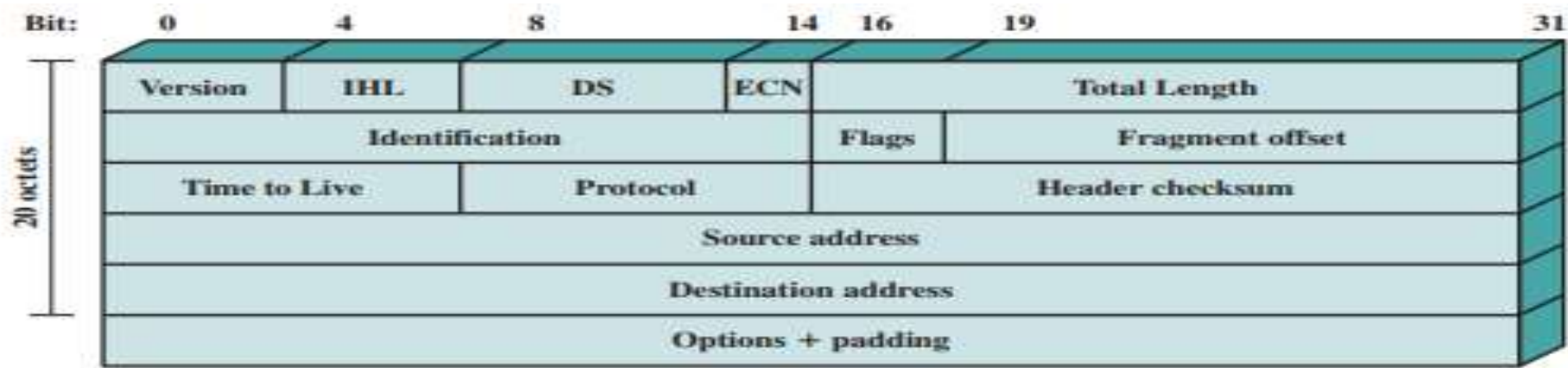
- **Padding**

- Addition of dummy data to fill up unused space in the transmission unit and make it conform to the standard size is called as padding.
- Options field is used for padding.

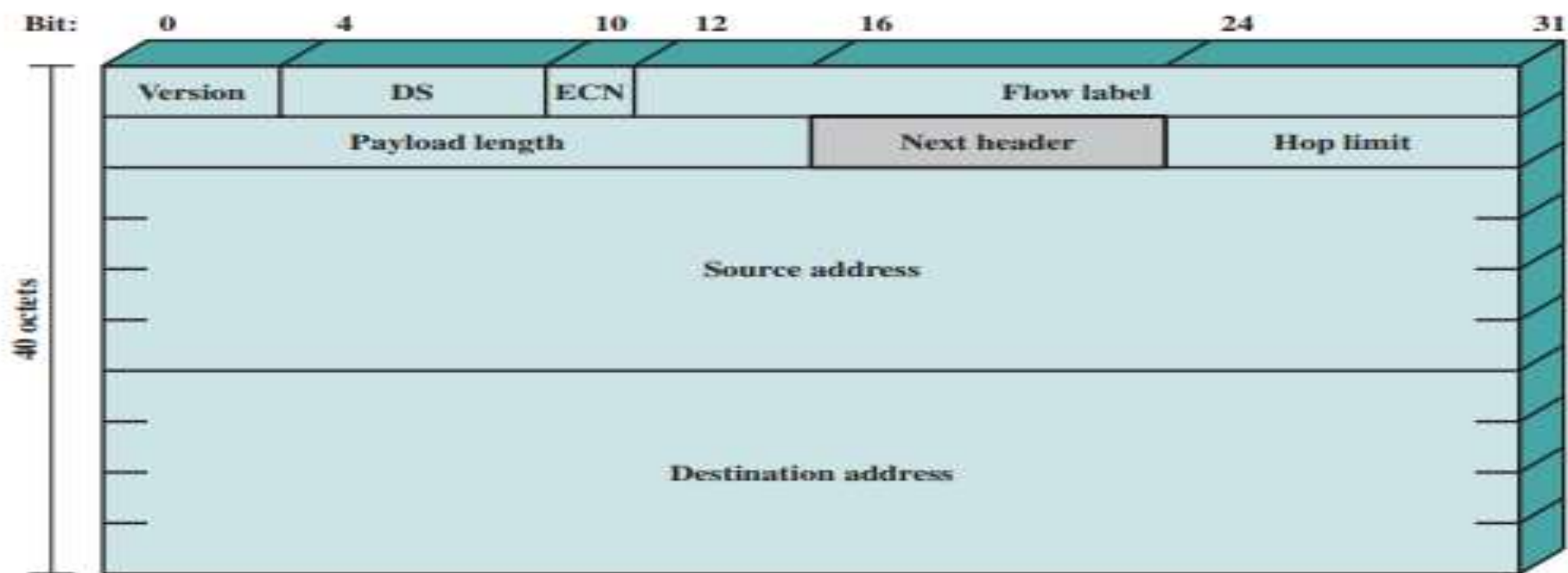
- **Example**

- When header length is not a multiple of 4, extra zeroes are padded in the Options field.
- By doing so, header length becomes a multiple of 4.
- If header length = 30 bytes, 2 bytes of dummy data is added to the header.
- This makes header length = 32 bytes.
- Then, the value $32 / 4 = 8$ is put in the header length field.
- In worst case, 3 bytes of dummy data might have to be padded to make the header length a multiple of 4.

- IP Headers



(a) IPv4 header

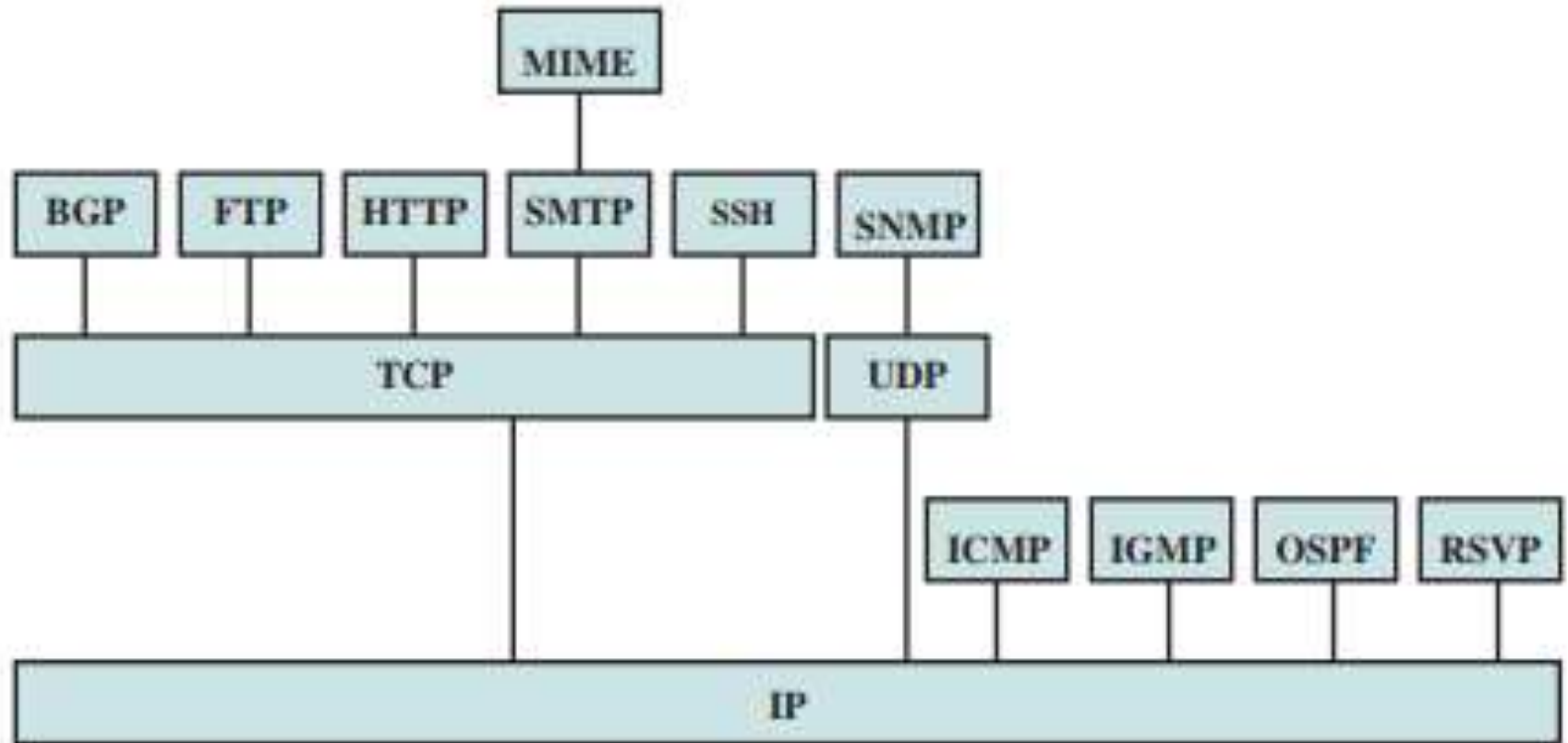


(b) IPv6 header

DS = Differentiated Services field
ECN = Explicit Congestion Notification field

Note: The 8-bit DS/ECN fields were formerly known as the Type of Service field in the IPv4 header and the Traffic Class field in the IPv6 header.

- Some Protocols in the TCP/IP Protocol Suite



BGP	=	Border Gateway Protocol
FTP	=	File Transfer Protocol
HTTP	=	Hypertext Transfer Protocol
ICMP	=	Internet Control Message Protocol
IGMP	=	Internet Group Management Protocol
IP	=	Internet Protocol
MIME	=	Multipurpose Internet Mail Extension

OSPF	=	Open Shortest Path First
RSVP	=	Resource ReSerVation Protocol
SMTP	=	Simple Mail Transfer Protocol
SNMP	=	Simple Network Management Protocol
SSH	=	Secure Shell
TCP	=	Transmission Control Protocol
UDP	=	User Datagram Protocol

Data Communication & Computer Network

Lecture – 6, 7 & 8

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Data Transmission Concepts & Terminology

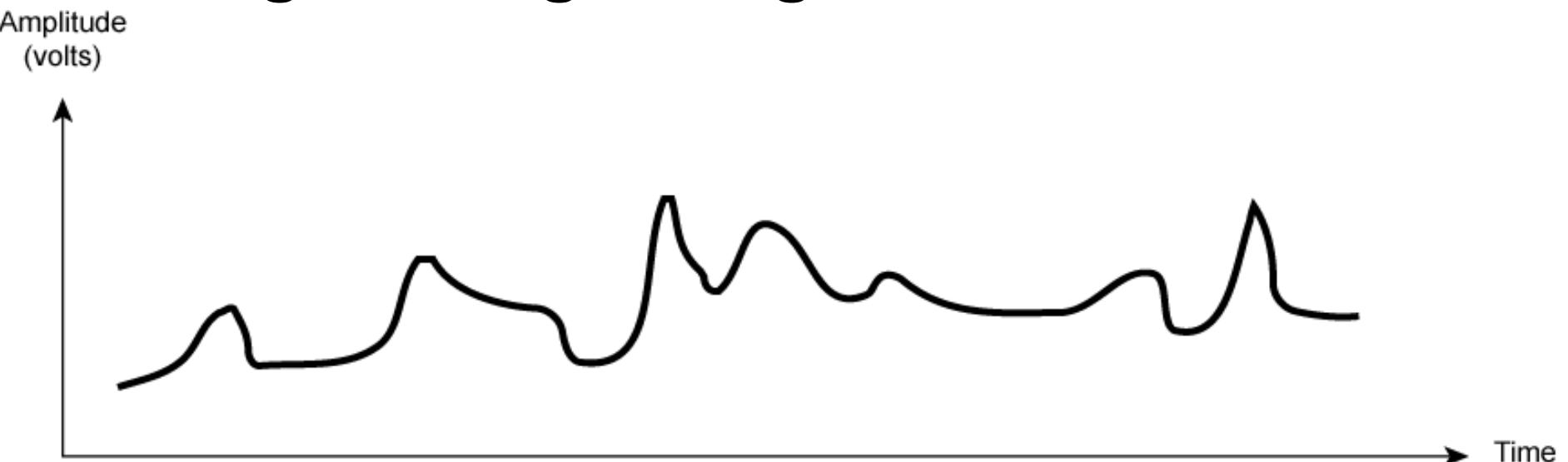
- **Data Transmission:** It occurs between Transmitter (Tx) and Receiver (Rx) over some transmission medium using electromagnetic waves.
- **Transmission medium**
 - **Guided medium:** waves guided along a physical path (e.g. twisted pair, optical fiber)
 - **Unguided medium (wireless)**
(e.g. air, water, vacuum, seawater)
- **Direct link (guided or unguided):** Transmission path between 2 devices in which signals propagate directly from Tx to Rx with no intermediate devices other than repeater or amplifier.

- **Point-to-point (Guided):** Provides a direct link between 2 devices and those are the only 2 devices sharing the medium.
- **Multi-point (Guided)**
 - More than two devices share the link
- **Simplex :** Transmission in one direction
 - e.g. Television
- **Half duplex:** Transmission in either direction, but only one way at a time
 - e.g. police radio
- **Full duplex:** Transmission in both directions at the same time
 - e.g. telephone

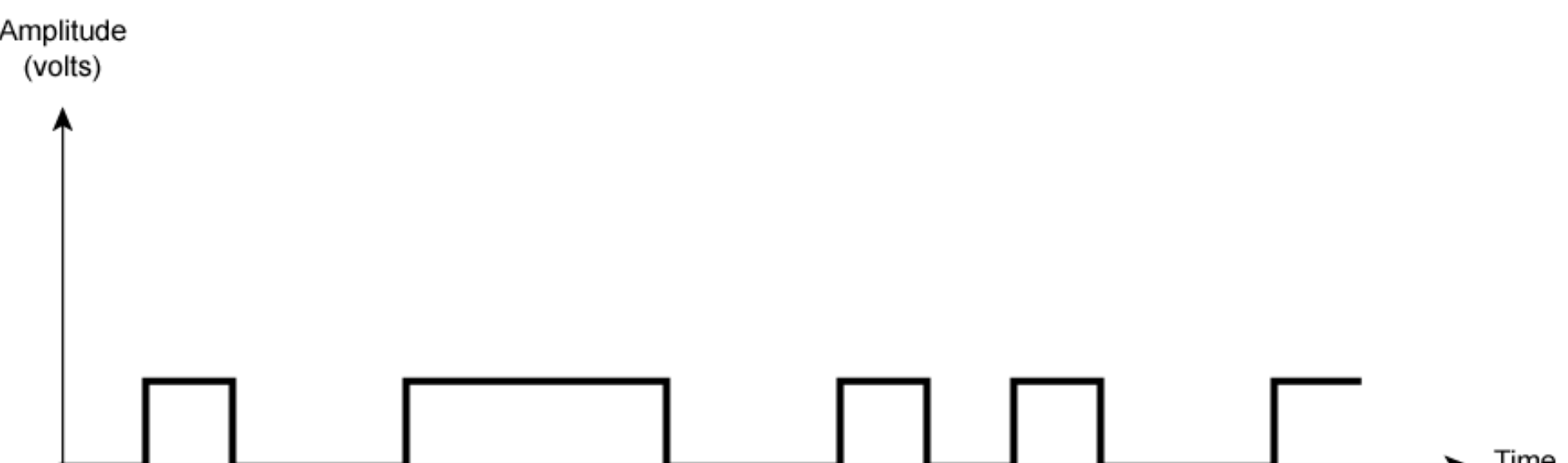
Frequency, Spectrum and Bandwidth

- **Signal is a function of time or frequency (Time Domain or Frequency Domain)**
- **Time domain concepts**
 - **Analog signal: Signal intensity varies in a smooth or continuous way over time**
 - **Digital signal: Signal intensity maintains a constant level for some period then changes to another constant level in a discrete way**
 - **Periodic signal**
 - **Pattern repeated over time**
 - **Aperiodic signal**
 - **Pattern not repeated over time**

- **Analog and Digital Signal**

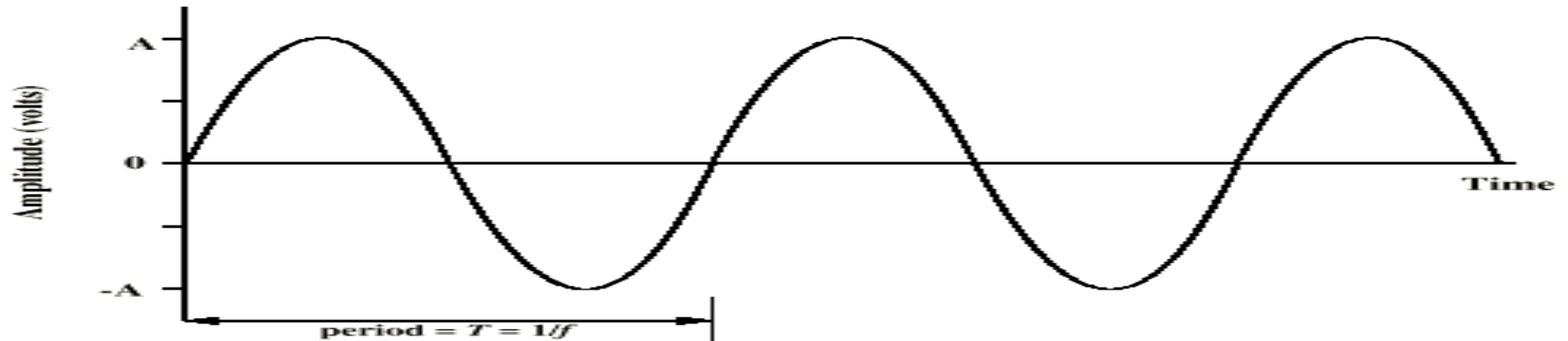


(a) Analog

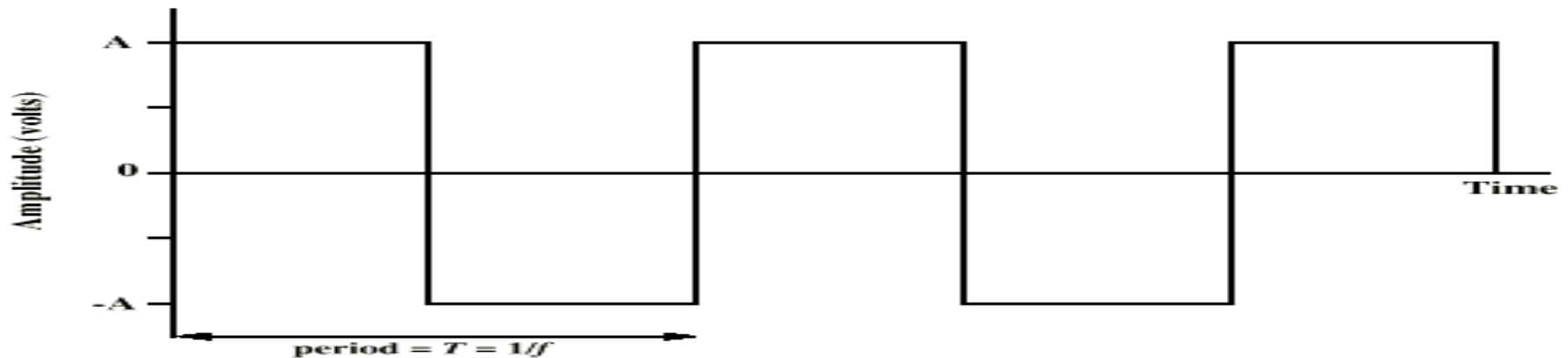


(b) Digital

- **Periodic Signal**



(a) Sine wave



(b) Square wave

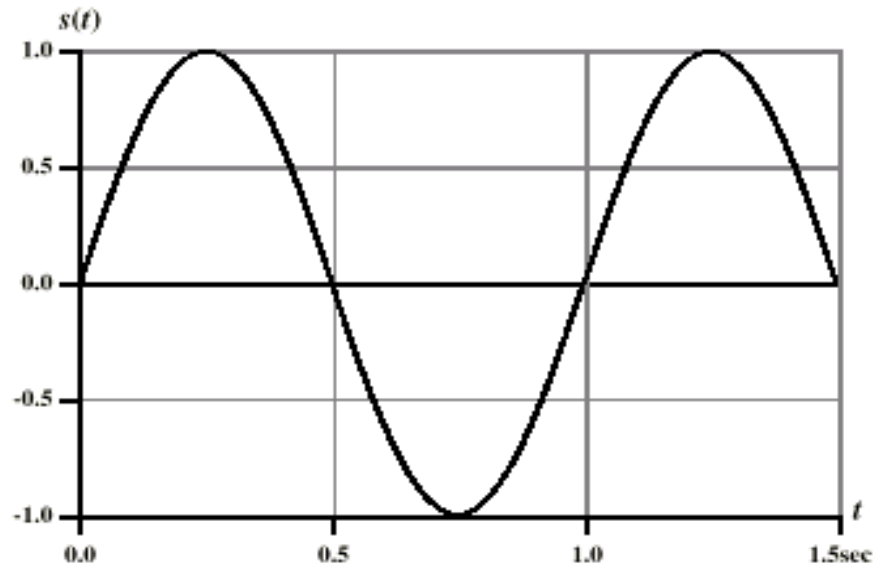
- A signal $s(t)$ is periodic if and only if

$$s(t + T) = s(t) \quad -\infty < t < +\infty$$

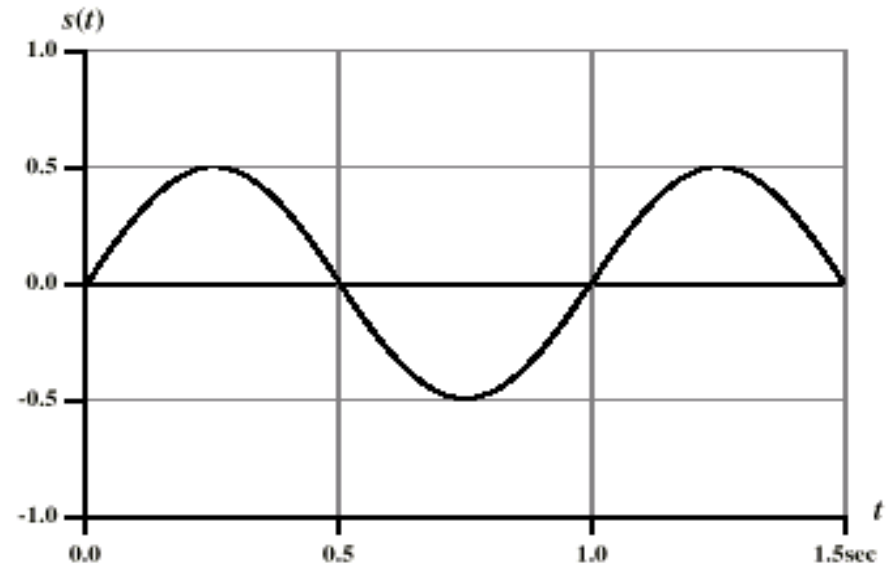
where the constant T is the period of the signal (T is the smallest value that satisfies the equation). Otherwise, a signal is **aperiodic**.

- **Sine Wave**
 - **Peak Amplitude (A)**
 - maximum strength of signal over time
 - Measured in volts
- **Frequency (f)**
 - Rate of change of signal
 - Measured in Hertz (Hz) or cycles per second
 - Period = time for one repetition (T)= $1/f$
- **Phase (ϕ)**
 - Measure of the relative position in time within a single period of signal

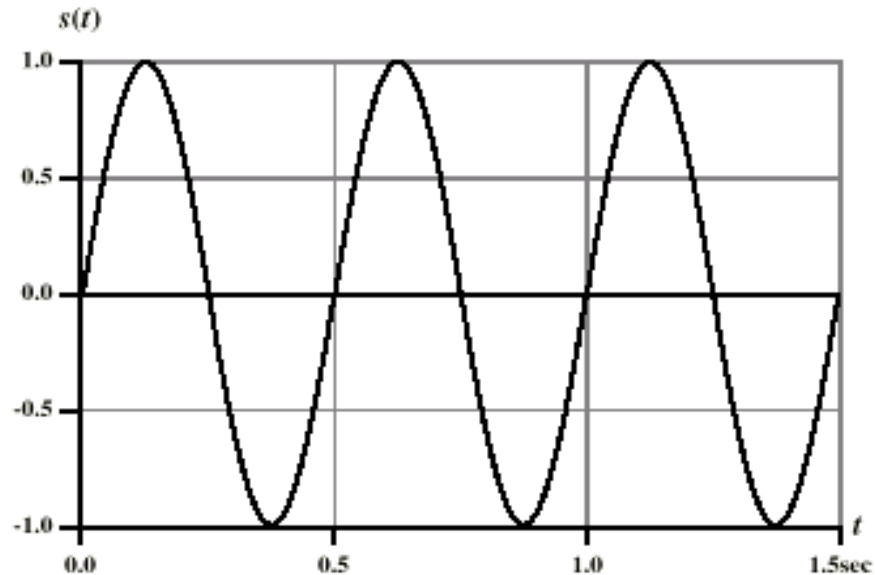
- Varying Sine Waves: $s(t) = A \sin(2\pi ft + \Phi)$**



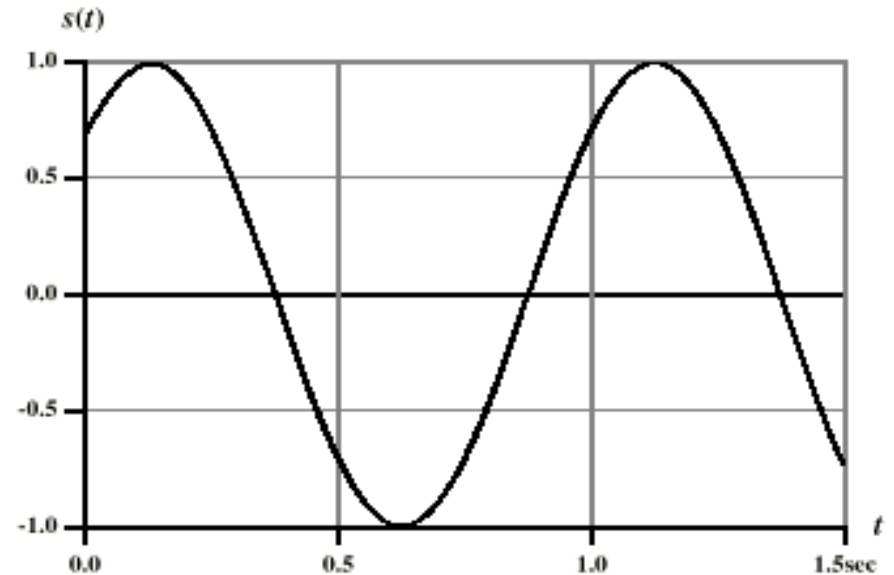
(a) $A = 1, f = 1, \phi = 0$



(b) $A = 0.5, f = 1, \phi = 0$



(c) $A = 1, f = 2, \phi = 0$



(d) $A = 1, f = 1, \phi = \pi/4$

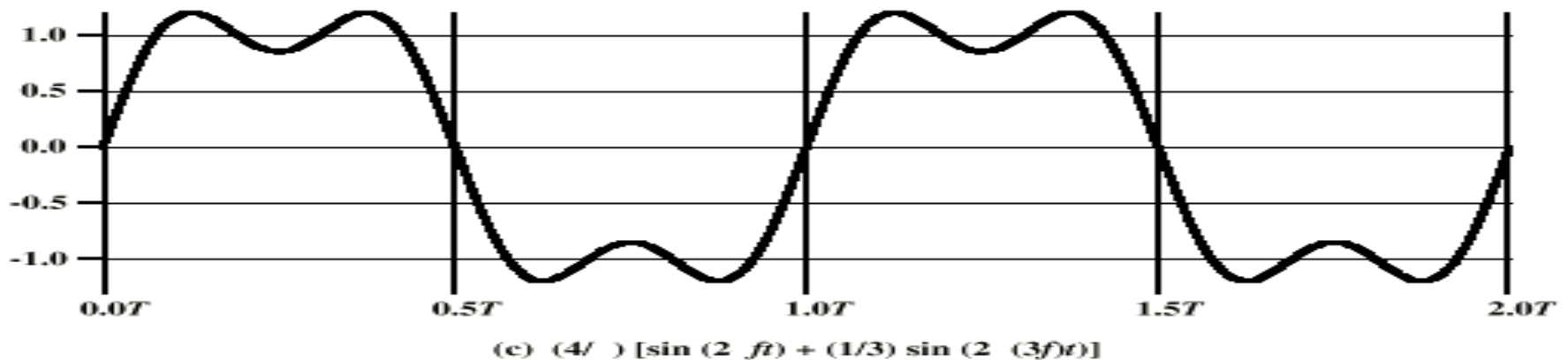
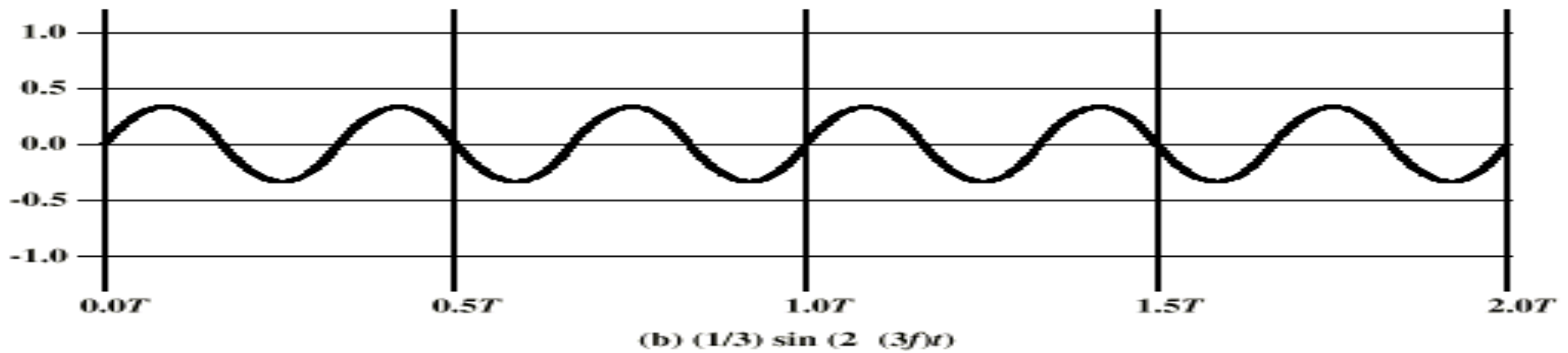
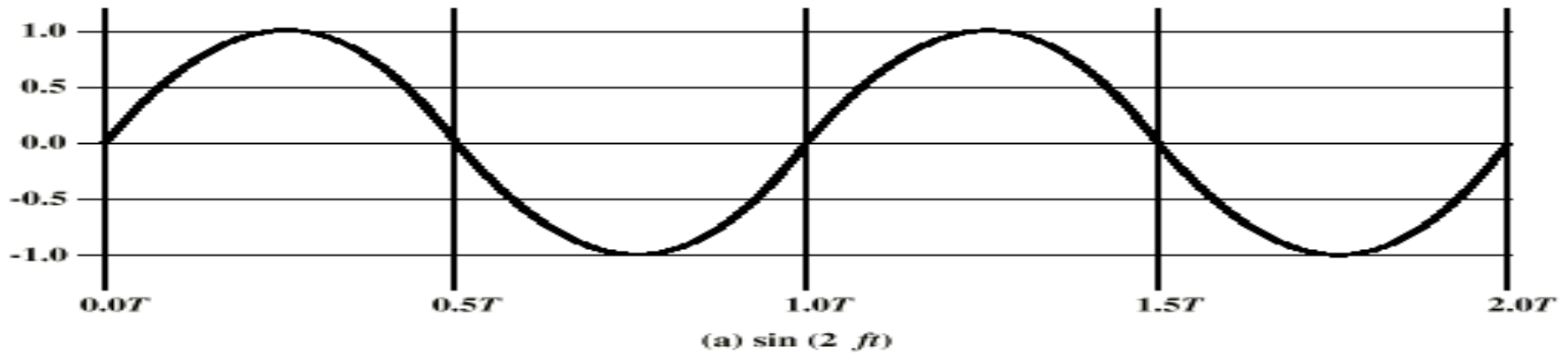
- **Wavelength (λ)**
 - **Distance occupied by one cycle**
 - **Distance between two points of corresponding phase in two consecutive cycles**
- **Assuming signal velocity v**
 - $\lambda = vT$
 - $\lambda f = v$
 - **When $v = c$ then $\lambda f = c$**
where $c = 3 \times 10^8 \text{ ms}^{-1}$ (speed of light in free space)

- **Frequency Domain Concepts**
 - **Signal usually made up of many frequencies**

$$s(t) = (4/\pi) \times [\sin(2\pi ft) + (1/3)\sin(2\pi(3f)t)]$$

- **Here, 2nd frequency (harmonic) is an integer multiple of 1st frequency (fundamental)**
 - **Components are sine waves**
 - **Can be shown (Fourier analysis) that any signal is made up of component sine waves**
 - **Can plot frequency domain functions**

- Addition of Frequency Components ($T = 1/f$)

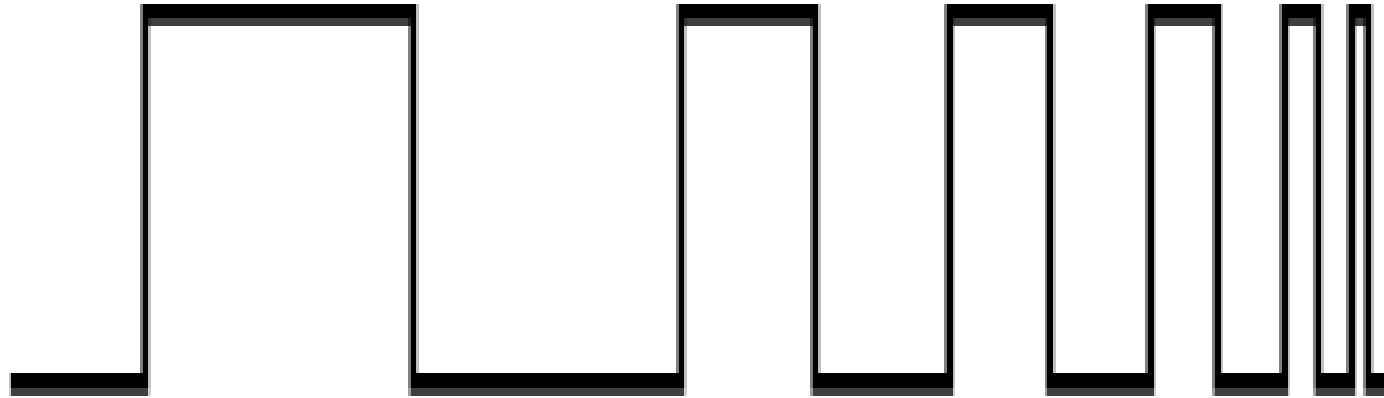


- **Spectrum and Bandwidth**
 - **Spectrum**
 - range of frequencies contained in signal
 - **Absolute bandwidth**
 - width of spectrum
 - **Effective bandwidth**
 - Narrow band of frequencies containing most of the energy
 - **DC Component**
 - Component of zero frequency
- **Data Rate and Bandwidth**
 - Any transmission system has a limited band of frequencies
 - This limits the data rate that can be carried

- **Analog and Digital Data Transmission**
 - **Data: Entities that convey meaning**
 - **Signals: Electric or electromagnetic representations of data**
 - **Transmission: Communication of data by propagation and processing of signals**
- **Analog and Digital Data**
 - **Analog: Continuous values within some interval (e.g. sound, video)**
- **Digital: Discrete values**
 - **e.g. text, integers**

- **Attenuation of Digital Signals**

Voltage at
transmitting end



Voltage at
receiving end

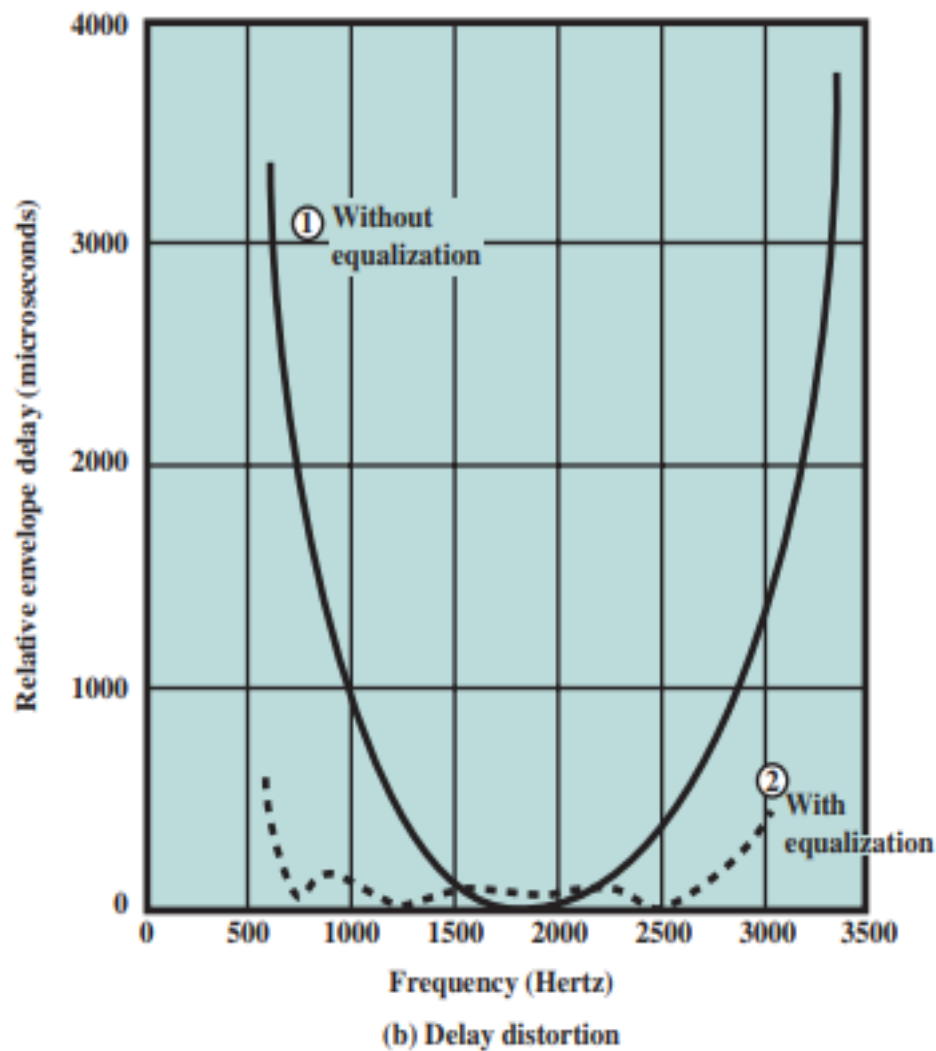
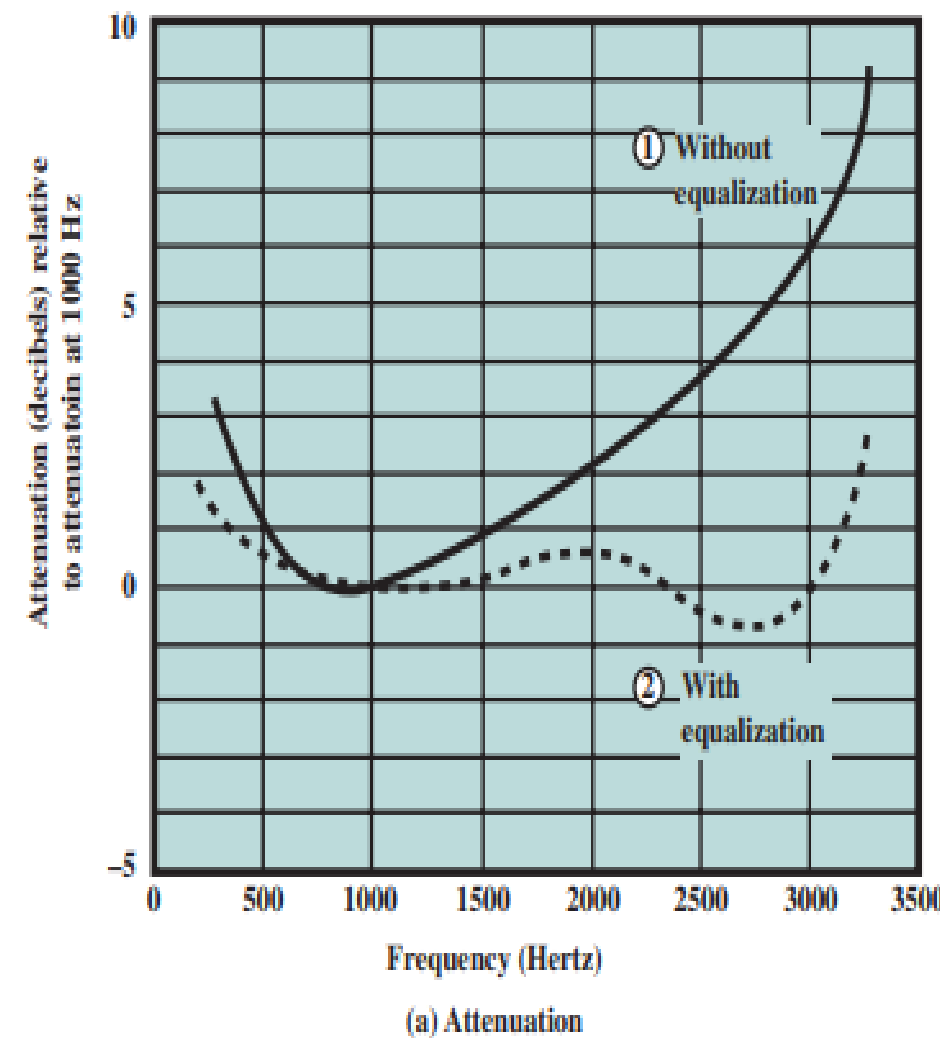


- **Data and Signals**
 - **Usually use digital signals for digital data and analog signals for analog data**
 - **Can use analog signal to carry digital data**
 - **Modem**
 - **Can use digital signal to carry analog data**
 - **Compact Disc audio**
 - **Analog Data -> Telephone -> Analog Signal**
 - **Digital data -> Modem -> Analog Signal**
 - **Analog Signal -> CODEC -> Digital Signal**
 - **Digital Data -> Digital Transceiver -> Digital Signal**

- **Transmission Impairments**
 - **Signal received may differ from signal transmitted**
 - **Analog: degradation of signal quality due to random modification**
 - **Digital: bit errors (1 -> 0 or 0 -> 1)**
 - **Significant Impairments**
 - **Attenuation and attenuation distortion**
 - **Delay distortion**
 - **Noise**
- **Attenuation (reduction in strength)**
 - **Signal strength falls off with distance over any transmission medium**

- **Depends on medium**
 - **Guided: exponential (expressed as a constant number of decibels per unit distance)**
 - **Unguided: Complex function of distance and the makeup of the atmosphere**
- **Received signal strength:**
 - **must be enough to be detected**
 - **must be sufficiently higher than noise to be received without error**
- **Attenuation is an increasing function of frequency (attenuation distortion)**
 - **Received signal is not only reduced in strength but is also distorted (Critical for analog signal)**

- **Attenuation & delay distortion for a typical leased line**
 - **Positive values on the y-axis represent attenuation greater than that at 1000 Hz**



- To show that a signal has lost or gained strength, we use the unit of the decibel. The decibel (dB) measures the relative strengths of two signals or one signal at two different points. Note that the decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$\text{dB} = 10 \text{ Log}_{10} (P_2 / P_1)$$

- Example:

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P_2 is $(1/2) P_1$. In this case, the attenuation (loss of power) can be calculated as follows:

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

- **Solutions: (1) Equalizing Attenuation**
(2) Use of amplifiers that amplify high frequencies more than lower frequencies

- **Attenuation distortion is not an issue with digital signal**
- **Delay Distortion**
 - **Occurs only in guided media**
 - **Propagation velocity varies with frequency**
 - **Various components of a signal will arrive at the receiver at different times**
 - **Critical for digital data**
 - **Solution: Equalizing technique**

- **Noise**
 - **Additional signals inserted between transmitter and receiver**
 - **Thermal**
 - **Due to thermal agitation of electrons**
 - **Uniformly distributed**
 - **White noise**

The amount of thermal noise to be found in a bandwidth of 1 Hz in any device or conductor is

$$N_0 = kT(\text{W/Hz})$$

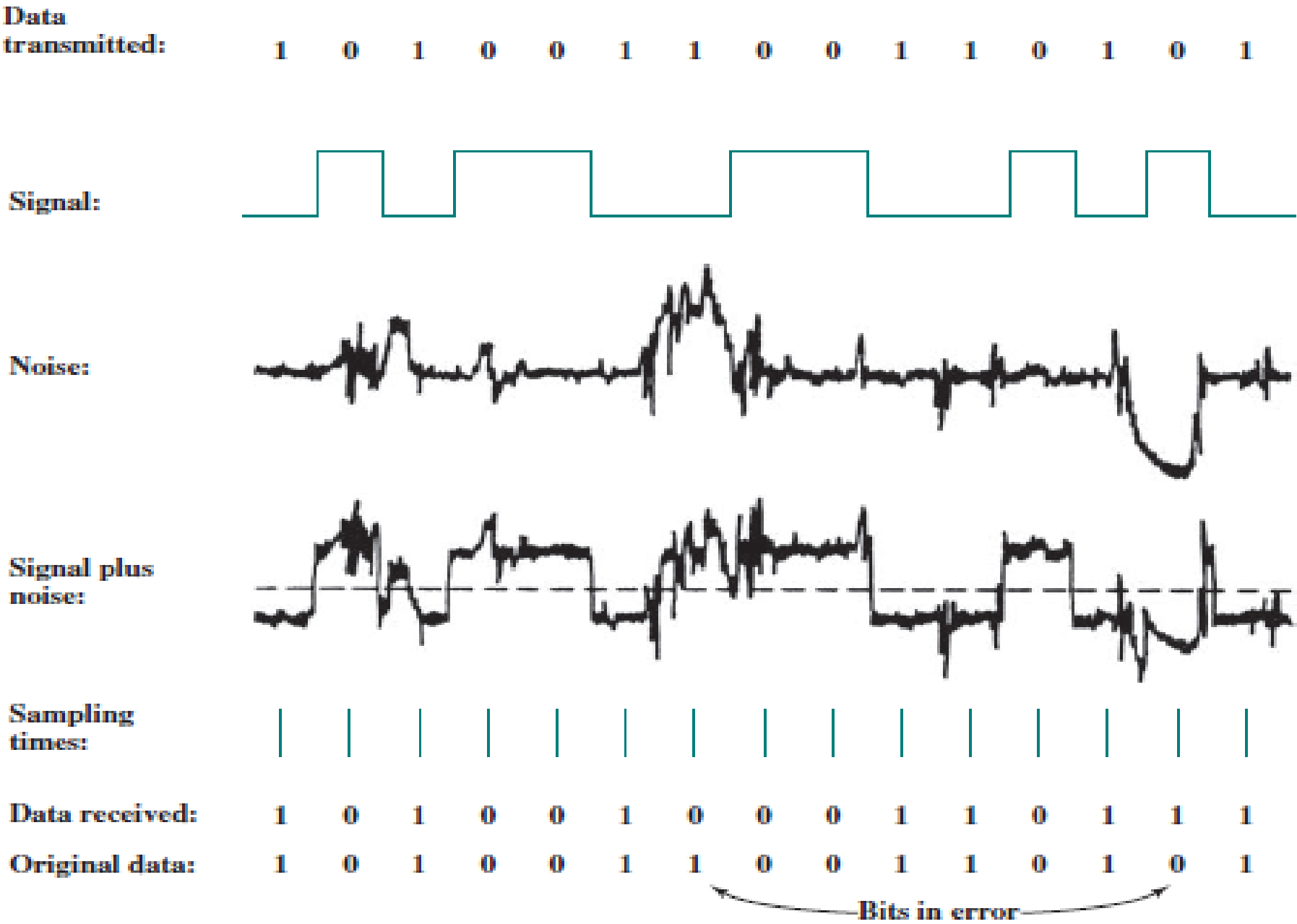
N_0 = noise power density in watts per 1 Hz of bandwidth

k = Boltzmann's constant = 1.38×10^{-23} J/K

T = temperature, in kelvins (absolute temperature) where the symbol K is used to represent 1 kelvin

- **Intermodulation**
 - **Signals that are the sum and difference of original frequencies sharing a medium**
 - **Crosstalk**
 - **A signal from one line is picked up by another**
- **Impulse noise**
 - **Irregular pulses or spikes of short duration and of relatively high amplitude**
 - **e.g. External electromagnetic interference**
 - **Primary source of error in digital data communication**

• Effect of Noise on a Digital Signal



- **Channel Capacity**

- **The maximum rate at which data can be transmitted over a given communication path or channel under given condition**

- **Data rate:** The rate, in bits per second (bps), at which data can be communicated
- **Bandwidth:** The bandwidth of the transmitted signal as constrained by the transmitter and the nature of the transmission medium, expressed in cycles per second, or hertz
- **Noise:** The average level of noise over the communications path
- **Error rate:** The rate at which errors occur, where an error is the reception of a 1 when a 0 was transmitted or the reception of a 0 when a 1 was transmitted

- **Nyquist Bandwidth**

- If rate of signal transmission is $2B$ then signal with frequencies no greater than B is sufficient to carry signal rate (In case of noise free channel)
- Given bandwidth B , highest signal rate is $2B$
- Given binary signal, data rate supported by B Hz is $2B$ bps
- Can be increased by using M signal levels
- For multilevel signalling, Nyquist formulation becomes $C = 2B \log_2 M$, where M is the no. of discrete signal or voltage levels
 - For a given bandwidth, the data rate can be increased by increasing the number of different signal elements

- **Shannon Capacity Formula**
 - Consider data rate, noise and error rate
 - Faster data rate shortens each bit so burst of noise affects more bits
 - At given noise level, high data rate means higher error rate
 - Signal to noise ratio (in decibels): Ratio of the power in a signal to the power contained in the noise that is present at a particular point in the transmission
 - $\text{SNR}_{\text{db}} = 10 \log_{10} (\text{signal power}/\text{noise power})$
 - A high SNR will mean a high quality signal and a low number of required intermediate repeaters
 - Capacity $C = B \log_2(1 + \text{SNR})$
 - This is error free capacity

Data Communication & Computer Network

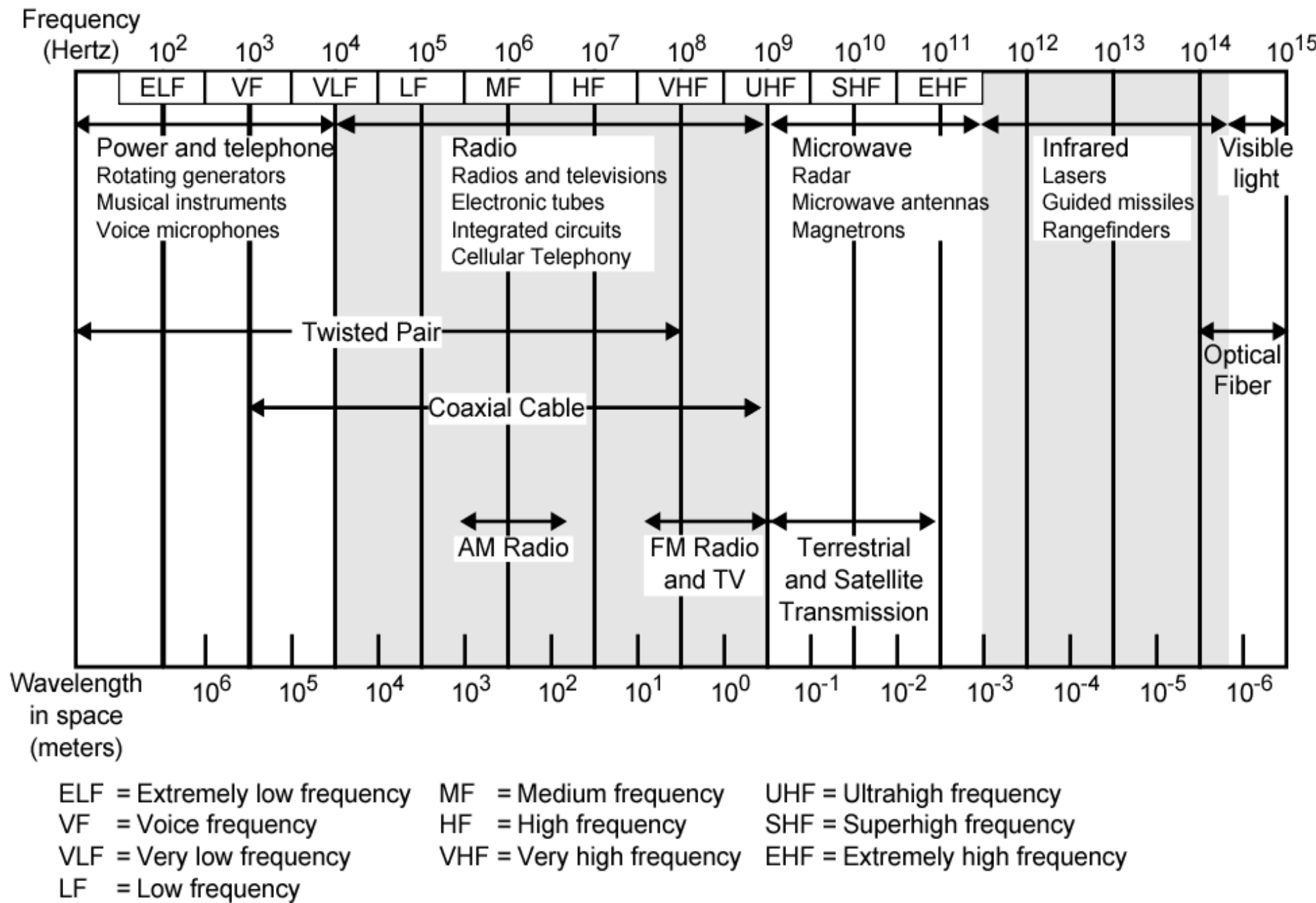
Lecture – 9, 10 and 11

Presented By
Dr. S.Kanungo

Transmission Media

- **In data transmission, key concerns are data rate and distance**
- **Objective: Achieving the highest data rates over the longest distances**
- **Design Factors**
 - **Bandwidth: Higher bandwidth gives higher data rate**
 - **Transmission impairments: Attenuation limit the distance; in guided media: twisted pair > coaxial cable > optical fiber**
 - **Interference: proper shielding in a guided medium can minimize this problem**
 - **Number of receivers**
 - **In guided media (point-to-point or multipoint)**
 - **More receivers (multi-point) introduce more attenuation**

• Electromagnetic Spectrum for Telecommunications



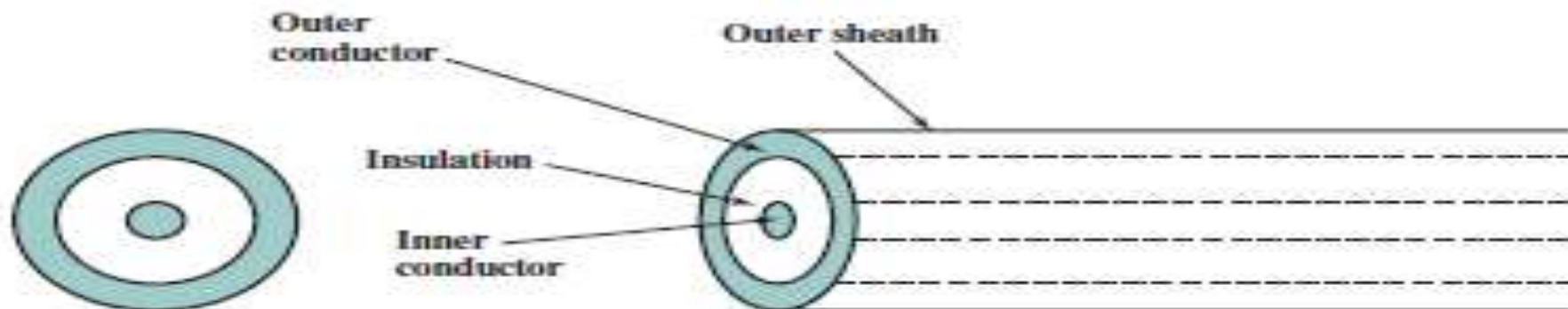
• Transmission Characteristics of Guided Media

	Frequency Range	Typical Attenuation	Typical Delay	Repeater Spacing
Twisted pair (with loading)	0 to 3.5 kHz	0.2 dB/km @ 1 kHz	50 μs/km	2 km
Twisted pairs (multi-pair cables)	0 to 1 MHz	0.7 dB/km @ 1 kHz	5 μs/km	2 km
Coaxial cable	0 to 500 MHz	7 dB/km @ 10 MHz	4 μs/km	1 to 9 km
Optical fiber	186 to 370 THz THz = terahertz = 10¹²Hz	0.2 to 0.5 dB/km	5 μs/km	40 km

- Separately insulated
- Twisted together
- Often "bundled" into cables
- Usually installed in building during construction

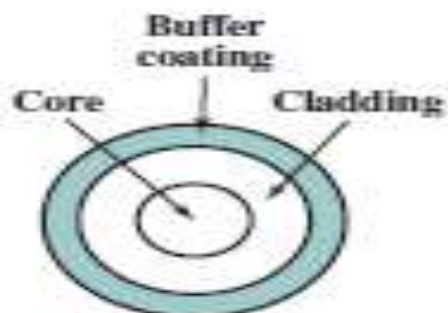


(a) Twisted pair

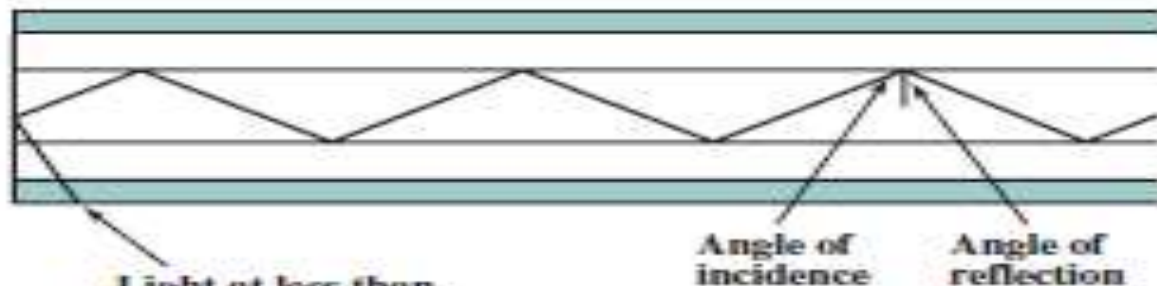


- Outer conductor is braided shield
- Inner conductor is solid metal
- Separated by insulating material
- Covered by padding

(b) Coaxial cable



- Glass or plastic core
- Laser or light-emitting diode
- Small size and weight



Light at less than critical angle is absorbed in buffer coating

(c) Optical fiber

- **Twisted Pair: Least expensive and widely used**
 - Easy to work with
 - Used to transmit both analog and digital transmission
 - It is limited in distance, bandwidth (1 MHz) and data rate(100 MHz)
 - Low data rate
 - Short range
 - Analog: Amplifiers every 5km to 6km
 - Digital: Use either analog or digital signals
 - repeater every 2km or 3km
 - Susceptible to interference and noise
 - Near End Crosstalk: Coupling of signal from one pair to another; Coupling takes place when transmit signal entering the link couples back to receiving pair; i.e. near transmitted signal is picked up by near receiving pair

- **Unshielded Twisted Pair (UTP):** consists of one or more twisted-pair cables, enclosed within an overall thermoplastic jacket, which provides no electromagnetic shielding
 - Ordinary telephone wire
 - Cheapest
 - Easiest to install
 - Suffers from external EM interference
- **Shielded Twisted Pair (STP):** shielding reduces interference and provides better performance at higher data rates
 - More expensive
 - Harder to handle (thick, heavy)

- **UTP Categories**
- **Cat 3**
 - up to 16MHz
 - Voice grade found in most offices
 - Twist length of 7.5 cm to 10 cm
- **Cat 4**
 - up to 20 MHz
- **Cat 5**
 - up to 100MHz
 - Commonly pre-installed in new office buildings
 - Twist length 0.6 cm to 0.85 cm
- **Cat 5E (Enhanced)**
- **Cat 6**
- **Cat 7**

• Twisted Pair Categories and Classes

	Category 3 Class C	Category 5 Class D	Category 5E	Category 6 Class E	Category 7 Class F
Bandwidth	16 MHz	100 MHz	100 MHz	200 MHz	600 MHz
Cable Type	UTP	UTP/FTP	UTP/FTP	UTP/FTP	SSTP
Link Cost (Cat 5 =1)	0.7	1	1.2	1.5	2.2

- **Coaxial Cable Applications**
 - **Most versatile medium**
 - **Television distribution**
 - **Ariel to TV**
 - **Cable TV**
 - **Long distance telephone transmission**
 - **Can carry 10,000 voice calls simultaneously**
 - **Being replaced by fiber optic**
 - **Short distance computer systems links**
 - **Local area networks**

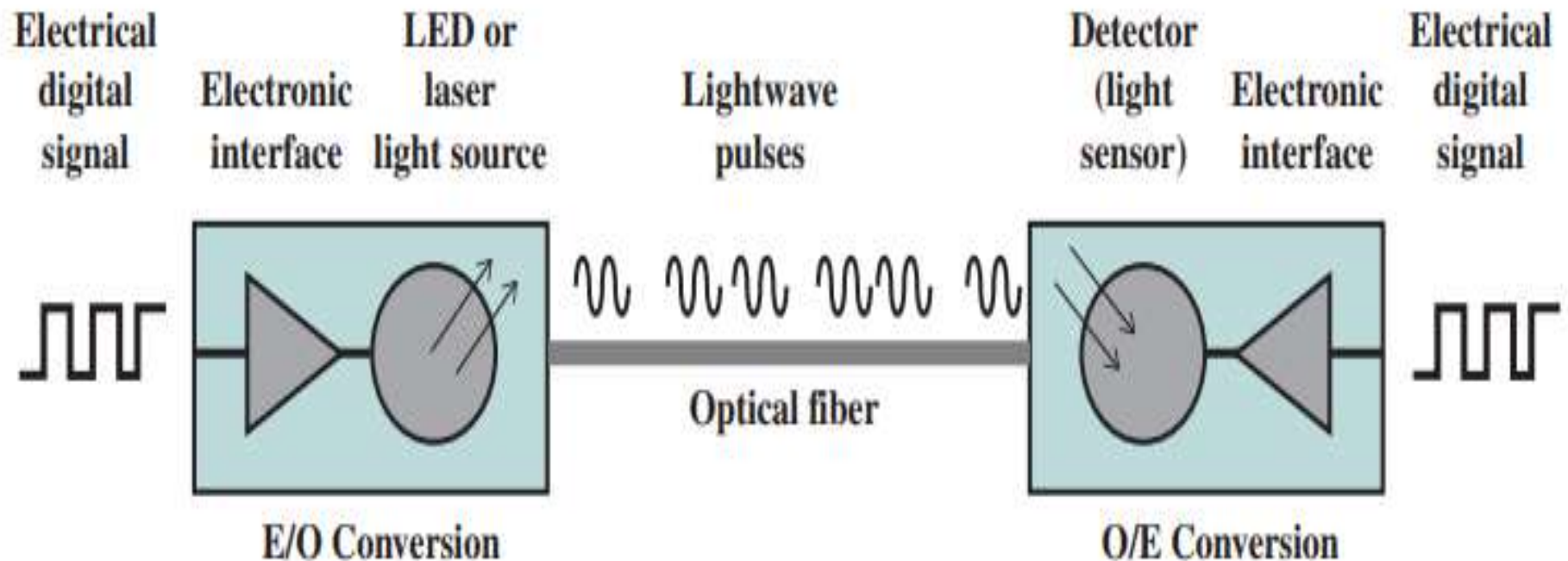
- **Coaxial Cable - Transmission Characteristics**
 - **Analog**
 - **Amplifiers are needed every few km**
 - **Closer spacing required if higher frequencies are used**
 - **Usable spectrum up to 500MHz**
 - **Digital**
 - **Repeater every 1km**
 - **Closer spacing required for higher data rates**
- **Optical Fiber – Benefits**
 - **Greater capacity**
 - **Data rates of hundreds of Gbps**
 - **Smaller size & weight**

- **Lower attenuation: constant over a wide range**
- **Electromagnetic isolation: not vulnerable to interference, impulse noise and crosstalk**
- **Greater repeater spacing**
 - **Tens of km at least is common**
- **Applications**
 - **Long-haul trunks**
 - **Metropolitan trunks**
 - **Rural exchange trunks**
 - **Subscriber loops**
 - **LANs**

- **Optical Fiber - Transmission Characteristics**
 - **Act as wave guide for 10^{14} to 10^{15} Hz**
 - **Portions of infrared and visible spectrum**
 - **Light Emitting Diode (LED)**
 - **Cheaper**
 - **Wider operating temp range**
 - **Last longer**
 - **Injection Laser Diode (ILD)**
 - **More efficient**
 - **Greater data rate**
 - **Wavelength Division Multiplexing**

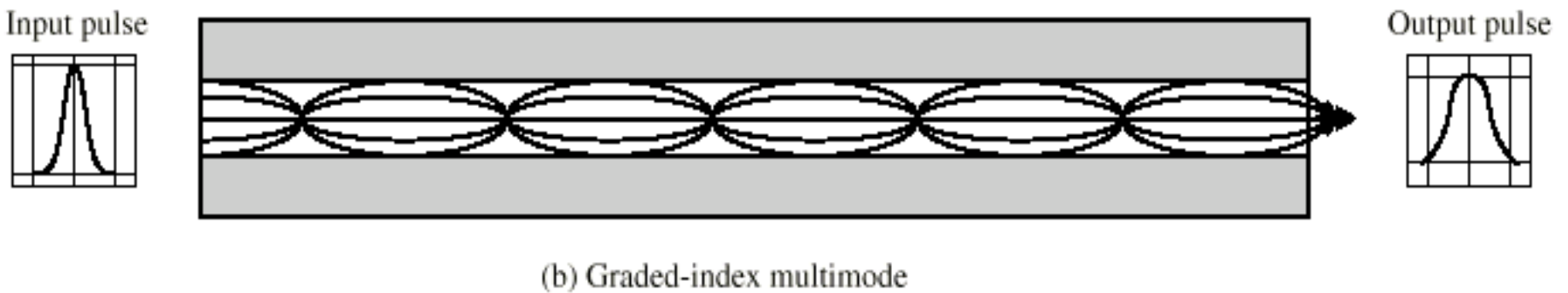
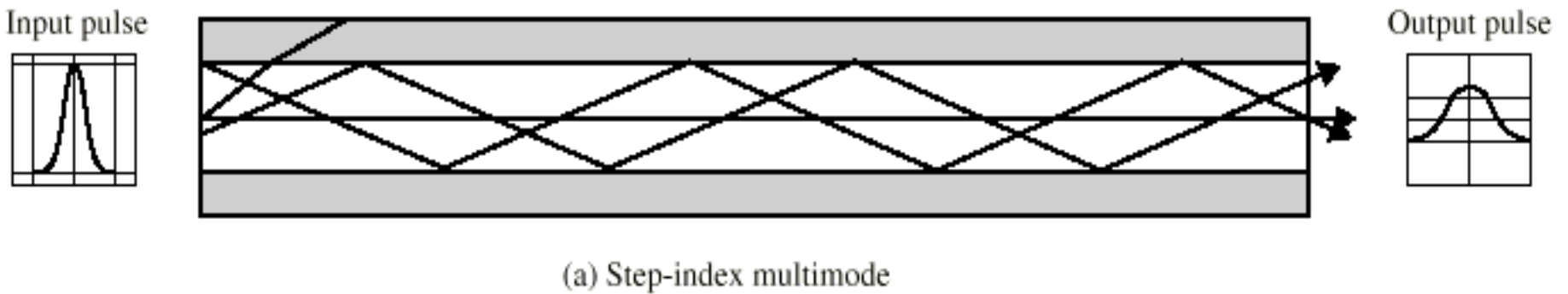
- **Optical Communication**

Optical fiber transmits a signal-encoded beam of light by means of total internal reflection. Total internal reflection can occur in any transparent medium that has a higher index of refraction than the surrounding medium. In effect, the optical



fiber acts as a waveguide for frequencies in the range of about 10^{14} to 10^{15} Hz; this covers portions of the infrared and visible spectra.

• Optical Fiber Transmission Modes



Wireless Transmission Frequencies

- **1GHz (10^9 Hz) to 40GHz**
 - Microwave
 - Highly directional beams are possible
 - Suitable for point-to-point transmission
 - Used for satellite communication
- **30MHz to 1GHz**
 - Suitable for omnidirectional applications
 - Broadcast radio
- **3×10^{11} to 2×10^{14}**
 - Infrared portion of the spectrum
 - For local applications

Antennas

- **Electrical conductor (or system of conductors) used to radiate electromagnetic energy or collect electromagnetic energy**
- **Transmission**
 - **Radio frequency electrical energy from transmitter converted to electromagnetic energy by antenna and radiated into surrounding environment**
- **Reception**
 - **Electromagnetic energy intersect on antenna**
 - **Converted to radio frequency electrical energy and fed to receiver**
- **Same antenna often used for both**

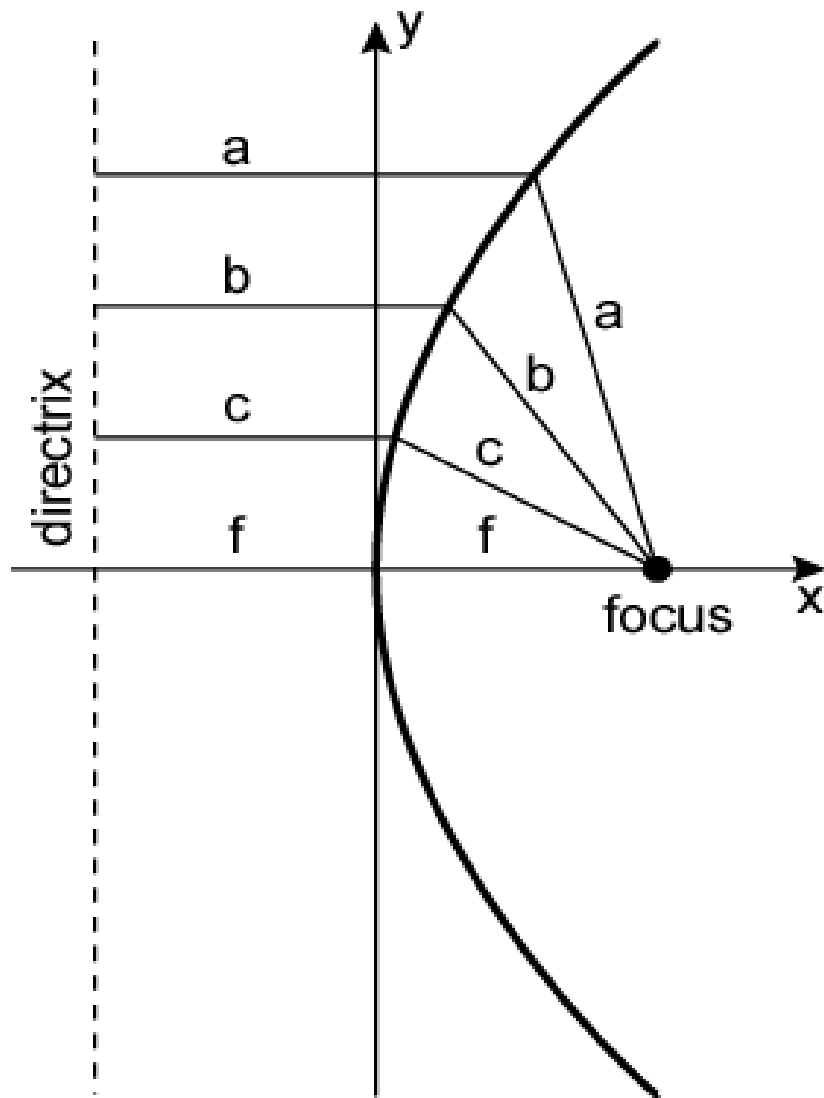
Radiation Pattern

- **Antenna characteristics are essentially the same whether an antenna is sending or receiving electromagnetic energy**
- **Power radiated in all directions**
- **Not same performance in all directions**
- **Isotropic antenna is (theoretical) point in space that radiates in all directions equally**
 - **Gives spherical radiation pattern**
 - **Also called omnidirectional antenna**

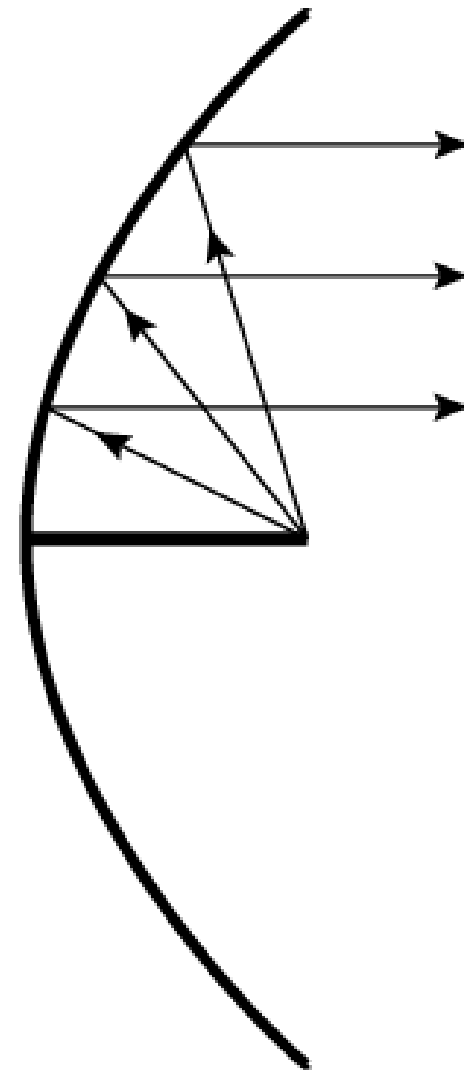
Parabolic Reflective Antenna

- **Used for terrestrial and satellite microwave**
- **Parabola is locus of point equidistant from a fixed line and a fixed point not on that line**
 - **Fixed point is focus**
 - **Fixed line is directrix**
- **Revolve parabola about axis to get paraboloid**
 - **Cross section parallel to axis gives parabola**
 - **Cross section perpendicular to axis gives circle**
- **Source placed at focus will produce waves reflected from parabola in parallel to axis**
 - **Creates (theoretical) parallel beam of light/sound/radio**
- **On reception, signal is concentrated at focus, where detector is placed**

Parabolic Reflective Antenna



(a) Parabola



(b) Cross-section of parabolic antenna showing reflective property

Antenna Gain

- **Measure of directionality of antenna**
- **Power output in particular direction compared with that produced by isotropic antenna**

$G_{\text{dB}} = 10 \log (P_2/P_1)$, where G is the antenna gain, P_1 is the radiated power of the directional antenna, and P_2 is the radiated power from the reference antenna.

- **Results in loss in power in another direction**
- **Effective area of an antenna relates to its size and shape**
 - **Related to gain**

Terrestrial Microwave

- **Most common type: Parabolic dish**
 - **3m in diameter**
 - **located at substantial heights**
- **Focuses a narrow beam to achieve line of sight transmission**
- **Used in long haul telecommunications**
- **Higher the frequency used, higher the bandwidth and thus give higher data rates**
- **The loss can be expressed as**

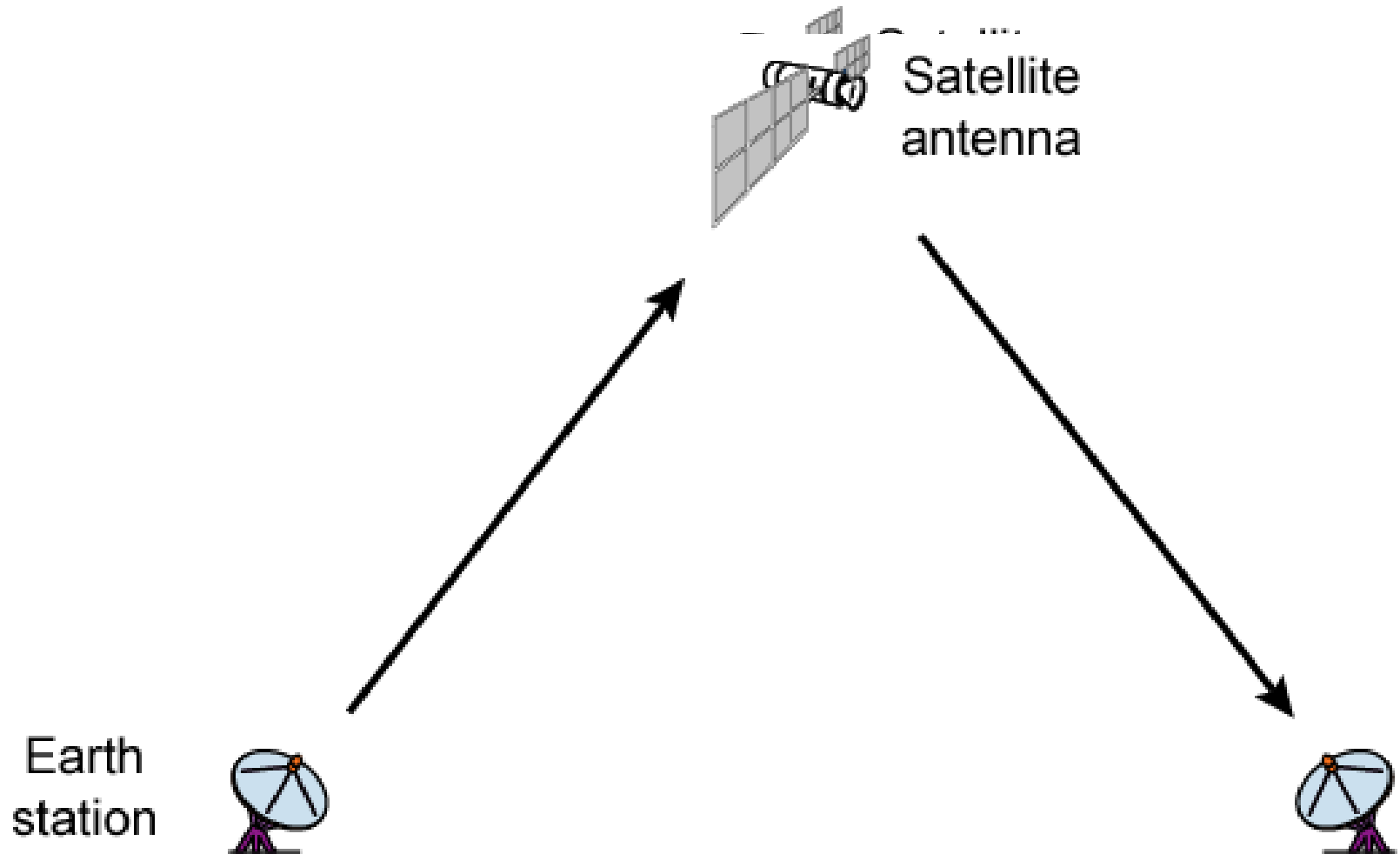
$$L = 10 \log \left(\frac{4\pi d}{\lambda} \right)^2 \text{ dB}$$

where d is the distance and λ is the wavelength, in the same units.

Satellite Microwave

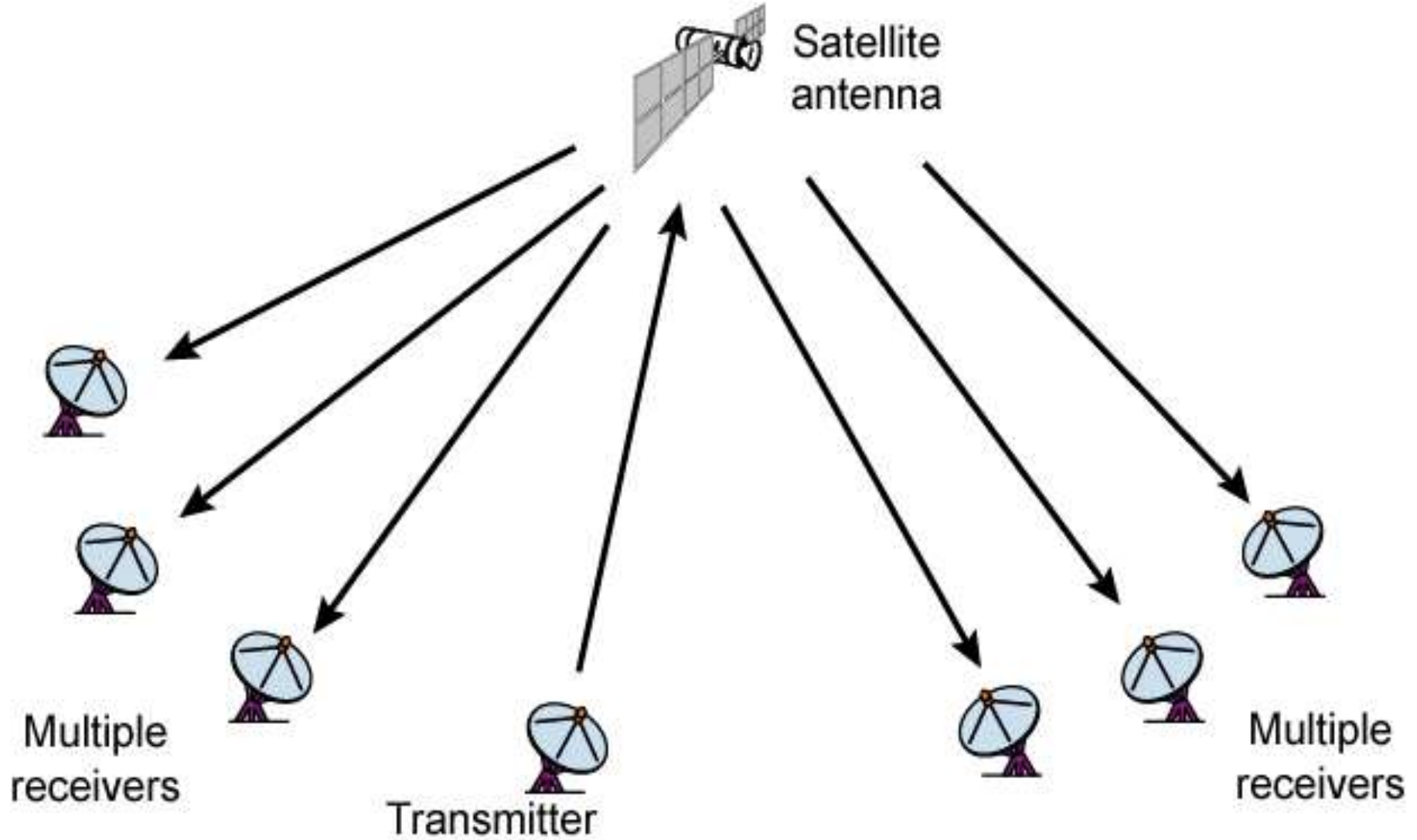
- **Satellite is relay station**
- **Satellite receives on one frequency, amplifies or repeats signal and transmits on another frequency**
- **Requires geo-stationary orbit**
 - **Height of 35,784km**
- **Television**
- **Long distance telephone**
- **Private business networks**

Satellite Point to Point Link



(a) Point-to-point link
(a) Point-to-point link

Satellite Broadcast Link



(b) Broadcast link

Broadcast Radio

- **Omnidirectional**
- **FM radio**
- **UHF and VHF television**
- **Line of sight**
- **Suffers from multipath interference**
 - **Reflections**

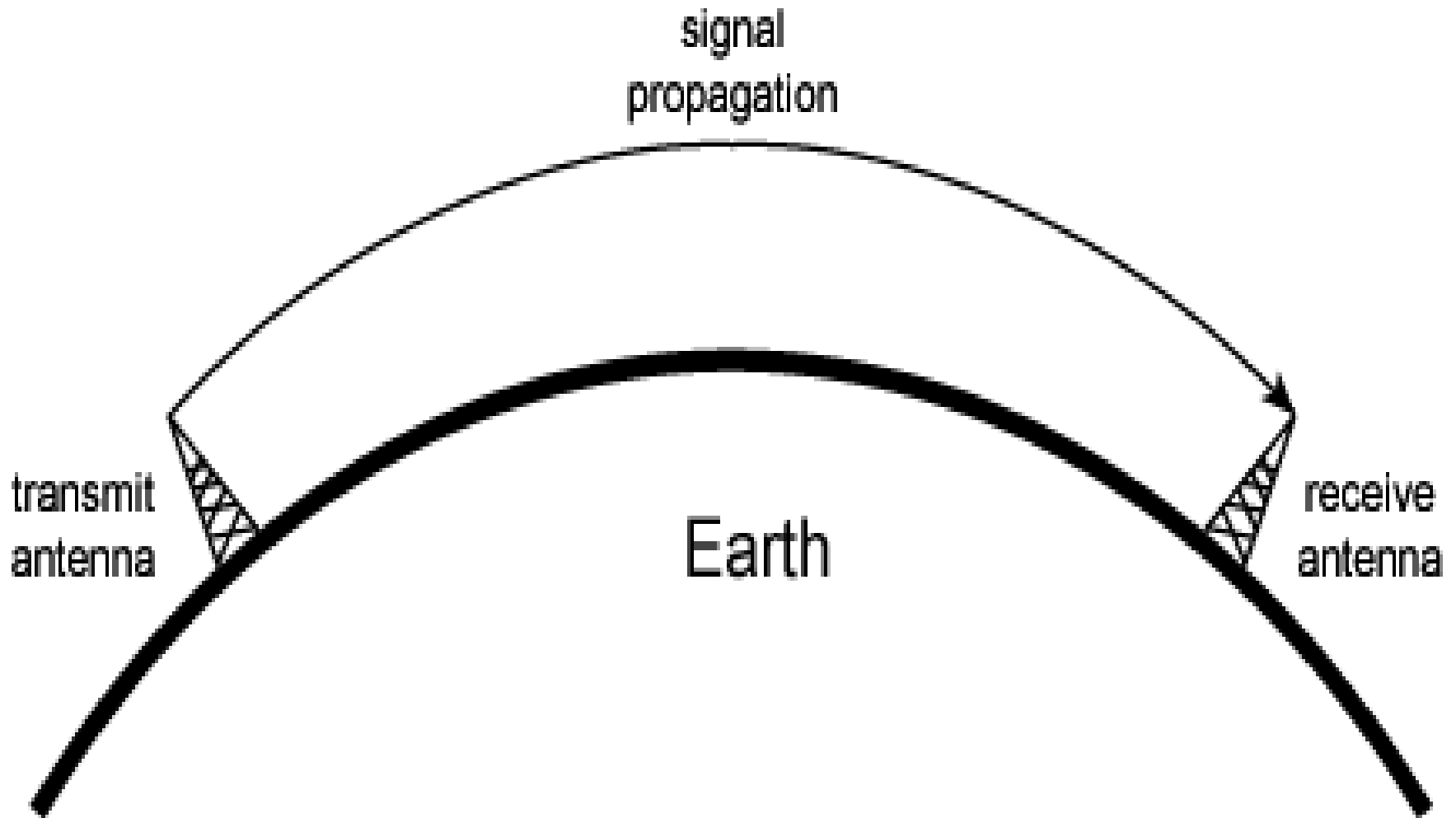
Infrared

- **Modulate noncoherent infrared light**
- **Line of sight (or reflection)**
- **Blocked by walls**
- **e.g. TV remote control, IRD port**

Wireless Propagation

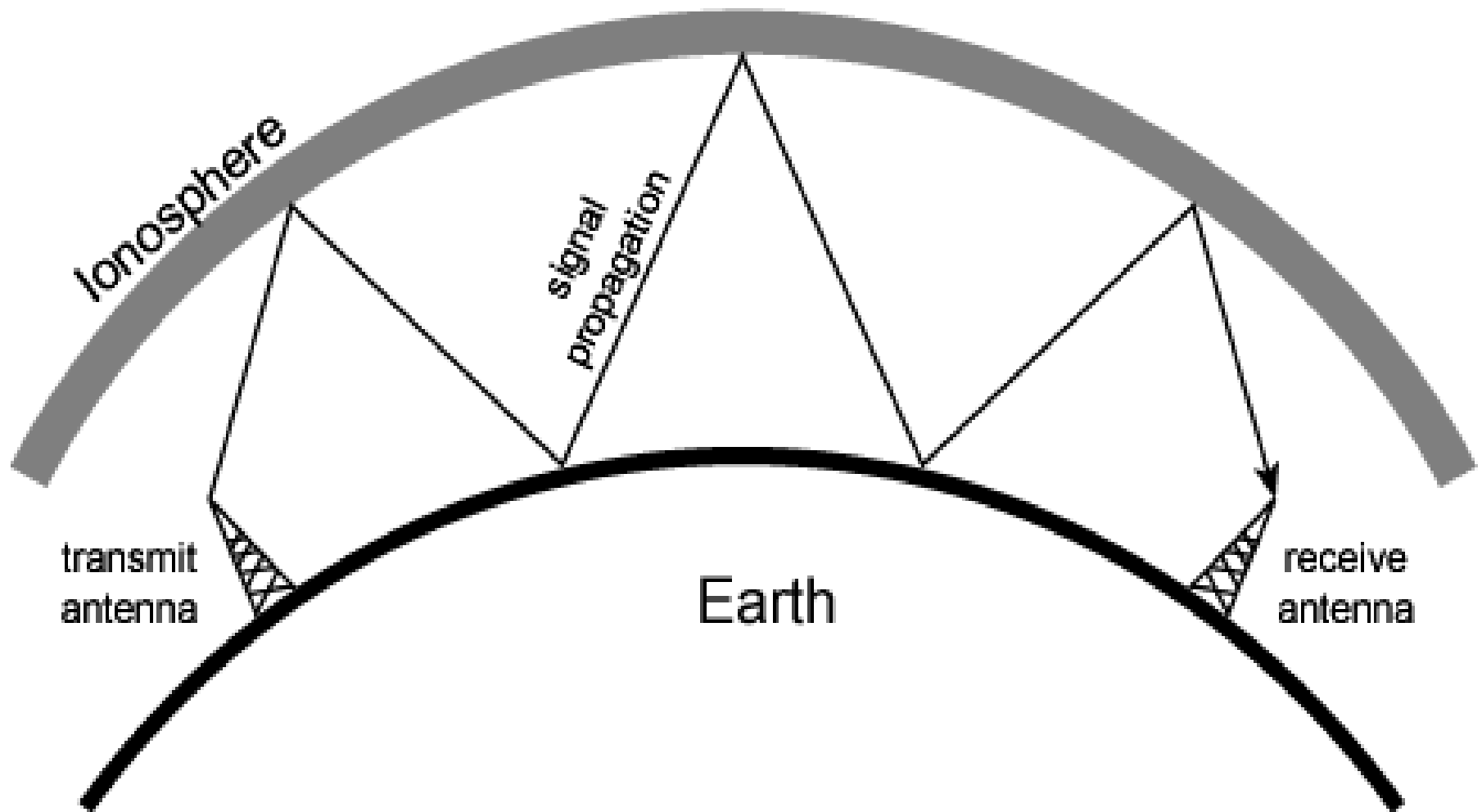
- **Signal travels along three routes**
 - **Ground wave**
 - **Follows contour of earth**
 - **Up to 2MHz**
 - **AM radio**
 - **Sky wave**
 - **Amateur radio, BBC world service, Voice of America**
 - **Signal reflected from ionosphere layer of upper atmosphere**
 - **(Actually refracted)**
 - **Line of sight**
 - **Above 30Mhz**
 - **May be further than optical line of sight due to refraction**

Ground Wave Propagation



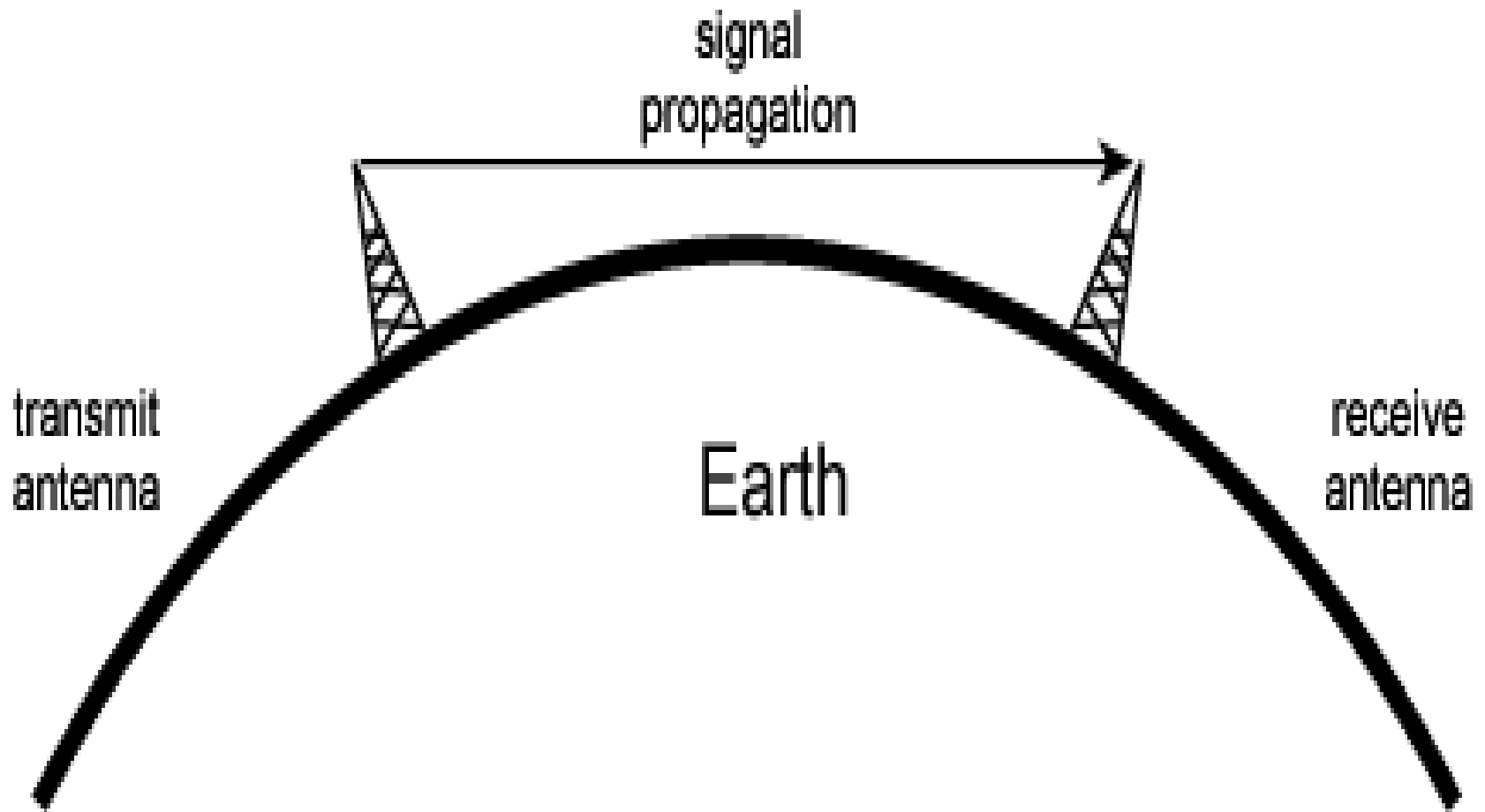
(a) Ground-wave propagation (below 2 MHz)

Sky Wave Propagation



(b) Sky-wave propagation (2 to 30 MHz)

Line of Sight Propagation

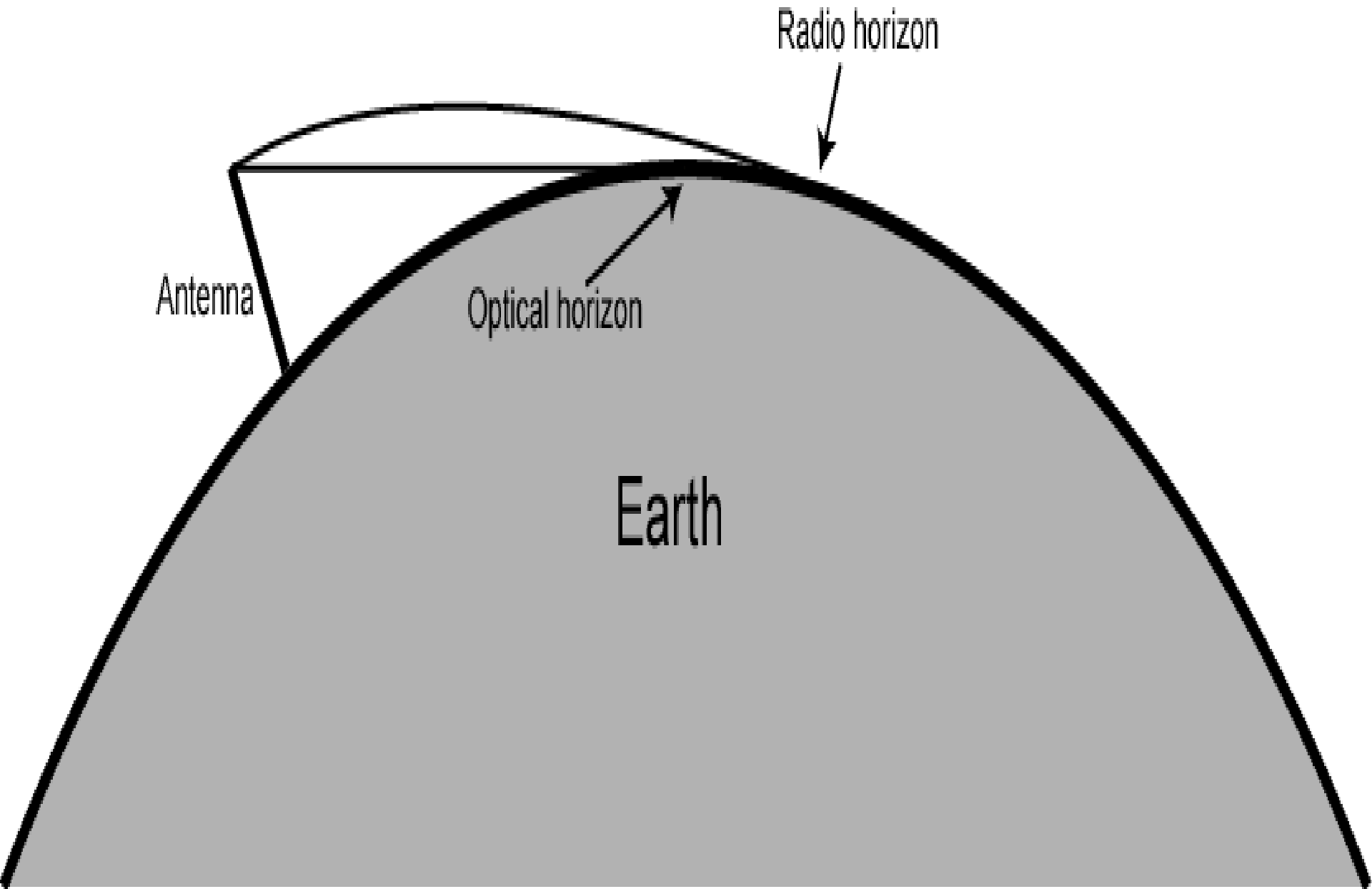


(c) Line-of-sight (LOS) propagation (above 30 MHz)

Refraction

- **Velocity of electromagnetic wave is a function of density of material**
 - **$\sim 3 \times 10^8$ m/s in vacuum, less in anything else**
- **As wave moves from one medium to another, its speed changes**
 - **Causes bending of direction of wave at boundary**
 - **Towards more dense medium**
- **Index of refraction (refractive index) is**
 - **$\sin(\text{angle of incidence})/\sin(\text{angle of refraction})$**
 - **Varies with wavelength**
- **May cause sudden change of direction at transition between media**
- **May cause gradual bending if medium density is varying**
 - **Density of atmosphere decreases with height**
 - **Results in bending towards earth of radio waves**

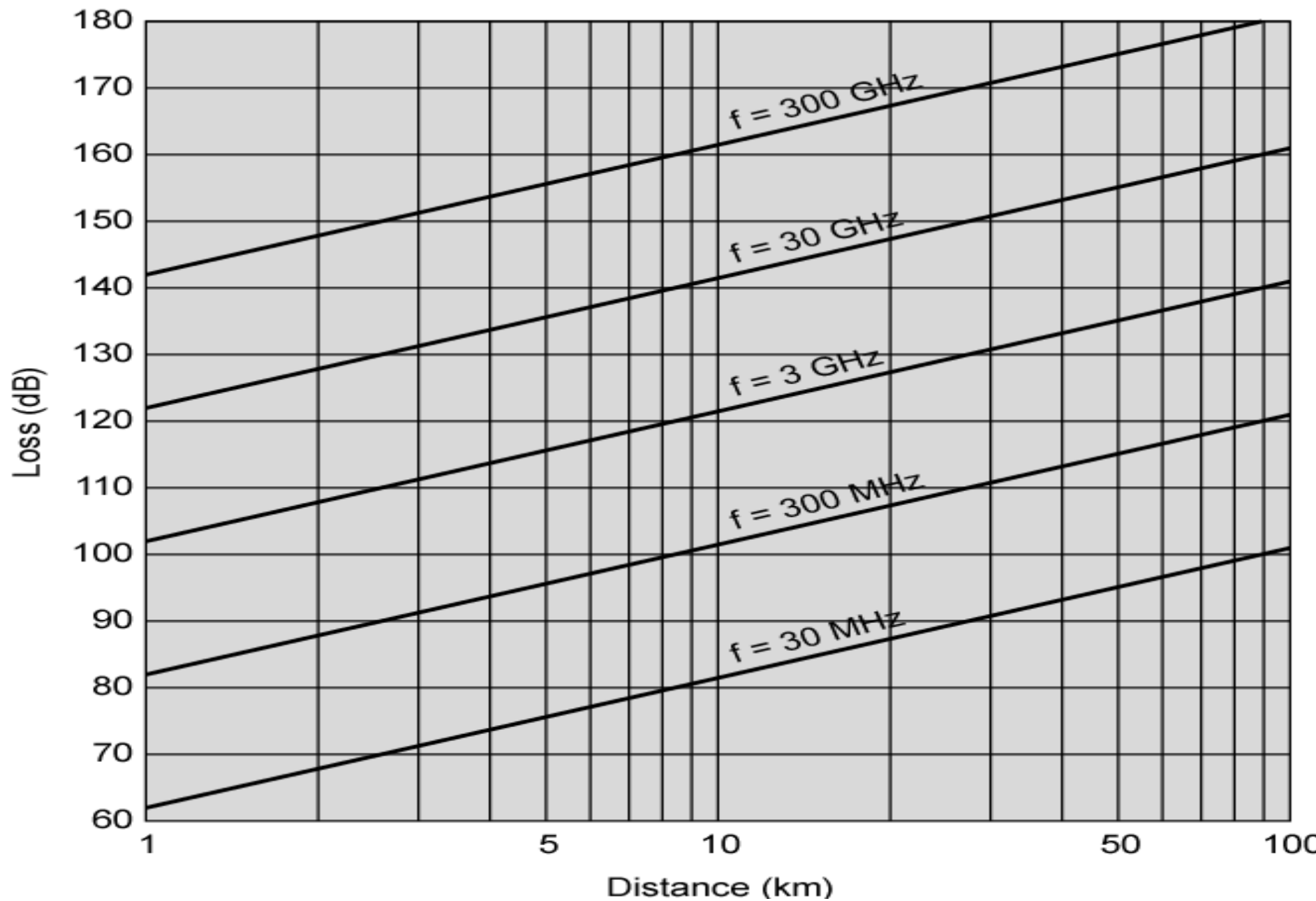
Optical and Radio Horizons



Line of Sight Transmission

- Free space loss
 - Signal disperses with distance
 - Greater for lower frequencies (longer wavelengths)
- Atmospheric Absorption
 - Water vapour and oxygen absorb radio signals
 - Water greatest at 22GHz, less below 15GHz
 - Oxygen greater at 60GHz, less below 30GHz
 - Rain and fog scatter radio waves
- Multipath
 - Better to get line of sight if possible
 - Signal can be reflected causing multiple copies to be received
 - May be no direct signal at all
 - May reinforce or cancel direct signal
- Refraction
 - May result in partial or total loss of signal at receiver

Free Space Loss



Revisit to Data and Signal

- **To be transmitted, data must be transformed to electromagnetic signals**
- **Both data and the signals that represent them can be either Analog or Digital in form**
- **Data can be Analog (continuous and take continuous values) or Digital (has discrete states and take discrete values)**
- **Analog signals can have an infinite number of values in a range**
- **Digital signals can have only a limited number of values**
- **In data communication, we commonly use periodic analog signals(because they need less bandwidth) and nonperiodic digital signals(because they can represent variation in data)**
- **A complete sine wave in the time domain can be represented by one single spike in the frequency domain**

- A single-frequency sine wave is not useful in data communication; we need to send a composite signal, a signal made up of many simple sine waves
- $\text{Wavelength} = \text{propagation speed} \times \text{period} = \text{propagation speed} / \text{frequency}$
- The propagation speed of electromagnetic signals depends on the medium and frequency
- If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies
- If the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies
- The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal
- $\text{Bit length} = \text{propagation speed} \times \text{bit duration}$

Numerical Problems

- Q1.** If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.
- Q2.** A digital signal has eight levels. How many bits are needed per level?
- Q3.** Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?
- Q4.** What is the bit rate for high-definition TV (HDTV)?

Solutions

Q1. Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz

Amplitude

10+-----]

100

300

500

700

900

Frequency

Bandwidth = $900 - 100 = 800 \text{ Hz}$

Q2. Number of bits per level = $\log_2 8 = 3$

Q3. A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1,636,000 \text{ bps} = 1.636 \text{ Mbps}$$

Solution

Q4. HDTV uses digital signals to broadcast high quality video signals. The HDTV Screen is normally a ratio of 16 : 9 (in contrast to 4 : 3 for regular TV), which means the screen is wider. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel. We can calculate the bit rate as

$$1920 \times 1080 \times 30 \times 24 = 1,492,992,000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

- **Based on Fourier analysis a digital signal (periodic or nonperiodic) is a composite Analog signal with frequencies between 0 and infinity**
- **A digital signal can be transmitted by using two different approaches: baseband transmission or broadband transmission**
- **Baseband transmission**
 - **sending a digital signal over a channel without changing the digital signal to analog signal**
 - **requires a low-pass channel (bandwidth that starts from 0)**
 - **low-pass channel with wide bandwidth: with a wide bandwidth(Coaxial or Fiber optic) , two stations can communicate by using digital signals with good accuracy**

- low-pass channel with limited bandwidth:
approximate the digital signal with an analog signal
- Rough approximation: using first harmonic($N/2$)
frequency; Required Bandwidth = $N/2$ where N is bit rate
- If we need to send bits faster, we need more bandwidth

Q5. What is the required bandwidth of a low-pass channel if we need to send 1 Mbps by using base-band transmission?

Solution:

The answer depends on the accuracy desired.

- The minimum bandwidth, a rough approximation, is $B = \text{bit rate} / 2$, or 500 kHz. We need a low-pass channel with frequencies between 0 and 500 kHz.
- A better result can be achieved by using the first and the third harmonics with the required bandwidth $B = 3 \times 500 \text{ kHz} = 1.5 \text{ MHz}$.
- Still a better result can be achieved by using the first, third, and fifth harmonics with $B = 5 \times 500 \text{ kHz} = 2.5 \text{ MHz}$.

Q6. We have a low-pass channel with bandwidth 100 kHz. What is the maximum bit rate of this channel?

Solution

The maximum bit rate can be achieved if we use the first harmonic. The bit rate is 2 times the available bandwidth, or 200 kbps.

- **Broadband Transmission(Modulation)**
 - **Changing the digital signal to an analog signal**
 - **Use a band-pass channel (bandwidth that does not start from 0)**

Transmission Impairment

Q7. A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Q8. Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $\text{dB}_m = 10 \log_{10} P_m$ where P_m is the power in milliwatts. Calculate the power of a signal if its $\text{dB}_m = -30$.

Solution: We can calculate the power in the signal as

$$\begin{aligned} \text{dB}_m &= 10 \log_{10} P_m = -30 \\ \log_{10} P_m &= -3 \quad P_m = 10^{-3} \text{ mW} \end{aligned}$$

Q9. The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution:

The loss in the cable in decibels is $5 \times (-0.3) = -1.5$ dB. We can calculate the power as

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = -1.5$$

$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

$$P_2 = 0.71P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$

Data Rate Limits

- Data rate depends on three factors:
 1. The bandwidth available
 2. The level of the signals we use
 3. The quality of the channel (the level of noise)

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- Increasing the levels of a signal may reduce the reliability of the system

Does the Nyquist theorem bit rate agree with the intuitive bit rate described in baseband transmission?

Solution

They match when we have only two levels. We said, in baseband transmission, the bit rate is 2 times the bandwidth if we use only the first harmonic in the worst case. However, the Nyquist formula is more general than what we derived intuitively; it can be applied to baseband transmission and modulation. Also, it can be applied when we have two or more levels of signals.

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

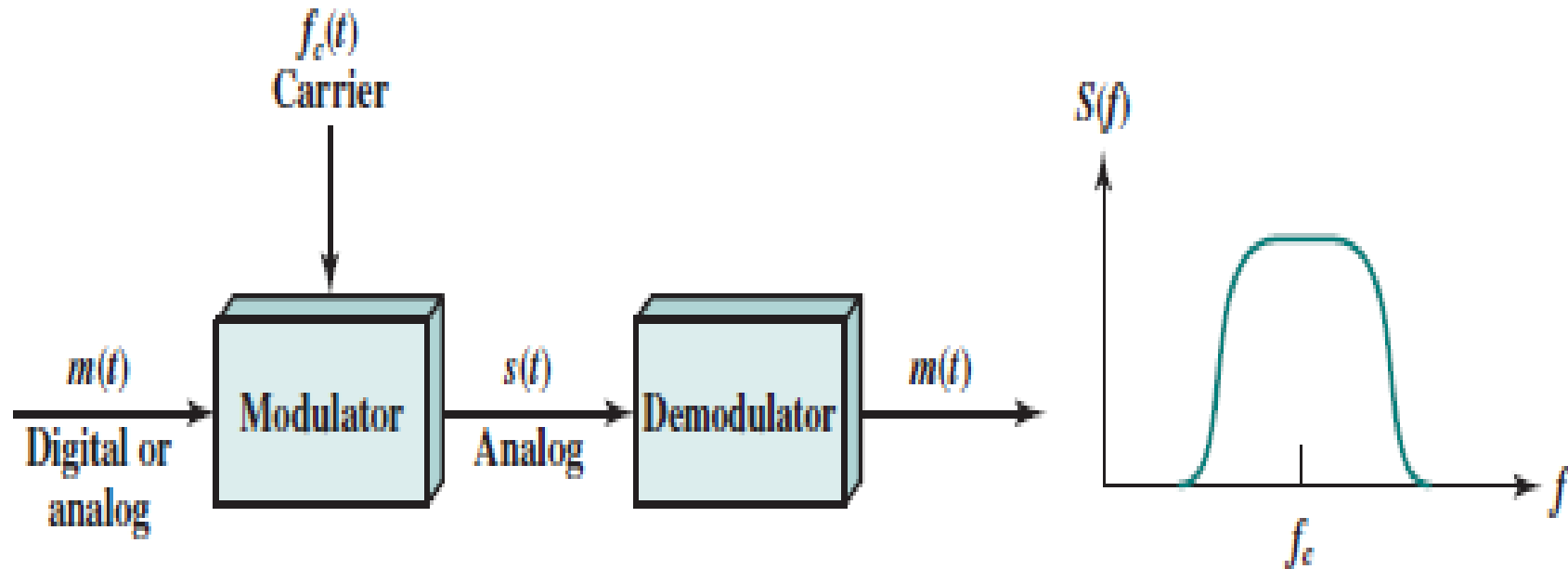
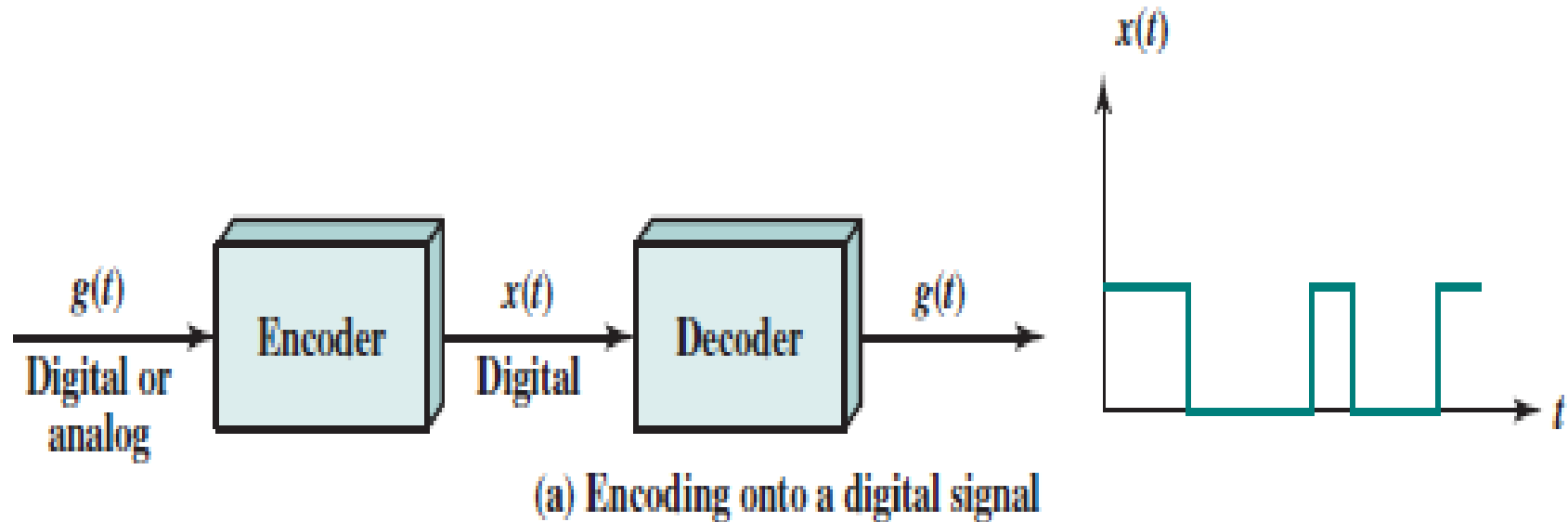
$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Data Communication & Computer Network

Lecture:12, 13, 14, 15 & 16

Presented By
Dr. S.Kanungo

Encoding and Modulation Techniques



Signal Encoding Techniques

- **Digital data, digital signal:** The equipment for encoding is less complex and less expensive than digital-to-analog modulation equipment
- **Analog data, digital signal:** Permits the use of modern digital transmission and switching equipment
- **Digital data, analog signal:** Some transmission media, such as optical fiber and unguided media, will only propagate analog signal
- **Analog data, analog signal:** Analog data in electrical form can be transmitted as baseband signals easily and cheaply; More complex mathematically

Digital Data, Digital Signal

- **Digital signal**
 - Discrete, discontinuous voltage pulses
 - Each pulse is a signal element
 - Binary data encoded into signal elements
- **Unipolar**
 - All signal elements have same sign
- **Polar**
 - One logic state represented by positive voltage the other by negative voltage
- **Data rate**
 - Rate of data transmission in bits per second

- **Duration or length of a bit**
 - Time taken for transmitter to emit the bit
- **Modulation rate**
 - Rate at which the signal level changes
 - Measured in baud = signal elements per second
- **Mark and Space**
 - Binary 1 and Binary 0 respectively

Interpreting Signals

- **Need to know**
 - Timing of bits - when they start and end
 - Signal levels
- **Factors affecting successful interpreting of signals**
 - Signal to noise ratio
 - Data rate
 - Bandwidth

Comparison of Encoding Schemes

- **Signal Spectrum**
 - Lack of high frequencies reduces required bandwidth
 - Lack of dc component allows ac coupling via transformer, providing isolation
 - Concentrate power in the middle of the bandwidth
- **Clocking**
 - Synchronizing transmitter and receiver
 - External clock
 - Sync mechanism based on signal

- **Error detection**
 - Can be built in to signal encoding
- **Signal interference and noise immunity**
 - Some codes are better than others
- **Cost and complexity**
 - Higher signal rate (& thus data rate) lead to higher costs
 - Some codes require signal rate greater than data rate

Encoding Schemes

- **Nonreturn to Zero-Level (NRZ-L)**
- **Nonreturn to Zero Inverted (NRZI)**
- **Bipolar -AMI**
- **Pseudoternary**
- **Manchester**
- **Differential Manchester**
- **B8ZS**
- **HDB3**

Nonreturn to Zero-Level (NRZ-L)

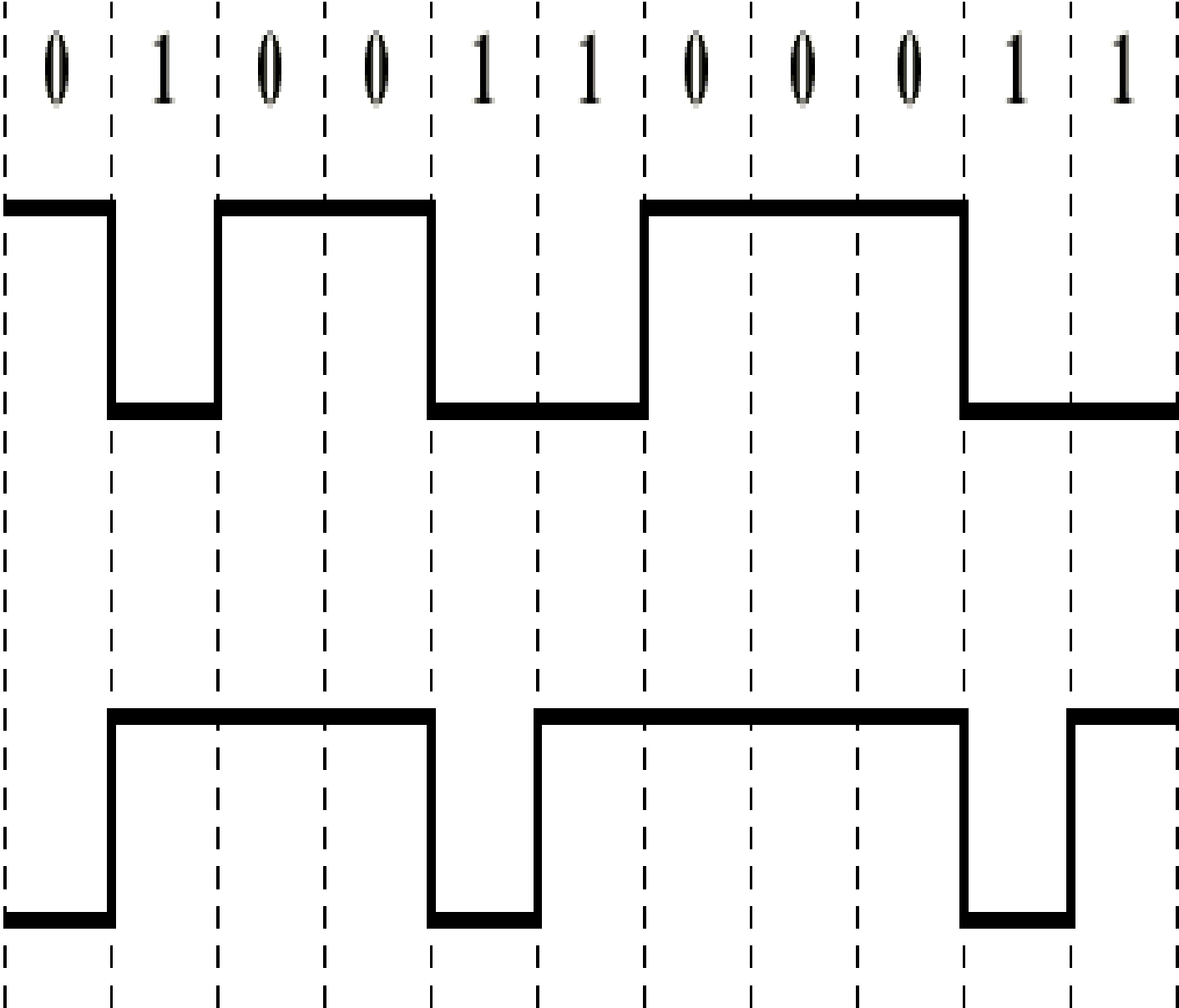
- **Two different voltages for 0 and 1 bits**
- **Voltage constant during bit interval**
 - **no transition i.e. no return to zero voltage**
- **e.g. Absence of voltage for zero, constant positive voltage for one**
- **More often, negative voltage for one value and positive for the other**

Nonreturn to Zero Inverted

- **Nonreturn to zero inverted on ones**
- **Constant voltage pulse for duration of bit**
- **Data encoded as presence or absence of signal transition at beginning of bit time**
- **Transition (low to high or high to low) denotes a binary 1**
- **No transition denotes binary 0**
- **An example of differential encoding**

• NRZ

NRZ-L



NRZI

Differential Encoding

- **Data represented by changes rather than levels**
- **More reliable detection of transition rather than level**
- **In complex transmission layouts it is easy to lose sense of polarity**

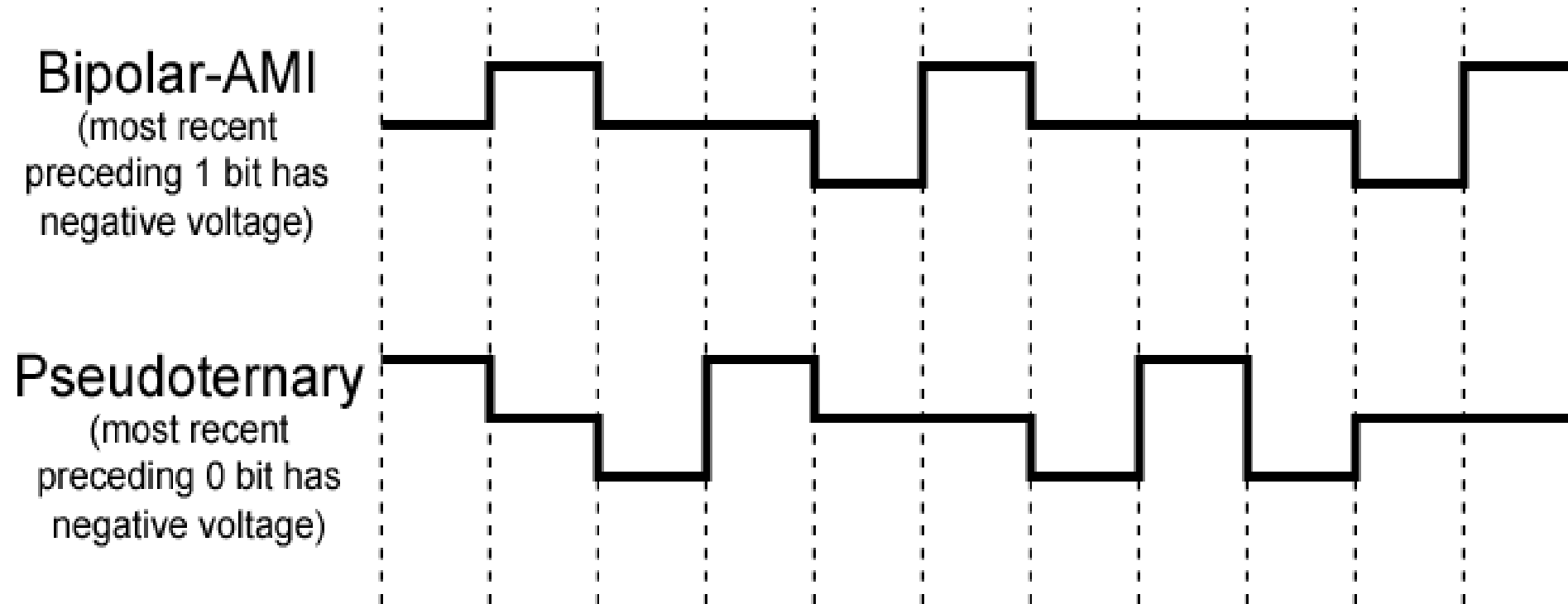
NRZ pros and cons

- **Pros**
 - Easy to engineer
 - Make good use of bandwidth
- **Cons**
 - dc component
 - Lack of synchronization capability
- **Used for magnetic recording**
- **Not often used for signal transmission**

Multilevel Binary

- **Use more than two levels**
- **Bipolar-AMI**
 - zero represented by no line signal
 - one represented by positive or negative pulse
 - one pulses alternate in polarity
 - No loss of sync if a long string of ones (zeros still a problem)
 - No net dc component
 - Lower bandwidth
 - Easy error detection

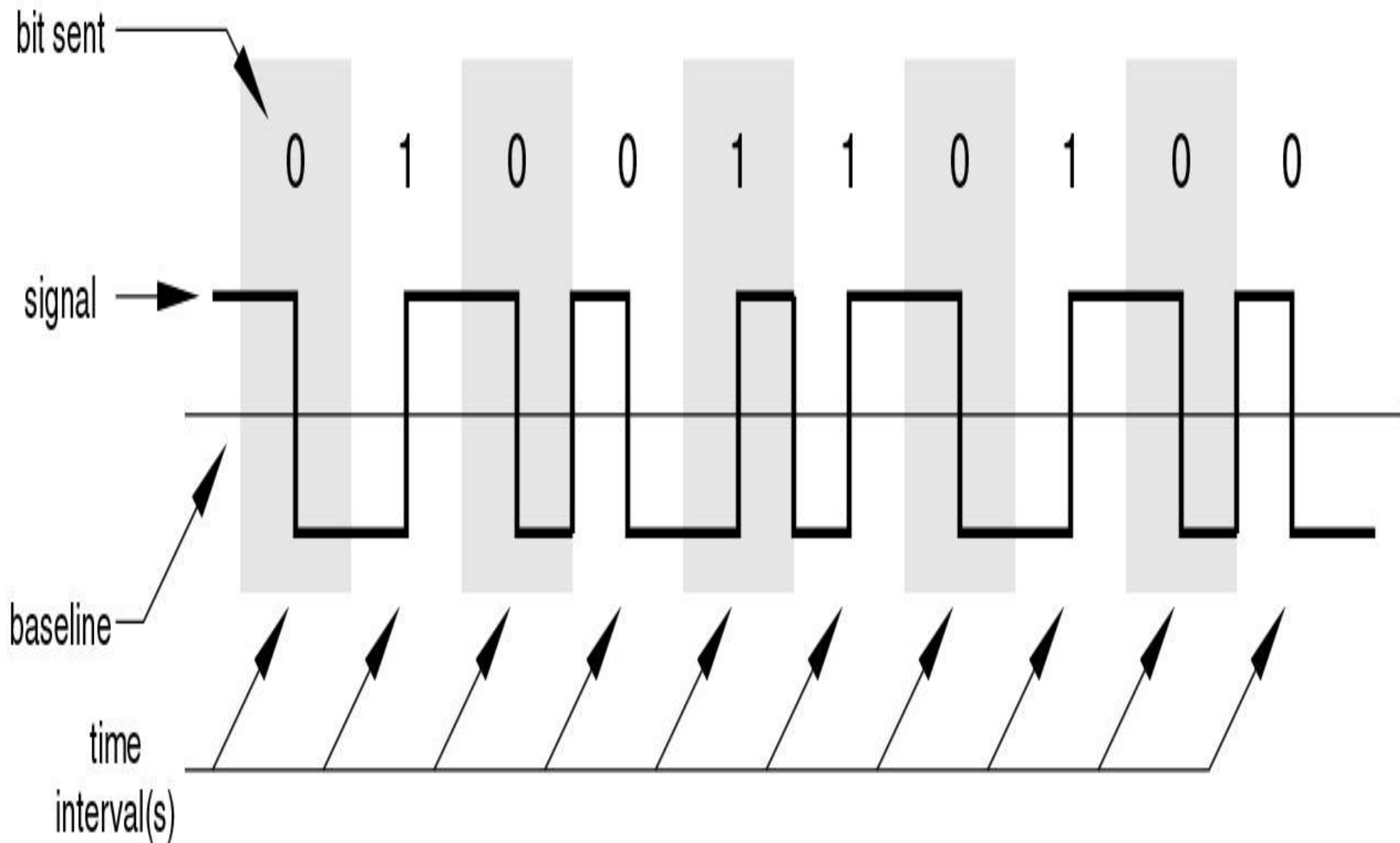
- **Pseudoternary**
 - One represented by absence of line signal
 - Zero represented by alternating positive and negative
 - No advantage or disadvantage over bipolar-AMI
- **Bipolar-AMI and Pseudoternary**



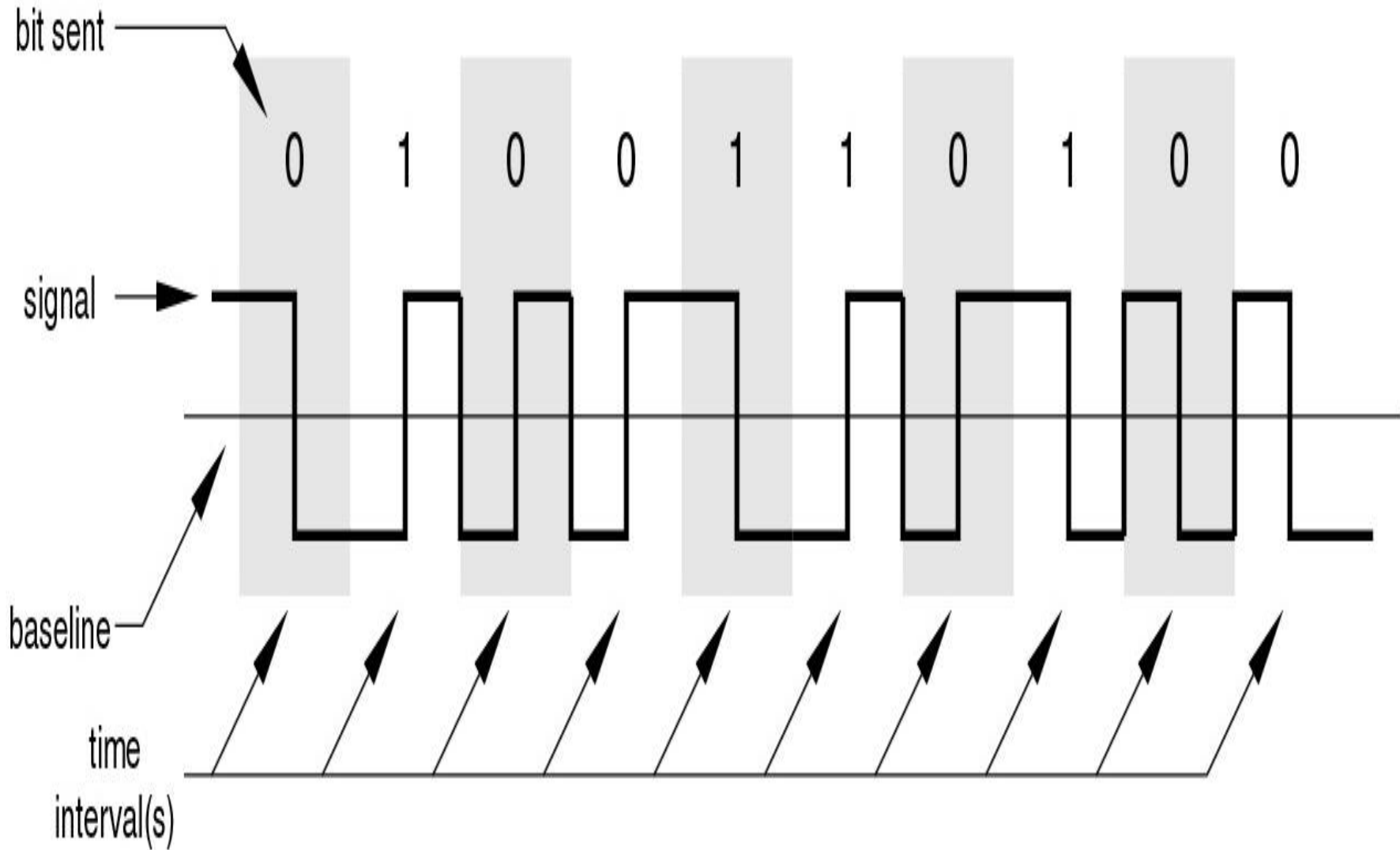
- **Trade Off for Multilevel Binary**
 - **Not as efficient as NRZ**
 - **Each signal element only represents one bit**
 - **In a 3 level system could represent $\log_2 3 = 1.58$ bits**
 - **Receiver must distinguish between three levels (+A, -A, 0)**
 - **Requires approx. 3dB more signal power for same probability of bit error**

- **Biphase**
 - **Manchester**
 - Transition in middle of each bit period
 - Transition serves as clock and data
 - Low to high represents one
 - High to low represents zero
 - Transition at start of bit period represents zero
 - Used by IEEE 802.3
 - **Differential Manchester**
 - Midbit transition in a bit period represents zero
 - No transition at start of a bit period represents one
 - Note: this is a differential encoding scheme
 - Used by IEEE 802.5

Manchester Encoding



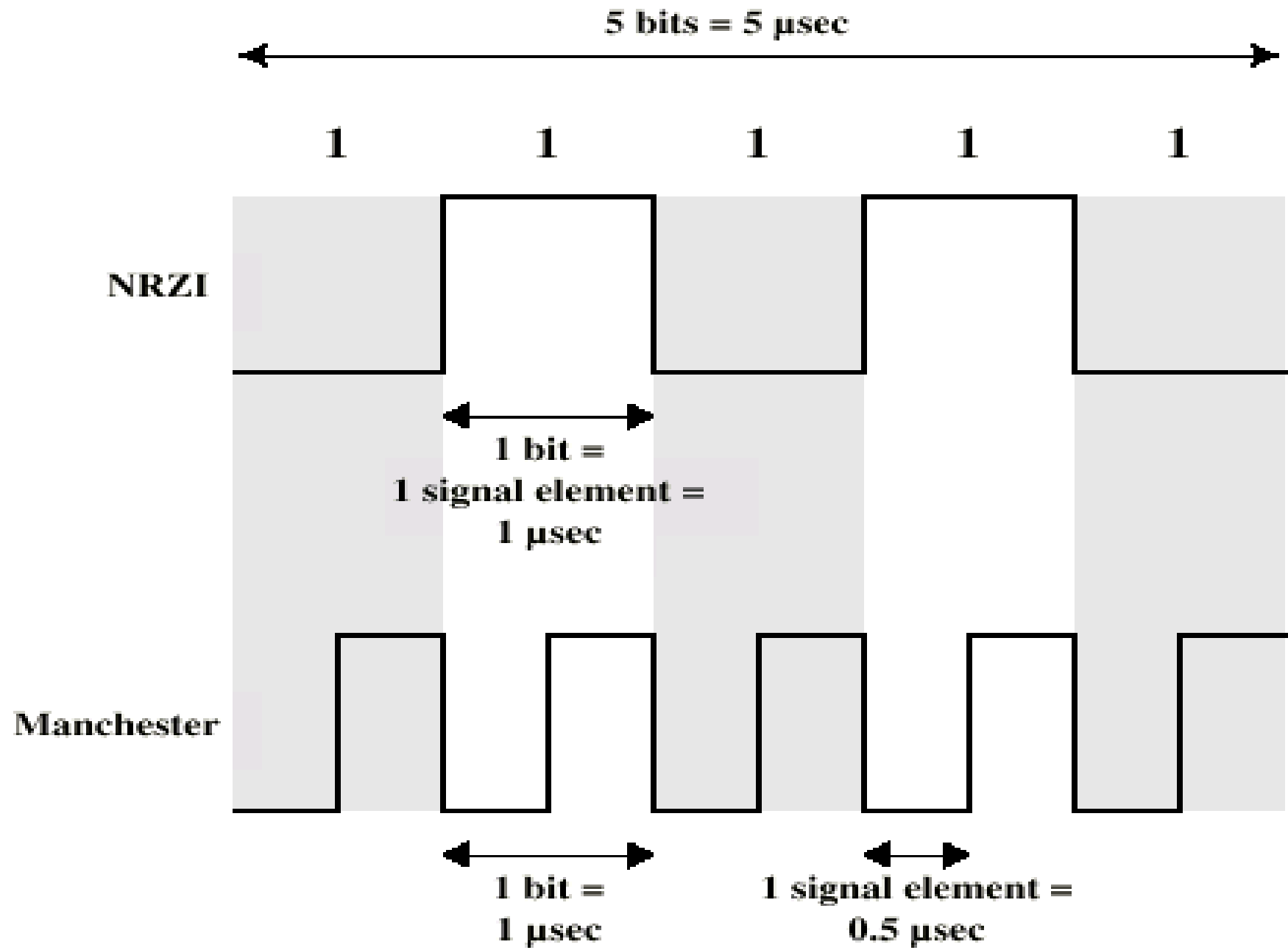
Differential Manchester Encoding



Biphase Pros and Cons

- **Con**
 - At least one transition per bit time and possibly two
 - Maximum modulation rate is twice NRZ
 - Requires more bandwidth
- **Pros**
 - Synchronization on mid bit transition (self clocking)
 - No dc component
 - Error detection
 - Absence of expected transition

- **Modulation Rate**

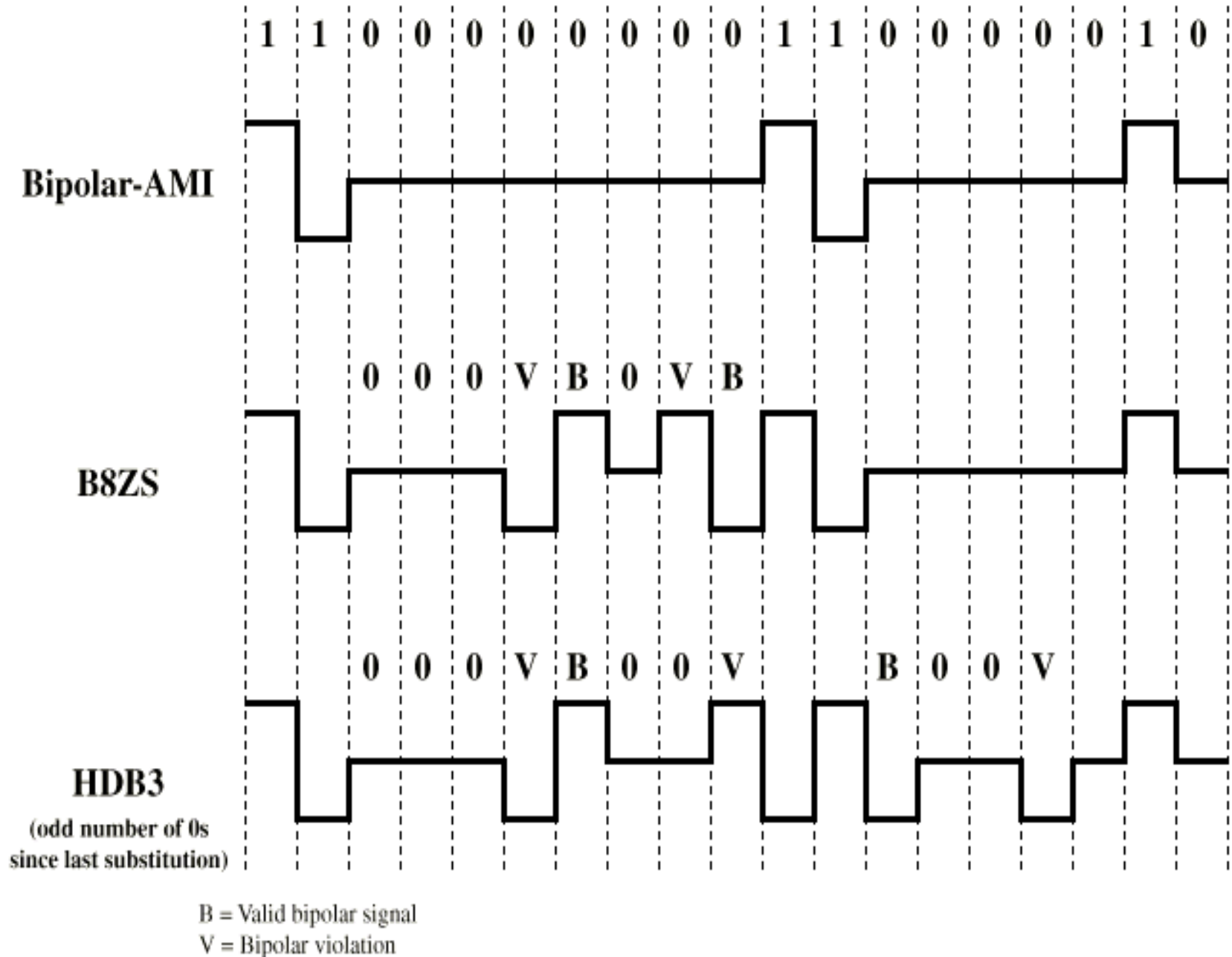


- **Scrambling**
 - **Use scrambling to replace sequences that would produce constant voltage**
 - **Filling sequence**
 - **Must produce enough transitions to sync**
 - **Must be recognized by receiver and replace with original**
 - **Same length as original**
 - **No dc component**
 - **No long sequences of zero level line signal**
 - **No reduction in data rate**
 - **Error detection capability**

- **B8ZS**
 - **Bipolar With 8 Zeros Substitution**
 - **Based on bipolar-AMI**
 - **If octet of all zeros and last voltage pulse preceding was positive encode as 000+-0-+**
 - **If octet of all zeros and last voltage pulse preceding was negative encode as 000-+0+-**
 - **Causes two violations of AMI code**
 - **Unlikely to occur as a result of noise**
 - **Receiver detects and interprets as octet of all zeros**

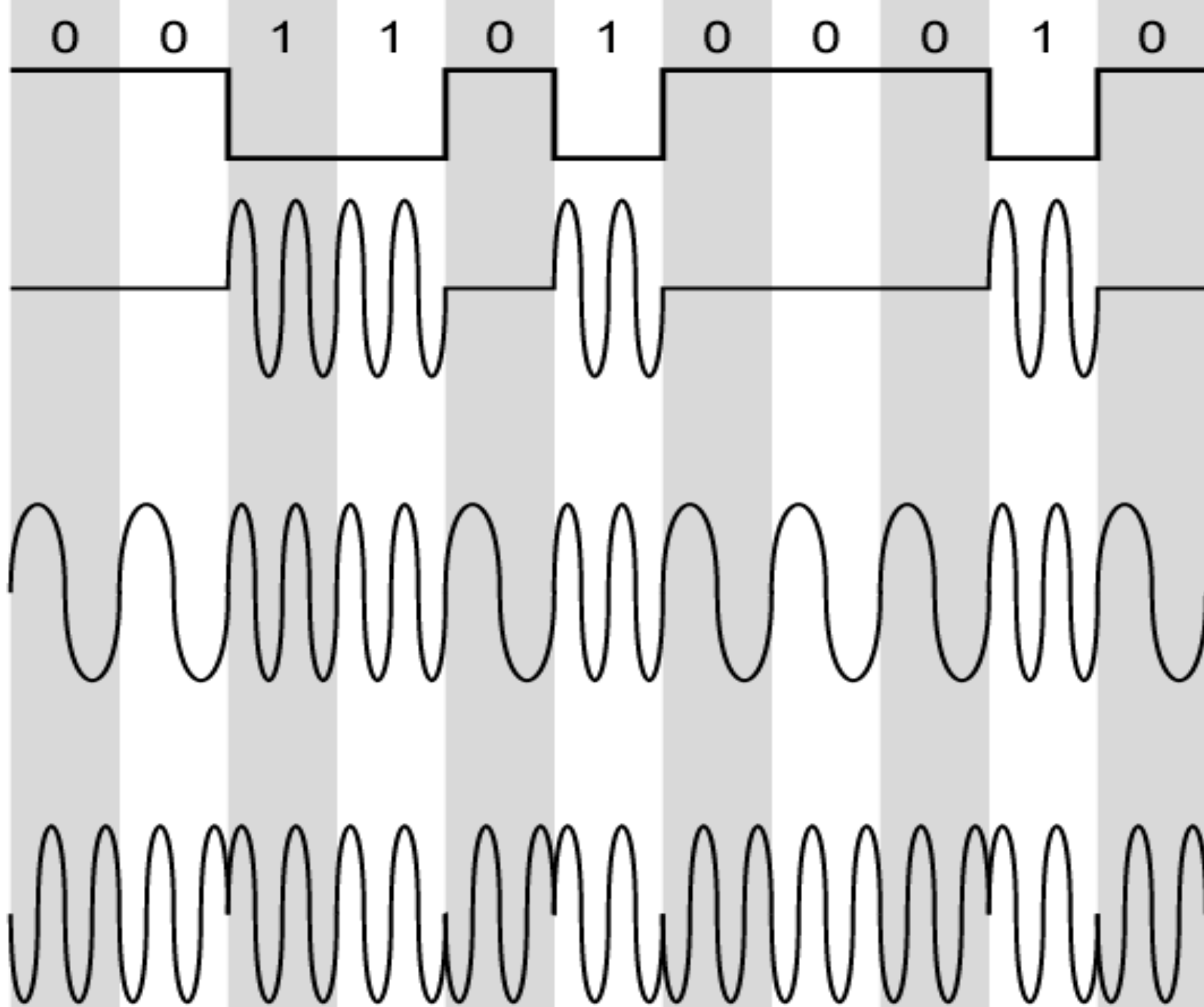
- **HDB3**
 - **High Density Bipolar 3 Zeros**
 - **Based on bipolar-AMI**
 - **String of four zeros replaced with one or two pulses**

- **B8ZS and HDB3**



Digital Data, Analog Signal

- **Public telephone system**
 - 300Hz to 3400Hz
 - Use modem (modulator-demodulator)
- **Amplitude shift keying (ASK)**
- **Frequency shift keying (FSK)**
- **Phase shift keying (PK)**



Amplitude Shift Keying

- Values represented by different amplitudes of carrier
- Usually, one amplitude is zero
 - i.e. presence and absence of carrier is used

$$\text{ASK } s(t) = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ 0 & \text{binary 0} \end{cases}$$

- Susceptible to sudden gain changes
- Inefficient modulation technique
- Used only up to 1200bps on voice grade lines
- Used over optical fiber

Binary Frequency Shift Keying

- Most common form is binary FSK (BFSK)
- Two binary values represented by two different frequencies (near carrier)

$$\text{BFSK } s(t) = \begin{cases} A \cos(2\pi f_1 t) & \text{binary 1} \\ A \cos(2\pi f_2 t) & \text{binary 0} \end{cases}$$

where f_1 and f_2 are typically offset from the carrier frequency f_c by equal but opposite amounts.

- Less susceptible to error than ASK
- Up to 1200bps on voice grade lines
- High frequency radio
- Even higher frequency on LANs using co-ax

Multiple FSK

- More than two frequencies used
- More bandwidth efficient
- More prone to error
- Each signalling element represents more than one bit

$$\text{MFSK} \quad s_i(t) = A \cos 2\pi f_i t, \quad 1 \leq i \leq M$$

where

$$f_i = f_c + (2i - 1 - M)f_d$$

f_c = the carrier frequency

f_d = the difference frequency

M = number of different signal elements = 2^L

L = number of bits per signal element

FSK on Voice Grade Line

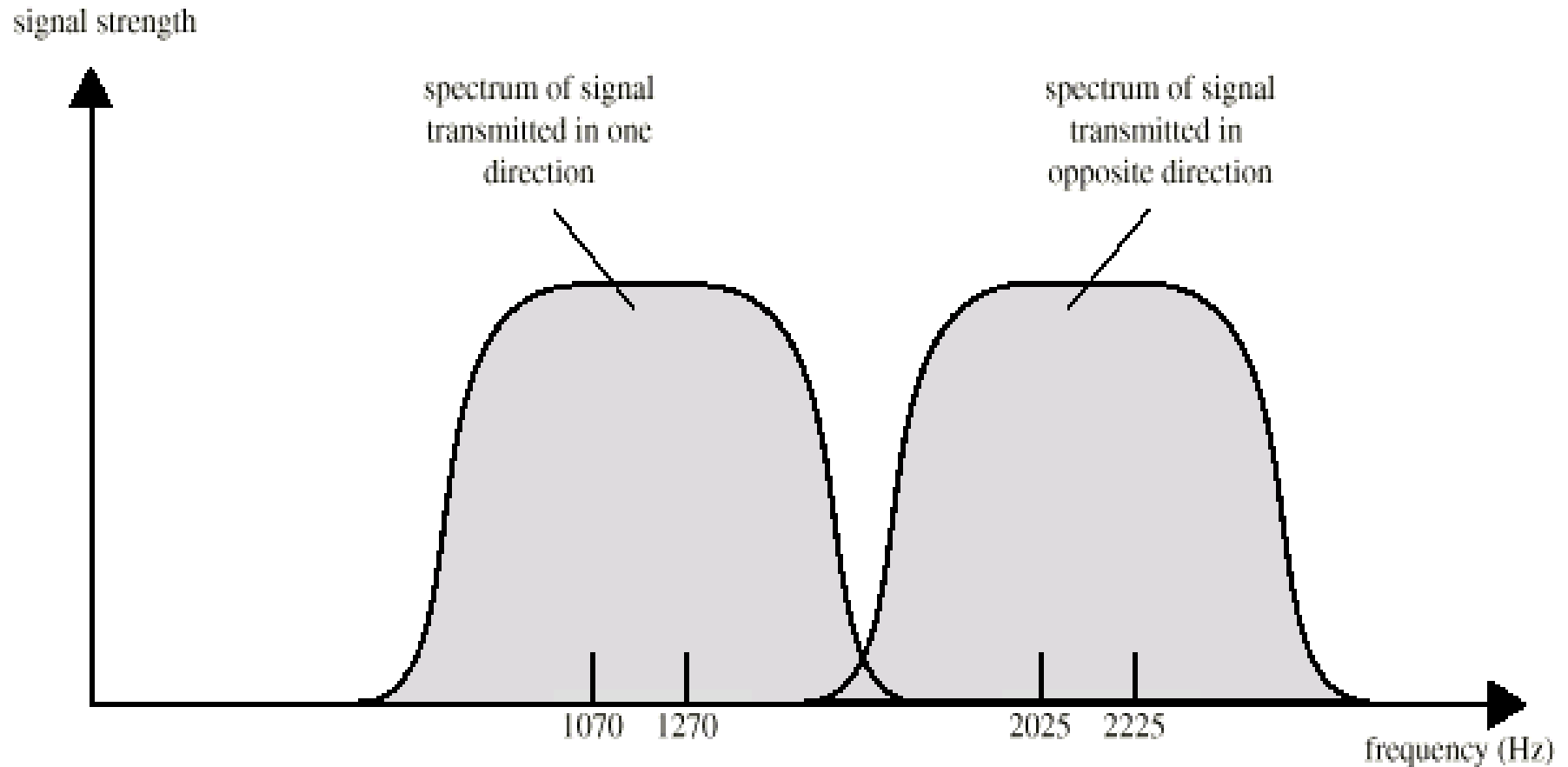


Figure 5.8 Full-Duplex FSK Transmission on a Voice-Grade Line

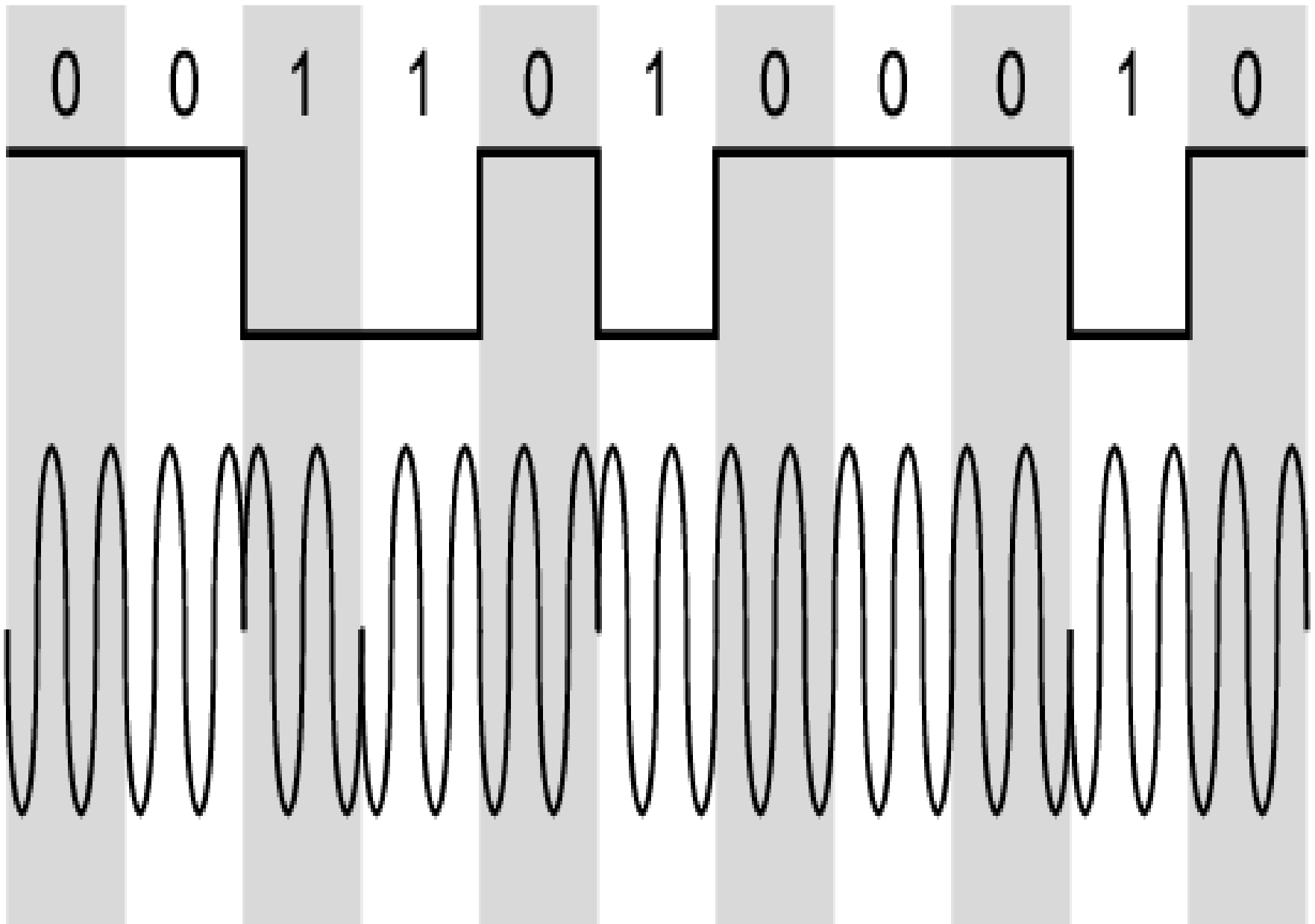
Phase Shift Keying

- **Phase of carrier signal is shifted to represent data**
- **Binary PSK**
 - **Two phases represent two binary digits**

$$\text{BPSK } s(t) = \begin{cases} A \cos(2\pi f_c t) \\ A \cos(2\pi f_c t + \pi) \end{cases} = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ -A \cos(2\pi f_c t) & \text{binary 0} \end{cases}$$

- **Differential PSK**
 - **Phase shifted relative to previous transmission rather than some reference signal**

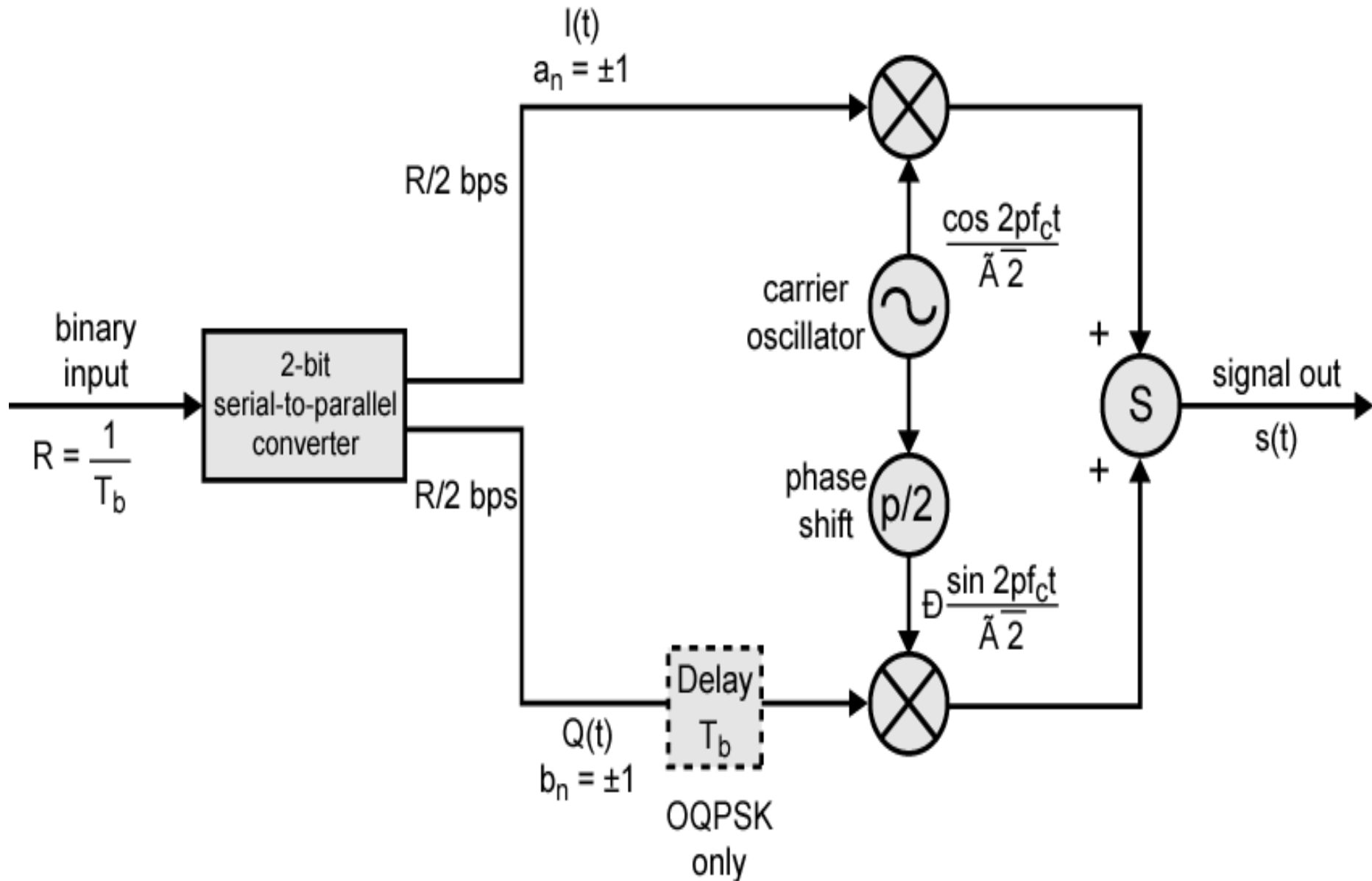
Differential PSK



Quadrature PSK

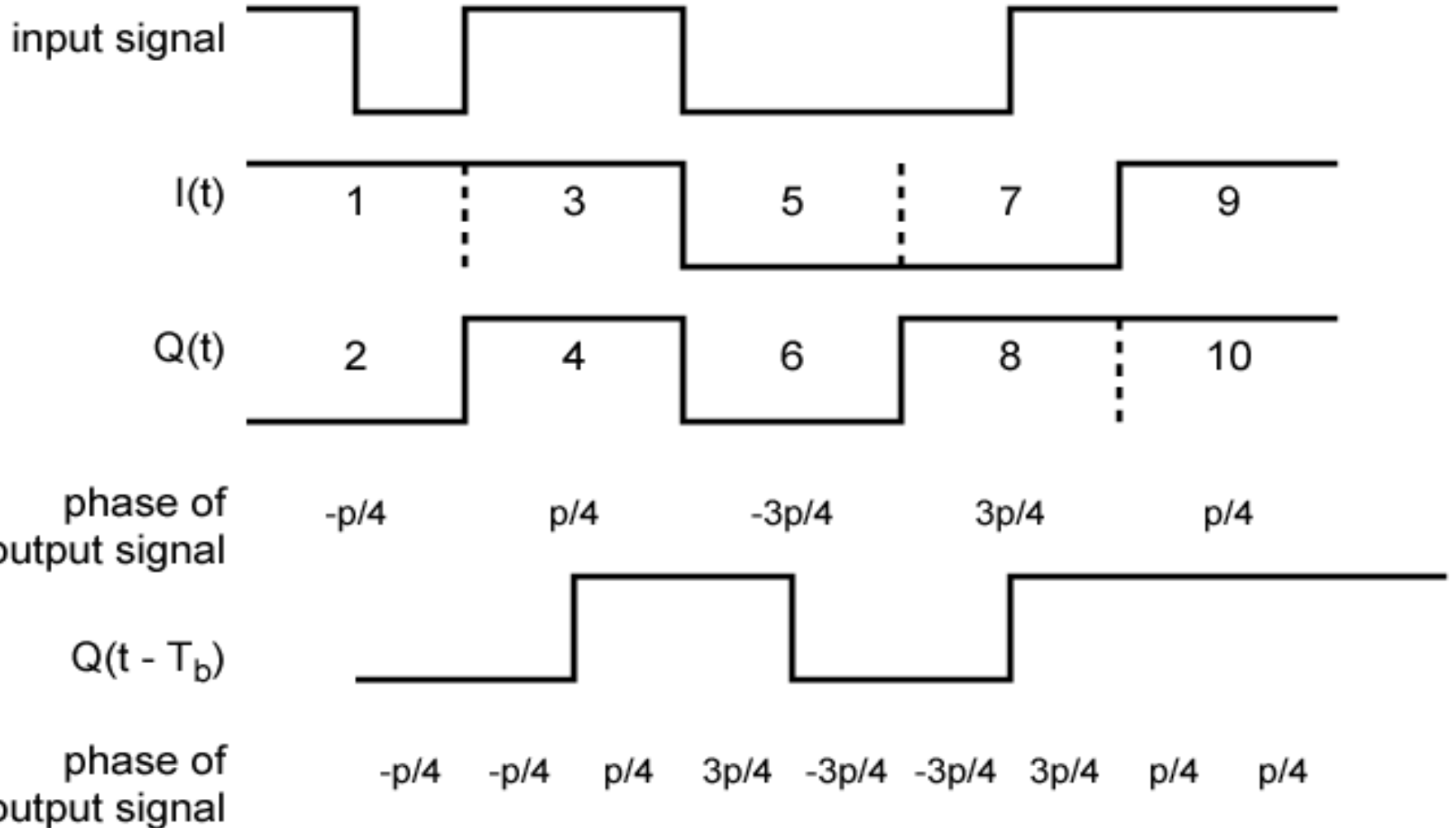
- **More efficient use by each signal element representing more than one bit**
 - e.g. shifts of $\pi/2$ (90°)
 - Each element represents two bits
 - Can use 8 phase angles and have more than one amplitude
 - 9600bps modem use 12 angles , four of which have two amplitudes
- **Offset QPSK (orthogonal QPSK)**
 - Differs from QPSK only by delay in the Q stream

QPSK and OQPSK Modulators



Examples of QPSF and OQPSK Waveforms

bit number	1	2	3	4	5	6	7	8	9	10
bit value	1	-1	1	1	-1	-1	-1	1	1	1
	I	Q	I	Q	I	Q	I	Q	I	Q



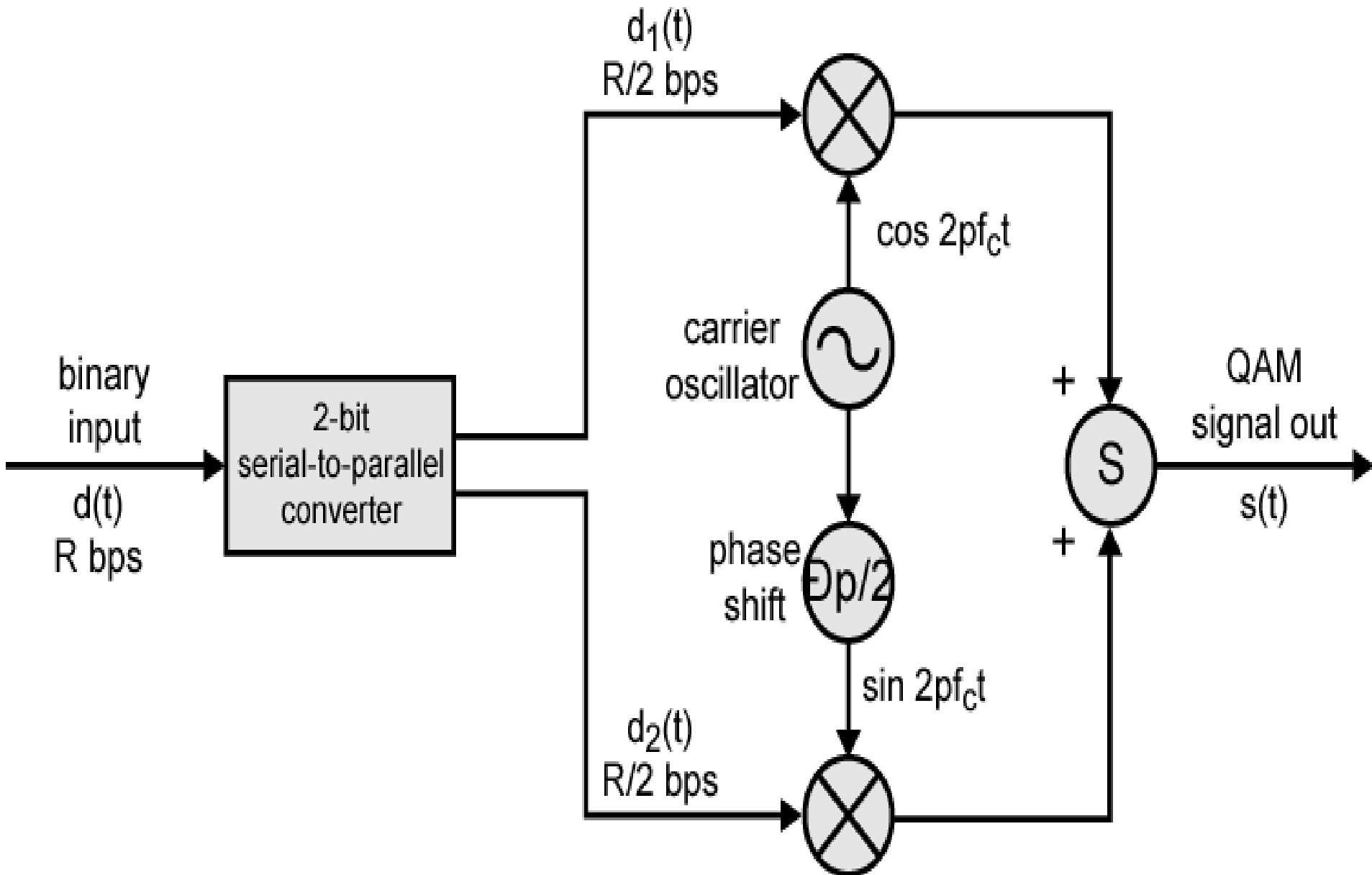
Performance of Digital to Analog Modulation Schemes

- **Bandwidth**
 - ASK and PSK bandwidth directly related to bit rate
 - FSK bandwidth related to data rate for lower frequencies, but to offset of modulated frequency from carrier at high frequencies
 - (See Stallings for math)
- In the presence of noise, bit error rate of PSK and QPSK are about 3dB superior to ASK and FSK

Quadrature Amplitude Modulation

- **QAM used on asymmetric digital subscriber line (ADSL) and some wireless**
- **Combination of ASK and PSK**
- **Logical extension of QPSK**
- **Send two different signals simultaneously on same carrier frequency**
 - **Use two copies of carrier, one shifted 90°**
 - **Each carrier is ASK modulated**
 - **Two independent signals over same medium**
 - **Demodulate and combine for original binary output**

QAM Modulator



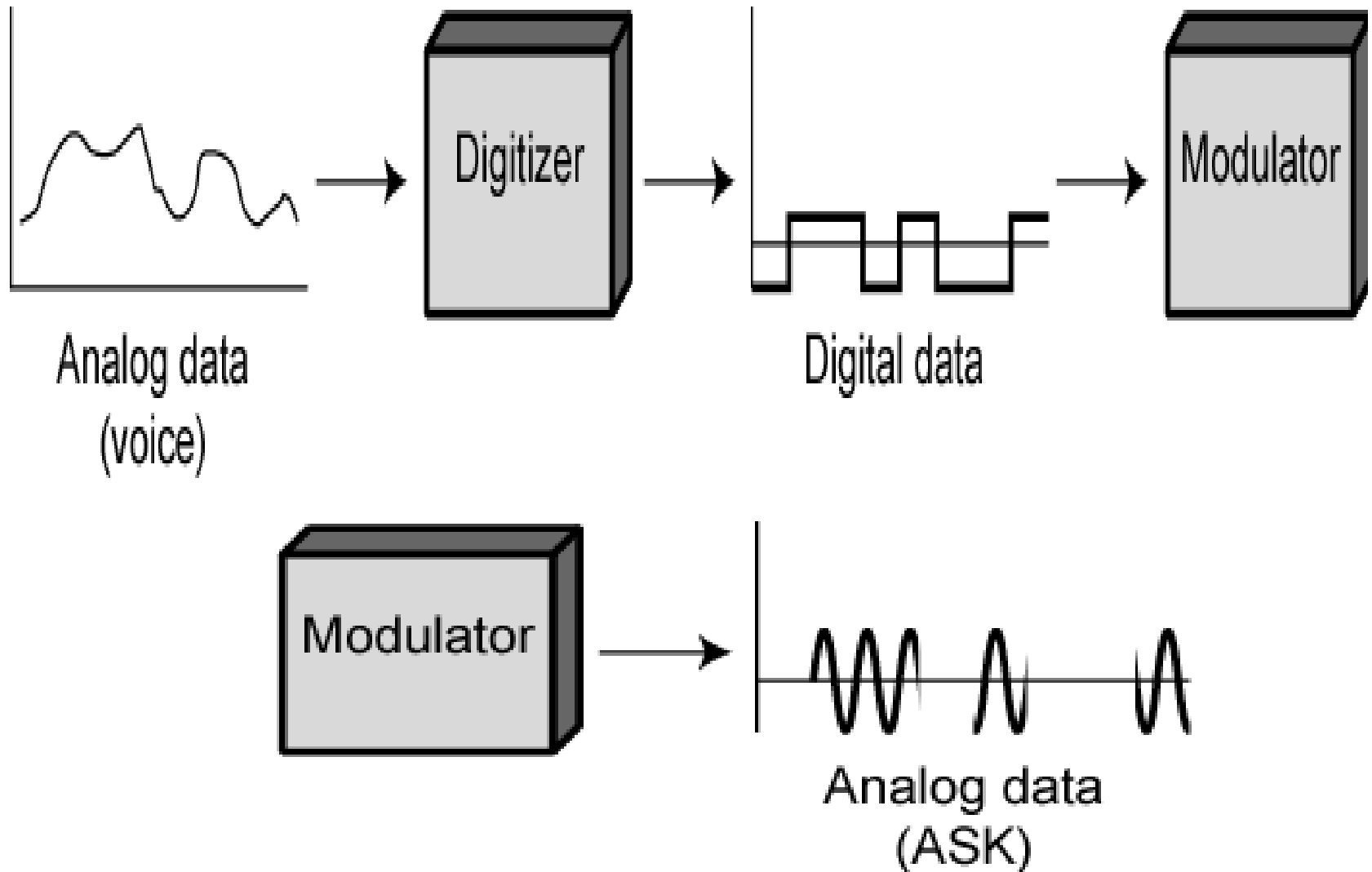
QAM Levels

- **Two level ASK**
 - Each of two streams in one of two states
 - Four state system
 - Essentially QPSK
- **Four level ASK**
 - Combined stream in one of 16 states
- **64 and 256 state systems have been implemented**
- **Improved data rate for given bandwidth**
 - Increased potential error rate

Analog Data, Digital Signal

- **Digitization**
 - Conversion of analog data into digital data
 - After this process several things can happen like:
 - Digital data can then be transmitted using NRZ-L
 - Digital data can then be transmitted using code other than NRZ-L
 - Digital data can then be converted to analog signal
 - Analog to digital conversion done using a codec(coder-decoder)
 - Two principal techniques used in codec:
 - Pulse code modulation
 - Delta modulation

Digitizing Analog Data



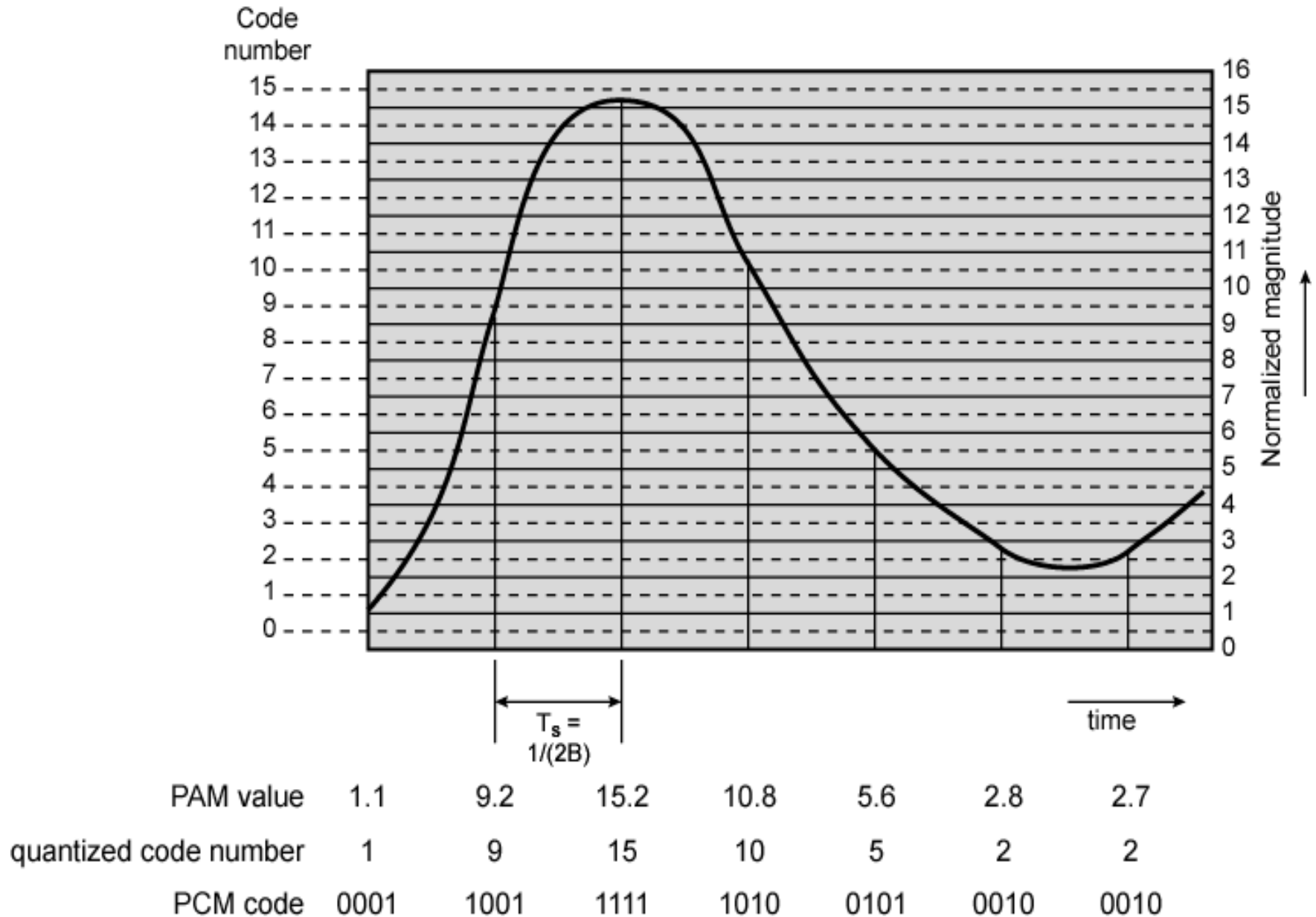
Pulse Code Modulation(PCM) (1)

- **Sampling Theorem: If a signal $f(t)$ is sampled at regular intervals at a rate higher than twice the highest signal frequency, the samples contain all the information of the original signal. $f(t)$ may be reconstructed from these samples using a low pass filter.**
 - (Proof - Stallings appendix 4A)
- **Voice data limited to below 4000Hz**
- **Require 8000 sample per second**
- **Analog samples called Pulse Amplitude Modulation (PAM) samples**
- **Each sample assigned digital value**

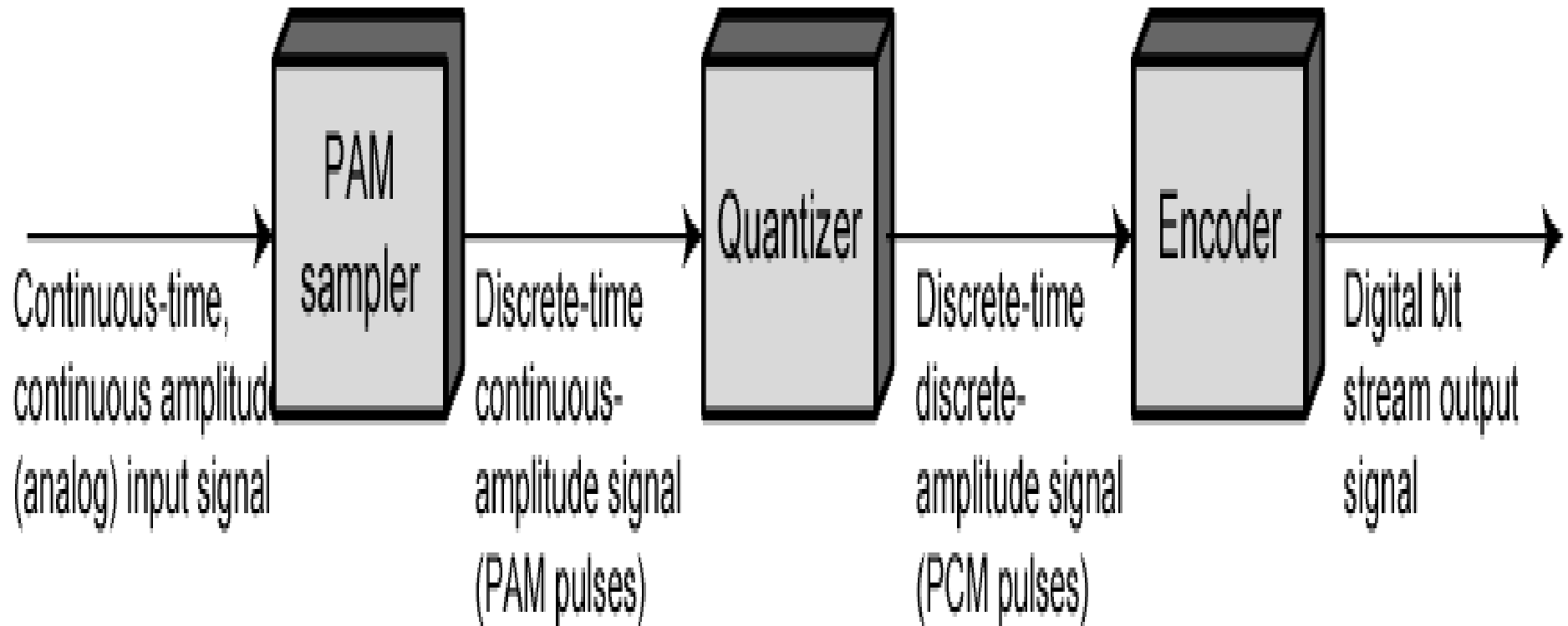
Pulse Code Modulation(PCM) (2)

- **4 bit system gives 16 levels**
- **Quantized**
 - **Quantizing error or noise**
 - **Approximations mean it is impossible to recover original exactly**
- **8 bit sample gives 256 levels**
- **Quality comparable with analog transmission**
- **8000 samples per second of 8 bits each gives 64kbps**

PCM Example



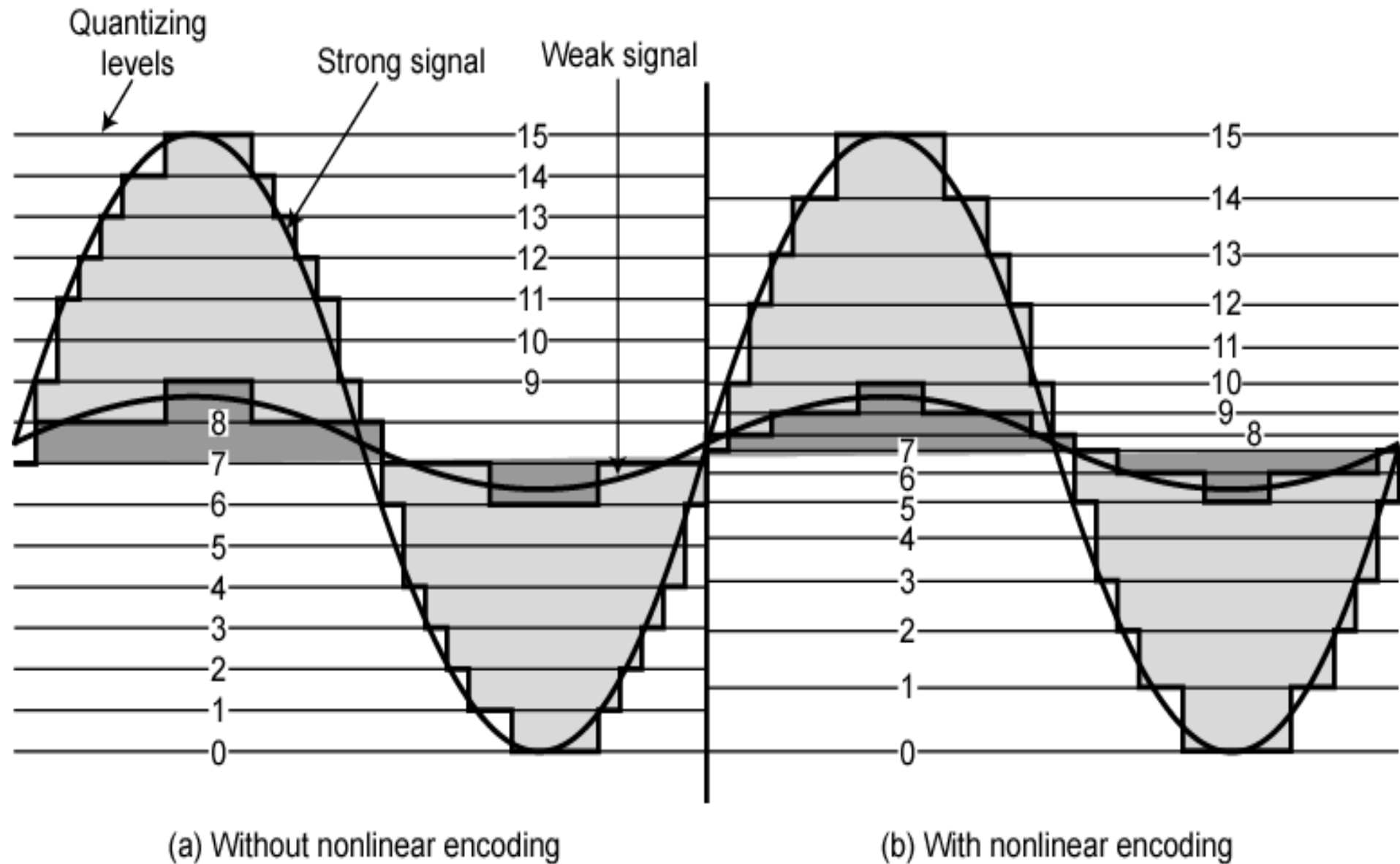
PCM Block Diagram



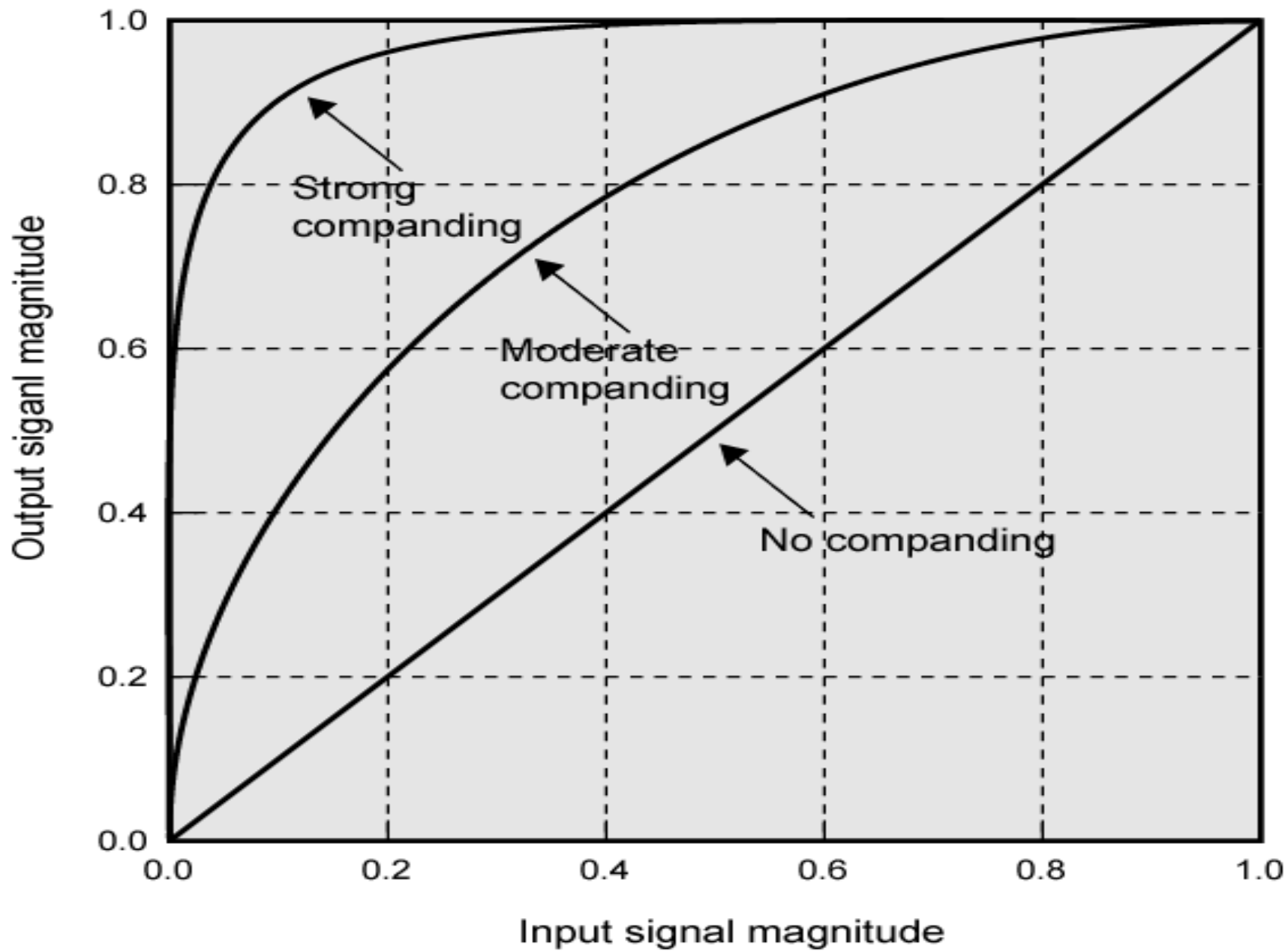
Nonlinear Encoding

- Quantization levels not evenly spaced
- Reduces overall signal distortion
- Can also be done by companding

Effect of Non-Linear Coding



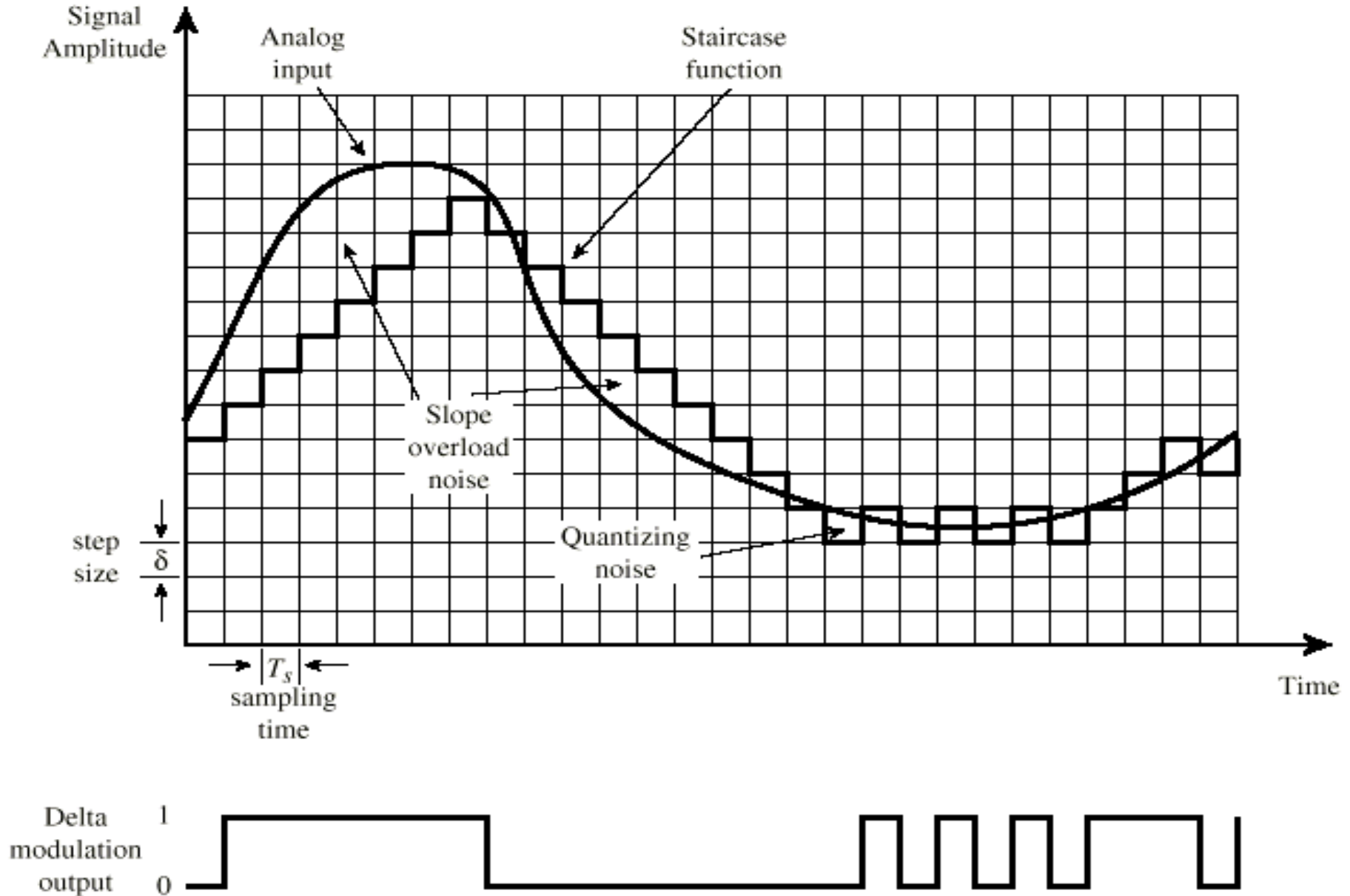
Typical Companding Functions



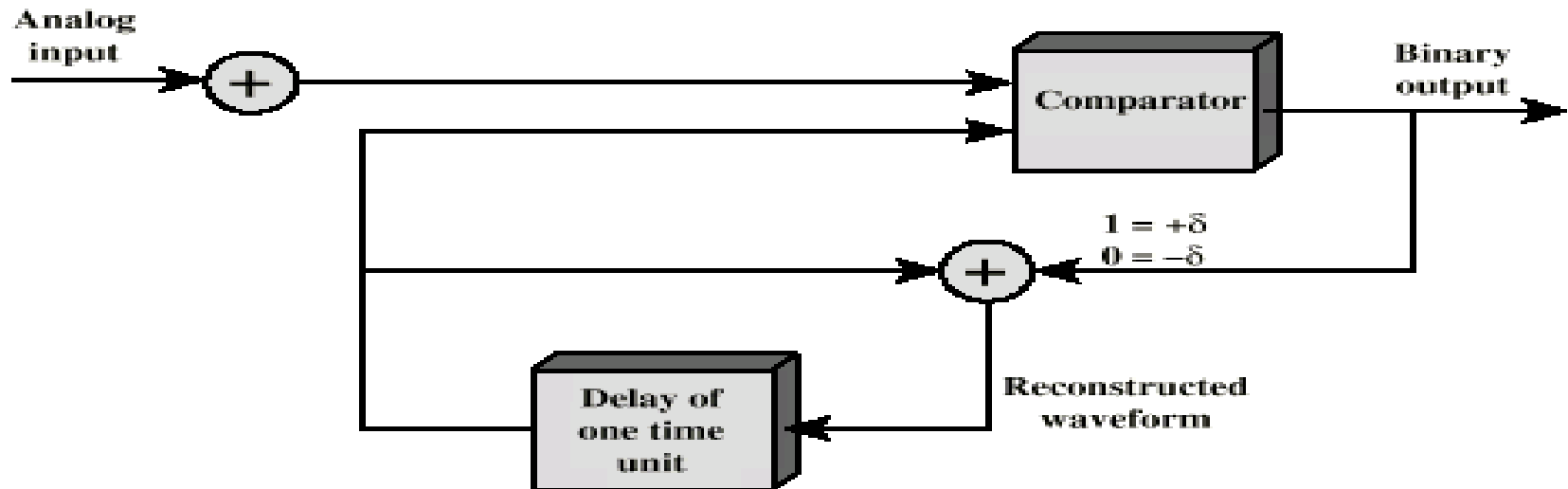
Delta Modulation

- **Analog input is approximated by a staircase function**
- **Move up or down one level (δ) at each sample interval**
- **Binary behavior**
 - **Function moves up or down at each sample interval**

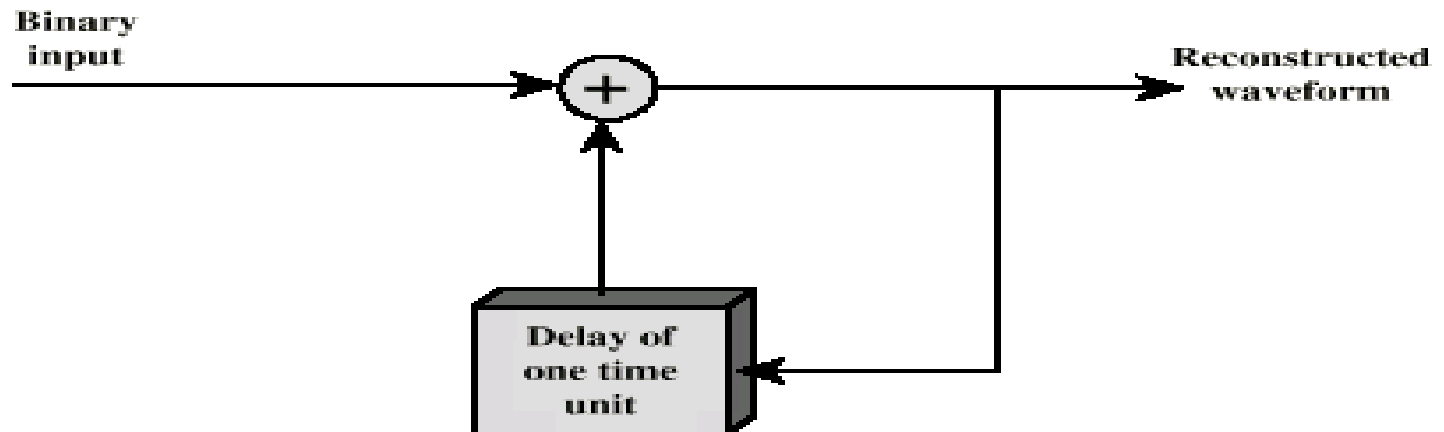
Delta Modulation – example



Delta Modulation – Operation



(a) Transmission



(b) Reception

Delta Modulation – Performance

- **Good voice reproduction**
 - **PCM - 128 levels (7 bit)**
 - **Voice bandwidth 4khz**
 - **Should be $8000 \times 7 = 56\text{kbps}$ for PCM**
- **Data compression can improve on this**
 - **e.g. Interframe coding techniques for video**

Analog Data, Analog Signals

- **Why modulate analog signals?**
 - **Higher frequency can give more efficient transmission**
 - **Permits frequency division multiplexing (chapter 8)**
- **Types of modulation**
 - **Amplitude**
 - **Frequency**
 - **Phase**

Analog Modulation

